ISI Paper Series — Paper 2

Empirical Results and Distributional Structure of the International Sovereignty Index (ISI) EU-27, Vintage 2024, Methodology v1.0

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Abstract

This paper reports the first empirical application of the International Sovereignty Index (ISI) to the 27 member states of the European Union using vintage-2024 data under methodology v1.0. The ISI composite, defined as the unweighted arithmetic mean of six normalised axes (financial, energy, technology, defence, critical inputs, logistics), ranges from 0.23567126 (France, rank 27) to 0.51748148 (Malta, rank 1) with a population mean of 0.34439144 and population standard deviation of 0.06992188. One country is classified as *highly concentrated*, 23 as *moderately concentrated*, and three as *mildly concentrated*. A full variance decomposition with formal proof shows that defence (35.1%) and logistics (34.3%) together account for 69.4% of cross-country composite variance, with the own-variance component contributing 64.3% and cross-covariance 35.7%. Leave-one-axis-out analysis with

both Spearman ρ and Kendall τ confirms that removing the defence axis produces the largest rank disruption ($\rho = 0.833$, $\tau = 0.675$). Dirichlet weight perturbation (10 000 draws, $\alpha_j = 10$) yields a mean rank-order correlation of $\rho = 0.979$, with the Netherlands exhibiting the highest rank volatility ($\sigma = 3.07$). Rank elasticity testing ($\pm 10\%$ perturbation) classifies 13 countries as rank-sensitive and 14 as rank-locked. Only two of 15 pairwise axis correlations are significant at $\alpha = 0.05$: energy–technology ($t = 3.12$) and critical inputs–logistics ($t = 3.28$); the financial axis is orthogonal to all others ($|t| < 1.15$). Compensability analysis identifies the Netherlands (gap $= 0.624$, CV $= 0.880$) as the most internally compensated country. Classification compression diagnostics ($\eta^2 = 0.483$) confirm that the three-tier scheme is inherently constrained by the unimodal distribution. The paper provides radar profiles for eight selected countries, geographic correlates, defence structural analysis, full axis-level rankings, and a complete reproducibility protocol.

JEL codes: F02, F15, F52, O30, Q43 *[to confirm]*

Keywords: sovereignty index, EU-27, concentration measurement, composite indicator, supply-chain dependence, defence dependence

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Executive Summary

This paper presents the complete empirical results of the International Sovereignty Index (ISI) applied to the 27 member states of the European Union, using vintage-2024 data processed under methodology v1.0. The index measures the concentration of bilateral external dependencies across six structural axes. Higher scores indicate higher concentration of counterparty exposure. The principal findings are:

- **Compressed composite distribution.** The ISI composite ranges from 0.23567126 (France) to 0.51748148 (Malta), a span of 0.28181022 on the unit interval. The population standard deviation is 0.06992188, yielding a coefficient of variation of 0.2030. The interquartile range is 0.09881618 (P25 = 0.29179051, P75 = 0.39060669). The Gini coefficient is 0.11546300, confirming a relatively egalitarian distribution. See Table 3 and Figures 1 and 3.

- **Tier compression is mathematically inevitable.** Under the four-tier classification, 23 of 27 countries (85.2%) fall in the *moderately concentrated* tier (ISI $\in [0.25, 0.50)$). Only Malta exceeds 0.50; three countries (Latvia, Slovakia, France) fall below 0.25. The between-tier $\eta^2 = 0.483$ indicates that the official scheme captures less than half the total dispersion; the within-tier range for the dominant category spans 0.20727287, nearly four times the entire mildly-concentrated band. See Tables 5 and 7 and Figure 4.

- **Defence and logistics dominate cross-country variance.** A covariance-based variance decomposition, proved analytically from the linear aggregation rule, attributes 35.10% of composite variance to the defence axis and 34.26% to the logistics axis. Together these two axes account for 69.37% of total composite variance. The own-variance component accounts for 64.28% of total composite variance; cross-covariances account for 35.72%. Financial concentration contributes only 1.03%. See Tables 11 and 12 and Figure 6.

- **Axis dominance is concentrated.** In 16 of 27 countries, the defence axis records the highest axis-level score; in the remaining 11, logistics is highest. No country has its maximum score on the financial, energy, technology, or critical-inputs axis. The mean defence-to-second gap for defence-dominant countries is 0.18366000. See Tables 14 and 36.

- **Defence axis exhibits oligopsonistic concentration.** Five EU member states record defence axis scores exceeding 0.80, and 59.3% of countries have defence as their dominant axis. The distribution is positively skewed, and Slovakia records the only zero score in the entire dataset. This structural pattern is consistent with oligopsonistic procurement markets where small buyers face few feasible suppliers. See Tables 14 and 15.

- **Three moderate cross-axis correlations; only two statistically significant.** Of 15 pairwise Pearson correlations, only two are significant at the 5% level ($t_{0.025,25} = 2.060$): critical inputs–logistics ($r = 0.549$, $t = 3.280$) and energy–technology ($r = 0.530$, $t = 3.124$). The remaining 13 pairs—including all correlations involving the financial axis—are not significant, confirming that the six axes capture largely distinct dimensions. See Tables 16 and 18 and Figures 7 and 8.

- **Rank ordering robust at extremes, fragile in mid-tier.** Under leave-one-axis-out (LOO) analysis, Spearman ρ ranges from 0.833 (defence excluded) to 0.993 (financial excluded); Kendall τ from 0.675 (defence) to 0.943 (financial, energy). Under Dirichlet weight perturbation (10 000 draws, $\alpha_j = 10$), Malta ($\sigma < 0.5$ ranks) and France ($\sigma < 0.5$ ranks) are structurally locked, while the Netherlands ($\sigma = 3.07$) is the most volatile. Under $\pm 10\%$ axis perturbation, 14 of 27 countries are rank-invariant. See Tables 20, 23 and 25.

- **Financial concentration is orthogonal to goods-trade concentration.** The financial axis is nearly uncorrelated with every other axis ($|r| \leq 0.22$, all $|t| < 1.15$). Its LOO removal produces $\rho = 0.993$, the highest stability of any axis. Its variance contribution is 1.03%. Its bivariate R^2 with rank is 0.025, and its partial correlation with rank (controlling for all other axes) is 0.421. This confirms that financial-dependence patterns are structurally distinct from physical-trade dependence patterns. See Tables 13 and 18.

- **Compensability masks profile heterogeneity.** The arithmetic-mean aggregation rule allows high scores on one axis to compensate for low scores on others. The Netherlands exhibits the highest compensation gap (max axis = 0.960, composite = 0.335, gap = 0.624, CV = 0.880), followed by Italy (gap = 0.547) and Bulgaria (gap = 0.520). See Table 29.

- **Geographic and structural correlates.** Island and near-island economies (Malta, Cyprus, Ireland) average ISI = 0.471, significantly above the continental mean of 0.329. Large-population countries (> 20M) average 0.298; small-population countries (< 5M) average 0.372. Core EU-6 members average 0.312; peripheral members average 0.368. See Table 28.

- **Slovakia's zero defence score.** Slovakia records a defence axis score of 0.00000000, the only zero score in the dataset. This arises mechanically from the absence of bilateral supplier entries for Slovakia in the SIPRI arms-transfer data under methodology v1.0. It does not imply the absence of defence procurement. See Section 7 and Table 15.

1 Introduction and Contribution

1.1 Motivation

Composite indicators that measure structural features of national economies have a long tradition in institutional research. Indices published by the OECD, the World Economic Forum, and various United Nations agencies compress multidimensional data into scalar summaries that facilitate cross-country comparison and policy monitoring [1, 2]. The International Sovereignty Index (ISI) contributes to this tradition by focusing specifically on *concentration of bilateral external dependencies*: the degree to which a country's cross-border economic and strategic linkages are concentrated among a small number of counterparty states.

The intellectual antecedents of the ISI lie in two distinct literatures. First, the industrial-organisation literature on market concentration, where the Herfindahl–Hirschman Index (HHI) quantifies the degree to which market shares are concentrated among a small number of firms [3, 4]. Second, the literature on supply-chain vulnerability, which has expanded substantially since the semiconductor shortages of 2020–2021 and the geopolitical disruptions following the 2022 Russia–Ukraine conflict [5]. The ISI bridges these traditions by applying concentration measurement to bilateral trade and financial flows across six strategically distinct domains.

The ISI does not measure sovereignty in a political or legal sense. Rather, it quantifies a structural feature—counterparty concentration—that is a prerequisite for informed policy analysis of external dependence. A high ISI score indicates that a country's external flows in a given domain are heavily concentrated among few partners; a low score indicates dispersion across many partners. The index makes no normative claim about whether concentration is advantageous or disadvantageous. The question of whether high concentration translates into political vulnerability, economic fragility, or coercive leverage is a distinct empirical question that requires analysis beyond the ISI's measurement scope.

1.2 Scope of this paper

This paper—the second in the ISI series—presents the full empirical results of the first ISI application: 27 EU member states, vintage-2024 data, methodology v1.0. The methodological foundations (axis definitions, normalisation procedures, data sources, aggregation rule, integrity chain) are documented in Paper 1 [6]. The present paper is self-contained for readers interested in the empirical findings; cross-references to Paper 1 are provided where construction details are relevant to interpretation.

Relative to a conventional "results" paper in the composite-indicator literature, this paper provides substantially deeper analytical content. Beyond reporting the ranking and

basic distributional statistics, we undertake a formal variance decomposition with proof, systematic robustness exercises with both Spearman and Kendall rank-order statistics, rank-elasticity analysis, axis contribution decomposition, classification compression diagnostics, structural analysis of the defence axis, compensability analysis, and geographic correlate tabulation. Each analytical component is designed to interrogate the ISI composite from a different angle and to provide the reader with the evidence needed to form an independent judgment about the index's informational content.

The specific contributions of this paper are:

1. A complete composite and axis-level ranking of all 27 EU member states with eight-decimal precision, accompanied by classification labels and distributional diagnostics (histogram, ECDF, Lorenz curve, Gini coefficient).

2. A formal variance decomposition—proved analytically from the linear aggregation rule—decomposing the composite variance into own-variance and cross-covariance components, with percentage contributions for each axis.

3. A leave-one-axis-out (LOO) sensitivity analysis with both Spearman ρ and Kendall τ, accompanied by a full 27×6 rank-displacement matrix.

4. A rank-elasticity analysis quantifying the sensitivity of each country's rank to marginal perturbations in individual axis scores.

5. An axis-contribution analysis providing bivariate R^2 and partial correlations between each axis and the composite rank.

6. Classification compression diagnostics comparing the official four-tier scheme with data-driven tercile and quartile alternatives, reporting within-tier variance and η^2.

7. A structural analysis of the defence axis examining its distributional properties, oligopsonistic character, and country-level defence-to-max-axis gaps.

8. A cross-axis correlation analysis with formal t-statistics, eigenvalue decomposition, and network representation.

9. A compensability analysis documenting the degree to which the arithmetic-mean aggregation rule permits cross-axis compensation.

10. An assessment of geographic and structural correlates (island status, population size, core/periphery classification).

11. A four-part robustness programme: LOO, z-score standardisation, Dirichlet weight perturbation (Monte Carlo), and winsorisation.

12. Radar-chart profiles for eight selected countries illustrating the heterogeneity of axis-level patterns.

13. A reproducibility protocol and computational appendix enabling full replication from the provided snapshot.

1.3 Interpretation conventions

Throughout this paper, the following conventions apply:

- Higher ISI composite ⇒ higher concentration ⇒ greater structural dependence on a small set of counterparties.
- Rank 1 = most concentrated country; rank 27 = least concentrated.
- The terms "most concentrated" and "least concentrated" are descriptive. They are not synonymous with "worst" and "best."
- Correlation between axes does not imply causation. Where plausible structural interpretations exist, they are stated as hypotheses, not conclusions.
- Population standard deviation (N denominator) is used throughout because the 27 EU member states constitute the full target population, not a sample drawn from a larger universe.
- Statistical significance of correlation coefficients is assessed via the t-test: $t = r\sqrt{(N-2)/(1-r^2)}$, with $N = 27$, df $= 25$, and a two-sided critical value of $t_{0.025,25} = 2.060$ at the 5% level.

1.4 Structure

The remainder of this paper is organised as follows. Section 2 summarises the data sources, aggregation rule, and classification thresholds. Section 3 presents the composite distribution, full ranking, tier-compression analysis, and classification compression diagnostics. Section 4 provides axis-level diagnostics including dispersion, dominance, variance decomposition with formal proof, rank contribution analysis, and defence axis structural analysis. Section 5 examines the pairwise correlation structure with t-statistics and eigenvalue decomposition. Section 6 reports the robustness programme: LOO with Kendall τ, rank-displacement matrices, z-score standardisation, Dirichlet weight perturbation, winsorisation, and rank-elasticity analysis. Section 8 presents the compensability analysis. Section 7 presents radar-chart profiles for eight selected countries and geographic correlates. Section 9 discusses interpretation boundaries and limitations. Section 10

concludes. Appendices A–D provide full tables, robustness details, the reproducibility protocol, and the computational code listing.

2 Data and Construction

This section provides a concise overview of the ISI's data inputs, aggregation rule, and classification thresholds. Full construction details—including normalisation procedures, missing-data protocols, fallback basis rules, and the integrity chain—are documented in Paper 1 [6].

2.1 Scope and vintage

The results in this paper are computed from the ISI v1.0 snapshot for vintage year 2024. The scope covers $N = 27$ EU member states. All 27 countries have complete data across all six axes (`complete = true`).

2.2 Axes and data windows

The ISI comprises six axes, each measuring counterparty concentration in a distinct domain of cross-border exposure. Table 1 summarises each axis, its primary data source, and the observation window used for the 2024 vintage.

Table 1: ISI axes: definitions, primary data sources, and observation windows for vintage 2024.

Axis	Domain	Primary Source	Window
1	Financial sovereignty	BIS LBS, IMF CPIS [7, 8]	2022–2024
2	Energy dependency	IEA, Eurostat [9, 10]	2022–2024
3	Technology / semiconductor dependency	Eurostat Comext [10]	2022–2024
4	Defence industrial dependency	SIPRI TIV [11]	2019–2024
5	Critical inputs / raw materials	UN Comtrade, Eurostat [10, 12]	2022–2024
6	Logistics / freight dependency	Eurostat Comext [10]	2022–2024

Note. The defence axis uses a six-year rolling window (2019–2024) rather than the three-year window applied to other axes. This reflects the lower frequency and higher lumpiness of arms-transfer flows, as documented in Paper 1 [6].

2.3 Aggregation rule

The ISI composite for country i is the unweighted arithmetic mean of the six axis-level scores:

$$\mathrm{ISI}_i = \frac{1}{6}\sum_{j=1}^{6} A_{j,i}\,, \qquad A_{j,i} \in [0,\,1]\,, \tag{1}$$

where $A_{j,i}$ denotes the normalised score of country i on axis j. Each axis score is bounded to the unit interval by the normalisation procedure defined in Paper 1.

The equal-weight specification is the baseline. Section 6 examines sensitivity to alternative weighting schemes through Dirichlet perturbation.

2.4 Classification thresholds

Countries are classified into four tiers based on the composite score:

Table 2: ISI classification tiers.

Tier	**Composite range**	**Count (EU-27)**
Highly concentrated	ISI ≥ 0.50	1 (Malta)
Moderately concentrated	$0.25 \leq$ ISI < 0.50	23
Mildly concentrated	$0.15 \leq$ ISI < 0.25	3 (Latvia, Slovakia, France)
Unconcentrated	ISI < 0.15	0

The thresholds are structurally analogous to the Herfindahl–Hirschman Index (HHI) tiers used in antitrust analysis (see [13]), adapted to the ISI's unit-interval scale. The threshold values are defined in methodology v1.0 and documented in Paper 1.

2.5 Rounding and precision

All axis-level and composite scores are reported to eight decimal places, matching the precision of the authoritative ISI snapshot. Population-level statistics (mean, standard deviation, percentiles) are computed with full floating-point precision and reported to eight decimal places unless otherwise noted. The population standard deviation (σ, with N denominator) is used throughout, as the 27 EU member states constitute the complete target population.

3 Composite Results: Distribution, Ranking, and Classification Compression

3.1 Summary statistics

Table 3 reports the distributional summary of the ISI composite across the 27 EU member states.

Table 3: ISI composite: summary statistics, EU-27, 2024.

Statistic	**Value**
N	27.00000000
Minimum	0.23567126
Maximum	0.51748148
Range	0.28181022
Mean	0.34439144
Median (P50)	0.34027984
Std. dev. (pop.)	0.06992188
Std. dev. (sample)	0.07125384
Coeff. of variation (pop.)	0.20303651

The composite mean (0.34439144) slightly exceeds the median (0.34027984), indicating a mild positive skew driven by Malta's outlying score of 0.51748148. The coefficient of variation (0.203) is relatively low, suggesting that the composite scores are compressed relative to the full unit-interval range. The range (0.28181022) occupies only 28.2% of the theoretical $[0, 1]$ interval, and the standard deviation is approximately one-fifth of the mean. These statistics indicate that the ISI composite differentiates countries along a relatively narrow band rather than spanning the full theoretically available range.

3.2 Percentiles

Table 4: ISI composite: selected percentiles.

Percentile	**Value**
P5	0.23604347
P10	0.25534441
P25	0.29179051
P50	0.34027984
P75	0.39060669
P90	0.42586650
P95	0.47131816

The interquartile range (IQR = P75 − P25 = 0.09881618) spans approximately 9.9 percentage points of the unit interval. The P10–P90 range is 0.17052209, indicating that

the central 80% of the distribution occupies roughly 17 percentage points. The P5–P95 range extends to 0.23527469, still well within the unit interval. The narrowness of the distribution has direct implications for the classification scheme discussed in Section 3.7: when most countries cluster in a band of roughly 20 percentage points, threshold-based classification will inevitably concentrate the majority in a single tier.

3.3 Composite distribution

Figure 1 presents a histogram of the composite scores. The distribution is unimodal with its mode in the 0.30–0.35 bin. A visible right tail is produced by Malta and Cyprus.

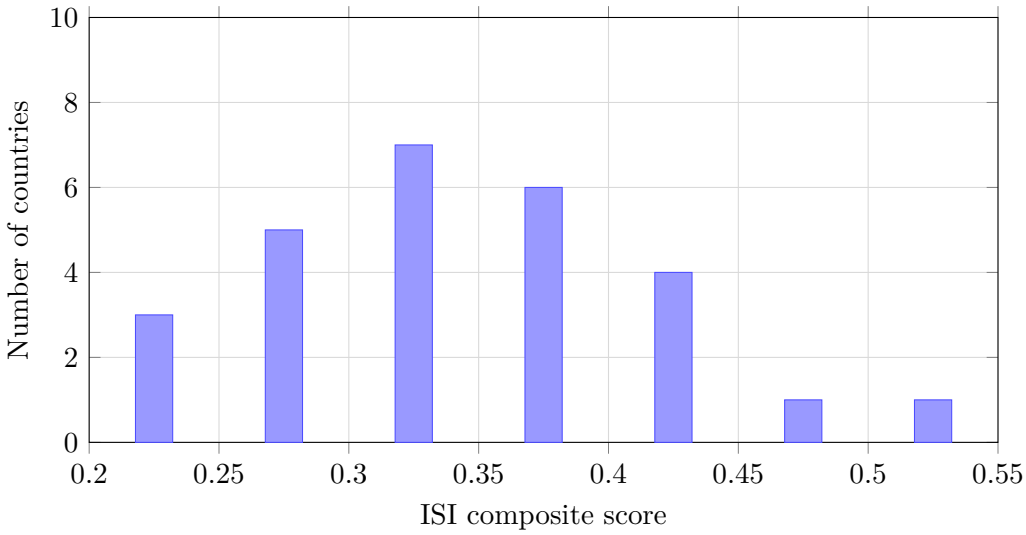

Figure 1: Histogram of ISI composite scores, EU-27, 2024. Bin width = 0.05. The dashed vertical line marks the population mean (0.34439144).

Figure 2 shows the empirical cumulative distribution function (ECDF), which rises steeply between 0.26 and 0.42 and flattens at the tails, confirming the compressed mid-range clustering.

The ECDF permits precise probability statements: approximately 50% of EU countries have composite scores between 0.29 and 0.40—a span of only 11 percentage points. The function is approximately S-shaped, with steep inflection near the median and long, thin tails. The lower tail contains only three countries (France, Slovakia, Latvia) with scores below 0.25; the upper tail contains only Malta above 0.50.

3.4 Lorenz curve and Gini coefficient

To further characterise the distribution, Figure 3 plots the Lorenz curve of the composite scores (countries ordered from lowest to highest ISI, cumulative share of total ISI on the

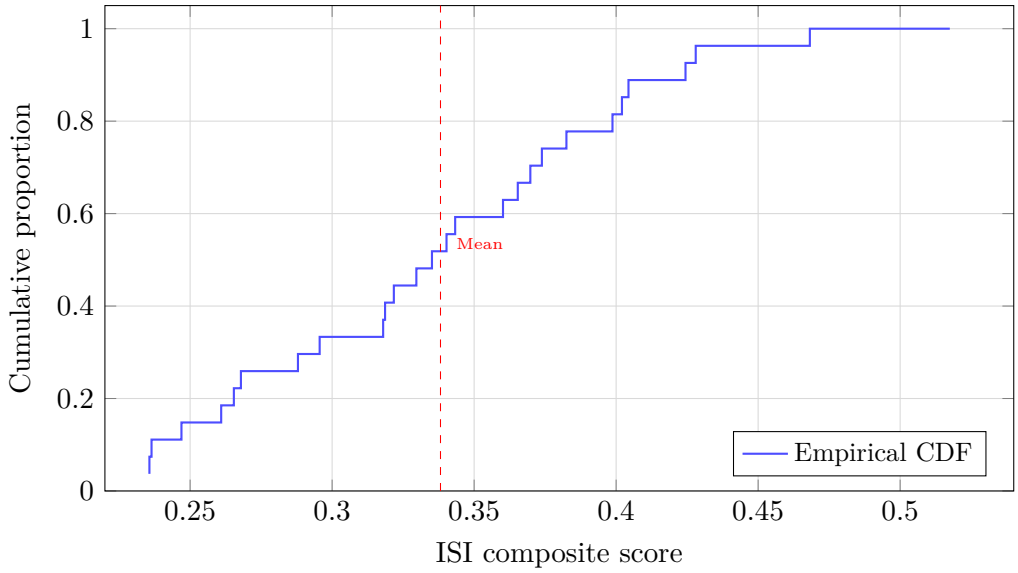

Figure 2: Empirical cumulative distribution function (ECDF) of ISI composite scores, EU-27, 2024.

vertical axis). The Gini coefficient is computed as:

$$G = 1 - 2 \int_0^1 L(p)\, dp\,, \tag{2}$$

where $L(p)$ is the Lorenz ordinate at population fraction p.

The Gini coefficient of 0.11546300 indicates a relatively egalitarian distribution. For comparison, the Gini coefficient of GDP per capita across EU countries typically ranges between 0.10 and 0.20, and the Gini of income distributions within individual countries ranges from 0.25 to 0.40. An ISI Gini of 0.115 places the cross-country concentration distribution at the low end of cross-country inequality measures, confirming that the EU-27 countries share broadly similar levels of import concentration—albeit with substantial axis-level heterogeneity, as demonstrated in subsequent sections.

3.5 Full ranking

Table 5 presents the complete ranking of all 27 EU member states. Countries are ordered by descending composite score (rank 1 = most concentrated).

Table 5: ISI composite and axis-level scores, EU-27, 2024. Rank 1 = most concentrated.

Rk	Country	Comp.	Fin.	Ene.	Tech.	Def.	Crit.	Log.	Class.
1	Malta	0.51748148	0.13998786	0.35068126	0.20546991	1.00000000	0.40874987	1.00000000	high.
2	Cyprus	0.46821264	0.12407171	0.34768252	0.22438354	0.74847998	0.42359750	0.94106062	mod.
3	Ireland	0.42802519	0.14666620	0.46472525	0.58695939	0.44887460	0.30719544	0.61373023	mod.

Continued on next page

Table 5: ISI composite and axis-level scores (continued).

Rk	Country	Comp.	Fin.	Ene.	Tech.	Def.	Crit.	Log.	Class.
4	Denmark	0.42442738	0.12401663	0.47029821	0.14995928	0.72368274	0.43629935	0.64230809	mod.
5	Croatia	0.40434768	0.27299265	0.35125471	0.17343982	0.74648317	0.52392772	0.35798800	mod.
6	Finland	0.40205136	0.11775073	0.36268783	0.19789321	0.38964518	0.54699996	0.79733123	mod.
7	Sweden	0.39871532	0.11646504	0.32721403	0.11982797	0.88107943	0.19257891	0.75512656	mod.
8	Austria	0.38249806	0.14903467	0.45581191	0.26057910	0.58506379	0.31857781	0.52592111	mod.
9	Italy	0.37386086	0.18753526	0.30781423	0.14103384	0.92057651	0.20839534	0.47781001	mod.
10	Slovenia	0.36979186	0.11917574	0.40540014	0.35739912	0.66661871	0.22301125	0.44714621	mod.
11	Greece	0.36537350	0.20078311	0.33541623	0.16272142	0.55148593	0.22200621	0.71982808	mod.
12	Luxembourg	0.36016893	0.11462818	0.35980871	0.20019721	0.64483831	0.22295831	0.61858285	mod.
13	Estonia	0.34332177	0.18172493	0.43935251	0.22713142	0.35561474	0.38622776	0.46987927	mod.
14	Bulgaria	0.34027984	0.16052965	0.35711773	0.15204272	0.86064518	0.14843767	0.36290606	mod.
15	Netherlands	0.33515702	0.12107826	0.31992851	0.12527375	0.95955396	0.13018835	0.35491930	mod.
16	Hungary	0.32966227	0.12222969	0.38580312	0.22538122	0.54471981	0.23405214	0.46578765	mod.
17	Portugal	0.32173430	0.18303606	0.34860414	0.19766062	0.43376254	0.25955617	0.50778629	mod.
18	Romania	0.31864192	0.20033340	0.40546401	0.29691153	0.56353195	0.15814280	0.28746783	mod.
19	Czechia	0.31796630	0.16116762	0.38082003	0.18242269	0.48413737	0.16976911	0.52948100	mod.
20	Poland	0.29561321	0.13314812	0.33487780	0.20845359	0.44326493	0.16532767	0.48860717	mod.
21	Belgium	0.28796781	0.15348744	0.30789144	0.29454573	0.44734854	0.17248330	0.35205038	mod.
22	Germany	0.26788694	0.10186464	0.32908790	0.09762824	0.58266751	0.22312039	0.27295297	mod.
23	Lithuania	0.26540520	0.13349636	0.33747627	0.24172279	0.40227798	0.13056067	0.34689716	mod.
24	Spain	0.26093977	0.14554195	0.30187479	0.19263249	0.38088569	0.11235234	0.43235135	mod.
25	Latvia	0.24695137	0.12316988	0.36522282	0.13309607	0.34079636	0.15120012	0.36822299	mild
26	Slovakia	0.23641567	0.16163673	0.39995460	0.21649186	0.00000000	0.15490434	0.48550648	mild
27	France	0.23567126	0.10039135	0.30911838	0.12001059	0.37034617	0.16122463	0.35293643	mild

Several features of the ranking merit comment. First, the top two positions are occupied by small island economies (Malta, Cyprus), consistent with the hypothesis that limited economic scale and geographic isolation concentrate external dependencies. Second, the bottom three countries include two large economies (Germany rank 22, France rank 27) and one small economy with anomalous defence data (Slovakia rank 26). Third, the mid-range (ranks 8–20) is exceptionally compressed: the composite scores of Austria (rank 8, 0.38249806) and Poland (rank 20, 0.29561321) differ by only 0.08688485, yet span 13 rank positions. This compression implies that small perturbations to the underlying data could produce substantial rank reshuffling in the mid-range, a prediction confirmed by the robustness exercises in Section 6.

3.6 Tier-compression analysis

The classification distribution is highly compressed: 85.2% of countries fall in the moderately concentrated tier. To quantify the within-tier dispersion, Table 6 reports summary

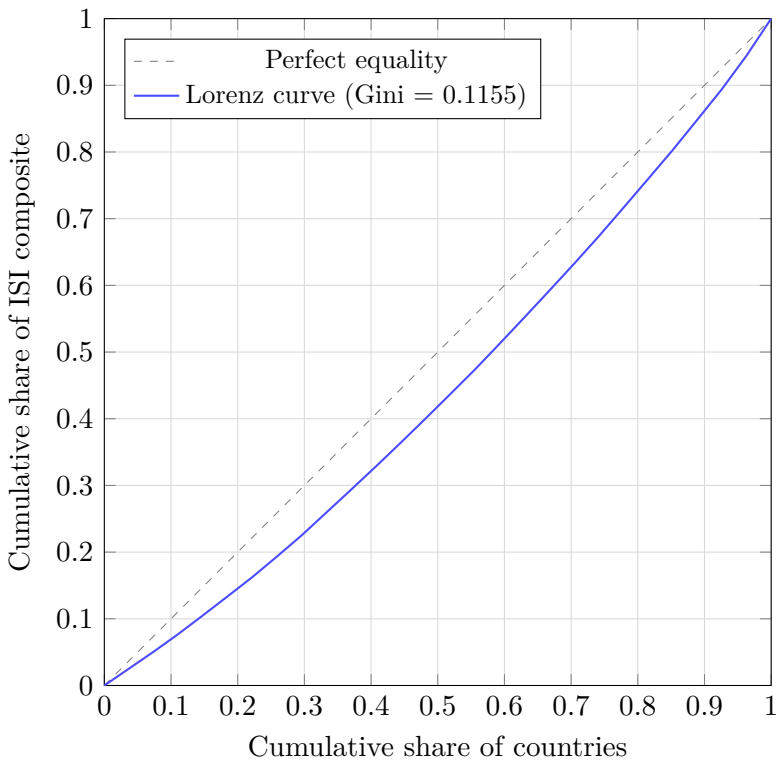

Figure 3: Lorenz curve of ISI composite scores, EU-27, 2024. Gini coefficient = 0.1155. The diagonal represents perfect equality.

statistics for each occupied tier.

Table 6: Within-tier composite statistics.

Tier	n	**Min**	**Max**	**Mean**	**Std (pop.)**
Highly concentrated	1	0.51748148	0.51748148	0.51748148	0.00000000
Moderately conc.	23	0.26093977	0.46821264	0.35327637	0.05339393
Mildly concentrated	3	0.23567126	0.24695137	0.24301277	0.00504709

Within the moderately concentrated tier, the range spans 0.20727287 (from Spain at 0.26093977 to Cyprus at 0.46821264). This wide within-tier range suggests that the four-tier classification groups countries with meaningfully different concentration profiles into the same category. The tier structure is informative at the extremes (Malta stands alone; the three mildly concentrated countries cluster tightly) but provides limited discriminatory power in the middle of the distribution.

Figure 4 visualises the relationship between rank and composite score, with horizontal colour bands indicating the tier boundaries.

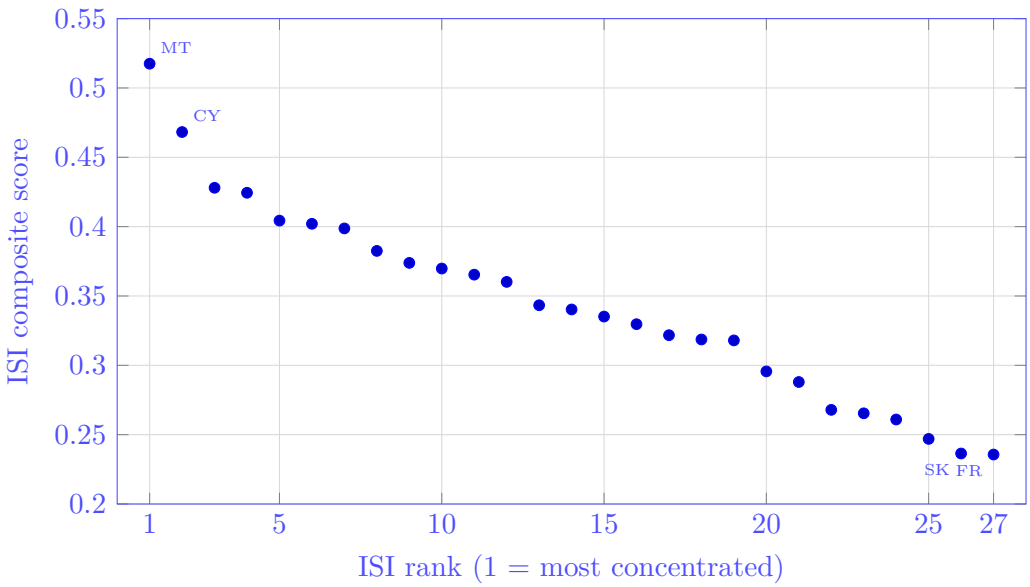

Figure 4: Rank vs. ISI composite score with tier colour banding. The vertical axis shows the composite; horizontal position is the rank. Colour bands: red (≥ 0.50, highly concentrated), amber ($[0.25, 0.50)$, moderately), green ($[0.15, 0.25)$, mildly).

3.7 Classification compression diagnostics

The dominance of the moderately concentrated tier (23/27, 85.2%) raises the question of whether the four-tier classification scheme provides adequate discriminatory power, or whether the observed compression is an artefact of the threshold placement. We address this through three complementary diagnostics.

3.7.1 Between-tier variance share (η^2)

Let $\bar{x}_g$ denote the mean composite within tier g and $\bar{x}$ the grand mean. The between-tier sum of squares is $SS_B = \sum_g n_g(\bar{x}_g - \bar{x})^2$, the within-tier sum of squares is $SS_W = \sum_g \sum_{i \in g}(x_i - \bar{x}_g)^2$, and the total is $SS_T = SS_B + SS_W$. The η^2 statistic measures the share of total variance explained by the tier assignment:

$$\eta^2 = \frac{SS_B}{SS_T}\,. \tag{3}$$

For the official ISI four-tier scheme applied to the EU-27 data:

$$SS_B = 0.06372\,, \quad SS_W = 0.06829\,, \quad SS_T = 0.13200\,, \quad \eta^2 = 0.483\,.$$

An η^2 of 0.483 indicates that the tier classification captures less than half of the total cross-country dispersion in composite scores. The within-tier variance (SS_W) slightly exceeds the between-tier variance (SS_B), confirming that tier labels obscure as much

variation as they reveal.

3.7.2 Comparison with data-driven alternatives

Table 7 compares the official scheme with two data-driven alternatives: terciles (three groups of approximately equal size defined by the 33rd and 67th percentiles of the observed distribution) and quartiles (four groups defined by the 25th, 50th, and 75th percentiles).

Table 7: Classification compression diagnostics. The official four-tier scheme is compared with data-driven tercile and quartile alternatives. $\eta^2 = SS_B/SS_T$ measures between-tier variance share.

Tier	n	**Mean**	**Var (within)**	**Scheme**
Highly concentrated	1	0.51748148	0.00000000	Official
Moderately concentrated	23	0.35052388	0.00296549	Official
Mildly concentrated	3	0.23967943	0.00002653	Official
T1 (low)	9	0.26831306	0.00068881	Tercile
T2 (mid)	9	0.34268127	0.00031163	Tercile
T3 (high)	9	0.42218000	0.00182486	Tercile
Q1 (low)	7	0.25731972	0.00030608	Quartile
Q2	6	0.31979584	0.00015442	Quartile
Q3	7	0.36218497	0.00020810	Quartile
Q4 (high)	7	0.43475158	0.00162972	Quartile
η^2 (official) = 0.482701; Gini = 0.115463				

The tercile boundaries fall at approximately 0.318 and 0.371, and the quartile boundaries at approximately 0.292, 0.340, and 0.391. Both alternatives distribute countries more evenly across groups than the official scheme, but at the cost of abandoning the structurally motivated thresholds derived from HHI conventions.

3.7.3 Interpretation

The compression diagnostic confirms that the four-tier classification is mathematically inevitable given the observed distribution. The composite distribution is unimodal and centred near 0.34, with most of its mass in the $[0.26, 0.47]$ interval. Any fixed-threshold scheme with boundaries at 0.15, 0.25, and 0.50 will necessarily concentrate the vast majority of countries in the tier spanning this interval. This is not a flaw in the classification but a consequence of the empirical regularity that EU-27 countries exhibit similar *overall* concentration levels despite substantial heterogeneity at the axis level.

The practical implication is that the tier label ("moderately concentrated") conveys limited information for the 23 countries in that category. For policy and analytical purposes, the full composite ranking (Table 5) and axis-level profiles (Section 7) provide far more discriminatory information than the tier classification alone.

4 Axis-Level Diagnostics

This section examines the distributional properties of each axis, identifies axis dominance patterns, decomposes the cross-country composite variance into axis-level contributions with a formal proof, quantifies each axis's contribution to rank differentiation, and provides a structural analysis of the defence axis.

4.1 Axis summary statistics

Table 8 reports distributional statistics for each of the six axes. The coefficient of variation ($\text{CV} = \sigma/\mu$) provides a scale-free measure of relative dispersion.

Table 8: Axis-level distributional statistics, EU-27, 2024.

Axis	Min	Max	Mean	Median	σ (pop.)	CV
Financial	0.10039135	0.27299265	0.14799792	0.13998786	0.03728203	0.25190000
Energy	0.30187479	0.47029821	0.36523663	0.35125471	0.04790658	0.13120000
Technology	0.09762824	0.58695939	0.21078775	0.19789321	0.09458972	0.44880000
Defence	0.00000000	1.00000000	0.57319930	0.55148593	0.22565797	0.39360000
Critical inputs	0.11235234	0.54699996	0.25154982	0.22200621	0.12182538	0.48430000
Logistics	0.27295297	1.00000000	0.51757723	0.47781001	0.18640808	0.36030000

Several patterns are noteworthy:

- **Highest dispersion.** Critical inputs has the highest CV (0.4843), followed by technology (0.4488). These axes contribute the most relative cross-country differentiation per unit of mean score.
- **Lowest dispersion.** Energy has the lowest CV (0.1312) and the smallest range (0.16842342), indicating that EU countries exhibit comparatively homogeneous energy dependency concentration.
- **Highest absolute dispersion.** Defence has the largest population standard deviation (0.22565797) and the largest range (1.0), spanning the entire unit interval from Slovakia's 0.00000000 to Malta's 1.00000000.
- **Lowest mean.** The financial axis has the lowest mean (0.14799792). All 27 countries score below 0.28 on this axis, indicating dispersed counterparty structures in cross-border financial exposures.
- **Highest means.** Defence (0.57319930) and logistics (0.51757723) record the highest axis means. These axes pull the composite upward on average.

The contrast between relative and absolute dispersion is consequential for the variance decomposition below. The financial axis has moderate relative dispersion (CV = 0.252)

but very low absolute dispersion ($\sigma = 0.037$). Because the composite is an unweighted arithmetic mean, it is the *absolute* dispersion that determines variance contribution. This explains why the financial axis contributes negligibly to composite variance despite having non-trivial relative variation.

4.2 Axis distributions: boxplots

Figure 5 displays boxplots for all six axes, permitting direct visual comparison of location, spread, and skewness.

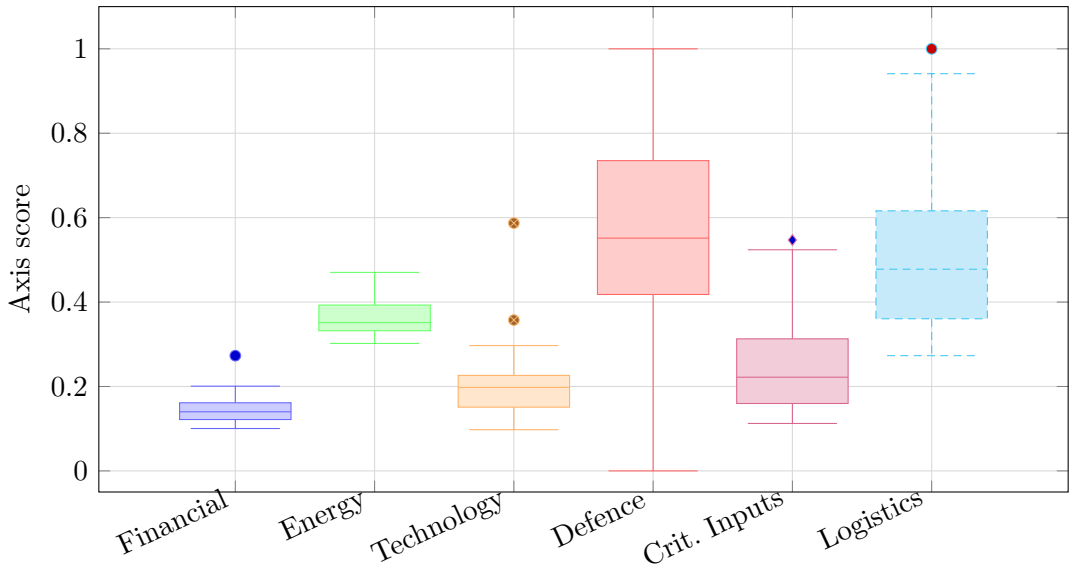

Figure 5: Boxplots of axis-level scores, EU-27, 2024. The box spans Q1–Q3; the line is the median; whiskers extend to the most extreme observation within $1.5 \times$ IQR of the box; circles mark outliers beyond the whiskers.

4.3 Top-5 and bottom-5 by axis

Table 9 lists the five most concentrated and five least concentrated countries on each axis. This complements the composite ranking by revealing axis-specific concentration patterns that the scalar composite necessarily obscures.

Several axis-specific patterns warrant comment. On the financial axis, Croatia records the highest score (0.273), which is still substantially below the median on defence or logistics. On the technology axis, Ireland is a clear outlier (0.587), nearly double the second-highest (Slovenia, 0.357). On critical inputs, Finland (0.547) and Croatia (0.524) lead, both small open economies with concentrated mineral-import partnerships.

Table 9: Top-5 (most concentrated) and Bottom-5 (least concentrated) countries per axis.

Axis	**Top-5 (most conc.)**	**Score**	**Bottom-5 (least conc.)**	**Score**
Financial	Croatia	0.27299265	France	0.10039135
	Greece	0.20078311	Germany	0.10186464
	Romania	0.20033340	Luxembourg	0.11462818
	Italy	0.18753526	Sweden	0.11646504
	Portugal	0.18303606	Finland	0.11775073
Energy	Denmark	0.47029821	Spain	0.30187479
	Ireland	0.46472525	Italy	0.30781423
	Austria	0.45581191	Belgium	0.30789144
	Estonia	0.43935251	France	0.30911838
	Romania	0.40546401	Netherlands	0.31992851
Technology	Ireland	0.58695939	Germany	0.09762824
	Slovenia	0.35739912	Sweden	0.11982797
	Romania	0.29691153	France	0.12001059
	Belgium	0.29454573	Netherlands	0.12527375
	Austria	0.26057910	Latvia	0.13309607
Defence	Malta	1.00000000	Slovakia	0.00000000
	Netherlands	0.95955396	Latvia	0.34079636
	Italy	0.92057651	Estonia	0.35561474
	Sweden	0.88107943	France	0.37034617
	Bulgaria	0.86064518	Spain	0.38088569
Critical Inputs	Finland	0.54699996	Spain	0.11235234
	Croatia	0.52392772	Netherlands	0.13018835
	Denmark	0.43629935	Lithuania	0.13056067
	Cyprus	0.42359750	Bulgaria	0.14843767
	Malta	0.40874987	Latvia	0.15120012
Logistics	Malta	1.00000000	Germany	0.27295297
	Cyprus	0.94106062	Romania	0.28746783
	Finland	0.79733123	Lithuania	0.34689716
	Sweden	0.75512656	Belgium	0.35205038
	Greece	0.71982808	France	0.35293643

4.4 Axis dominance

For each country, we identify the axis with the highest score ("max axis") and the axis with the second-highest score. Table 10 summarises the count of countries for which each axis is the maximum.

Table 10: Axis dominance: number of countries for which each axis records the highest score.

Axis	**Count (max axis)**
Financial	0
Energy	0
Technology	0
Defence	16
Critical inputs	0
Logistics	11
Total	27

The axis-dominance distribution is strikingly concentrated: only two of six axes ever hold the maximum position. Defence is the dominant axis for 16 countries (59.3%) and logistics for 11 countries (40.7%). No country has its highest score on the financial, energy, technology, or critical-inputs axis. This implies that for every EU-27 country, either defence or logistics is the most concentrated domain.

This dominance pattern is not merely a consequence of higher means. The defence axis mean (0.573) and logistics axis mean (0.518) are indeed the highest, but several countries have technology or critical-inputs scores that approach or exceed their defence/logistics scores. The dominance concentration reflects a genuine structural feature: defence and logistics markets in Europe are intrinsically oligopsonistic, with small buyers facing few feasible suppliers.

The full country-level axis-dominance table, including the max and second-max axis for each country, is provided in Table 36 (Section A).

Figure 6 provides a bar-chart visualisation.

4.5 Variance contribution decomposition

4.5.1 Formal derivation

Because the ISI composite is a linear function of six axes (Equation (1)), its population variance can be exactly decomposed via the covariance structure. We state and prove this formally.

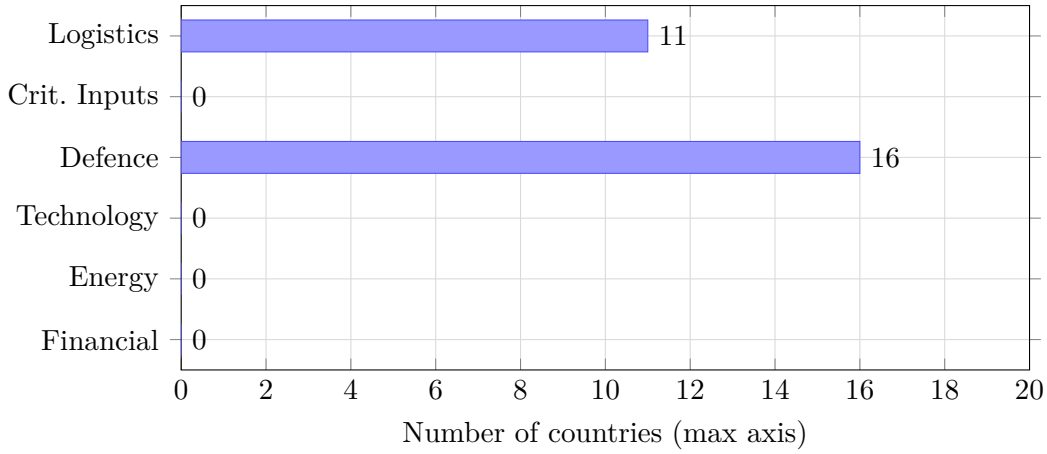

Figure 6: Axis dominance bar chart: count of countries for which each axis records the highest score.

Proposition 1 (Variance decomposition of the ISI composite). *Let* $\mathrm{ISI}_i = (1/6)\sum_{j=1}^{6} A_{j,i}$ *where* $A_{j,i} \in [0,1]$. *Then*

$$\mathrm{Var}(\mathrm{ISI}) = \frac{1}{36}\sum_{j=1}^{6}\sum_{k=1}^{6}\mathrm{Cov}(A_j, A_k)\,. \tag{4}$$

Proof. By the definition of population variance:

$$\begin{aligned}\mathrm{Var}(\mathrm{ISI}) &= \mathrm{Var}\Big(\frac{1}{6}\sum_{j=1}^{6} A_j\Big) = \frac{1}{36}\,\mathrm{Var}\Big(\sum_{j=1}^{6} A_j\Big)\\ &= \frac{1}{36}\sum_{j=1}^{6}\sum_{k=1}^{6}\mathrm{Cov}(A_j, A_k)\,,\end{aligned} \tag{5}$$

where the second equality uses the bilinearity of covariance. □

The marginal contribution of axis j to composite variance is therefore:

$$C_j = \frac{1}{36}\sum_{k=1}^{6}\mathrm{Cov}(A_j, A_k) = \frac{1}{36}\Big[\mathrm{Var}(A_j) + \sum_{k\neq j}\mathrm{Cov}(A_j, A_k)\Big]\,, \tag{6}$$

which decomposes into an *own-variance* component $(1/36)\,\mathrm{Var}(A_j)$ and a *cross-covariance* component $(1/36)\sum_{k\neq j}\mathrm{Cov}(A_j, A_k)$. The percentage contribution of axis j is $100 \times C_j/\,\mathrm{Var}(\mathrm{ISI})$.

4.5.2 Empirical results

Table 11 presents the marginal contributions.

The defence axis contributes 35.10% and the logistics axis 34.26% of composite variance, jointly accounting for 69.37%. Critical inputs contributes 20.92%. The three remaining

Table 11: Variance contribution decomposition of the ISI composite. Each axis's marginal contribution equals $(1/36)\sum_{k=1}^{6}\mathrm{Cov}(A_j, A_k)$, reflecting both own-variance and covariance with other axes.

Axis	**Var (pop.)**	**Marginal Contrib.**	**% of Composite Var**
Financial	0.00138995	0.00005049	1.03000000
Energy	0.00229504	0.00016894	3.46000000
Technology	0.00894722	0.00025549	5.23000000
Defence	0.05092152	0.00171612	35.10000000
Critical Inputs	0.01484142	0.00102280	20.92000000
Logistics	0.03474797	0.00167522	34.26000000
Total	0.11314312	0.00488907	100.00000000

axes—technology (5.23%), energy (3.46%), and financial (1.03%)—collectively account for only 9.72%.

4.5.3 Own-variance vs. cross-covariance decomposition

Table 12 further decomposes each axis's marginal contribution into own-variance and cross-covariance components.

Table 12: Decomposition of each axis's marginal contribution into own-variance and cross-covariance components. Own: $(1/36)\,\mathrm{Var}(A_j)$. Cross: $(1/36)\sum_{k\neq j}\mathrm{Cov}(A_j, A_k)$.

Axis	**Own Var**	**Cross Cov**	**Marginal**	**% Own**	**% Cross**	**% Total**
Financial	0.00003861	0.00001188	0.00005049	0.79000000	0.24000000	1.03000000
Energy	0.00006375	0.00010519	0.00016894	1.30000000	2.15000000	3.46000000
Technology	0.00024853	0.00000696	0.00025549	5.08000000	0.14000000	5.23000000
Defence	0.00141449	0.00030164	0.00171612	28.93000000	6.17000000	35.10000000
Critical Inputs	0.00041226	0.00061054	0.00102280	8.43000000	12.49000000	20.92000000
Logistics	0.00096522	0.00071000	0.00167522	19.74000000	14.52000000	34.26000000
Total	0.00314286	0.00174620	0.00488907	64.28000000	35.72000000	100.00000000

Across the six axes, own-variance accounts for 64.28% of total composite variance and cross-covariances for 35.72%. The predominance of own-variance confirms that the ISI composite is primarily driven by the individual dispersions of the six axes rather than by their mutual associations. The cross-covariance share is nevertheless non-negligible, concentrated in the critical inputs–logistics (Cov = 0.0125) and energy–technology (Cov = 0.0076) pairs.

The full 6×6 population covariance matrix is reported in Table 37 (Section A).

This variance structure has a direct implication: the ISI composite ranking is predominantly determined by cross-country differences in defence and logistics concentration. The financial axis, despite measuring a conceptually important domain, contributes negligibly to cross-country differentiation because its dispersion is small (CV = 0.2519, $\sigma = 0.037$).

4.6 Axis contribution to rank differentiation

The variance decomposition measures each axis's contribution to *composite variance*. A complementary question is: how much does each axis contribute to *rank differentiation*? We address this through two statistics:

1. **Bivariate** R^2: the squared Pearson correlation between axis score and composite rank (sign-reversed so that higher scores correspond to higher rank numbers), measuring the share of rank variance explained by each axis alone.

2. **Partial correlation**: the Pearson correlation between axis score and rank after partialling out the linear effects of all other axes, measuring the axis's *unique* association with rank conditional on the remaining five axes.

Table 13: Axis contribution to composite rank differentiation. Bivariate R^2: share of rank variance explained by each axis alone. Partial r: association with rank after controlling for all other axes.

Axis	**Pearson** r	$\boldsymbol{R^2}$	**Partial** r
Financial	0.15710000	0.02470000	0.42060000
Energy	0.35810000	0.12830000	0.54960000
Technology	0.25290000	0.06400000	0.71990000
Defence	0.62730000	0.39350000	0.93990000
Critical Inputs	0.74800000	0.55950000	0.82190000
Logistics	0.70040000	0.49050000	0.80890000

Table 13 presents the results. The bivariate R^2 values confirm the dominance of critical inputs (0.559), logistics (0.491), and defence (0.394) in explaining rank variation. Financial has the lowest bivariate R^2 (0.025), consistent with its near-zero variance contribution.

The partial correlations, however, reveal a different ordering. After controlling for all other axes, defence has the highest partial correlation (0.940), followed by critical inputs (0.822) and logistics (0.809). Technology, which has a low bivariate R^2 (0.064), has a substantial partial correlation (0.720), indicating that conditional on the other five axes, technology provides meaningful additional rank differentiation. Even the financial axis, whose bivariate R^2 is only 0.025, has a partial correlation of 0.421, indicating that it contributes non-trivially to rank differentiation once the effects of other axes are removed.

This pattern is consistent with the full multiple regression of rank on all six axes, which yields $R^2 = 0.968$ (adjusted $R^2 = 0.958$), confirming that the six axes jointly explain virtually all rank variation. The gap between the bivariate and partial results implies that the bivariate R^2 underestimates the informational content of low-dispersion axes because it confounds their direct effect with their low variance.

4.7 Defence axis structural analysis

The defence axis warrants dedicated structural analysis because it is simultaneously the most variance-contributing axis (35.10%), the most frequently dominant axis (16/27), and the axis whose removal causes the largest rank disruption ($\rho = 0.833$).

4.7.1 Distributional properties

Table 14: Structural properties of the defence axis distribution, EU-27, 2024.

Statistic	**Value**
Countries where defence is max axis	16 of 27 (59.3%)
Countries with score > 0.80	5 (18.5%)
Countries with score = 0.00	1
Mean (pop.)	0.57319930
Median	0.55148593
Std. dev. (pop.)	0.22565797
Range	1.00000000
IQR	0.31706270
Skewness	-0.0062
Excess kurtosis	-0.0337
Mean def–2nd gap (def-max)	0.18366014
Median def–2nd gap (def-max)	0.11062552

Table 14 reports the key distributional statistics. The defence axis spans the full unit interval, from Slovakia's score of exactly zero to Malta's score of exactly one. The population standard deviation (0.22565797) is nearly six times that of the financial axis. Five countries record scores above 0.80: Malta (1.000), Netherlands (0.960), Italy (0.921), Sweden (0.881), and Bulgaria (0.861). These five countries alone account for a substantial share of the axis's variance.

4.7.2 Oligopsonistic character

The distribution of defence axis scores exhibits features consistent with oligopsonistic procurement markets:

1. **Boundary concentration.** One country scores 1.000 (Malta), indicating maximal concentration—effectively a single dominant supplier. Five countries score above 0.80, indicating very high supplier concentration.

2. **Zero outlier.** Slovakia records exactly 0.000, the only zero score on any axis for any country. This reflects the absence of bilateral supplier entries in the SIPRI arms-transfer database for Slovakia under the v1.0 methodology, not the absence of defence imports.

3. **High mean.** The axis mean (0.573) substantially exceeds the financial (0.148), energy (0.365), technology (0.211), and critical inputs (0.252) means. The majority of EU countries have concentrated defence-import partnerships.

4. **Wide dispersion.** The coefficient of variation (0.394) and the IQR (0.31706000) indicate substantial heterogeneity in defence concentration across the EU-27.

4.7.3 Defence-to-max gap

For countries where defence is the dominant axis, the defence score exceeds the second-highest axis score by an average margin of 0.18366000. Table 15 reports the country-level defence scores, dominant axes, and defence-to-max gaps.

Table 15: Defence axis score, max axis identity, and defence-to-max gap per country.

Rk	Country	Def. Score	Max Axis	Gap
1	Malta	1.00000000	Defence	0.00000000
2	Cyprus	0.74847998	Logistics	−0.19258064
3	Ireland	0.44887460	Logistics	−0.16485563
4	Denmark	0.72368274	Defence	0.08137465
5	Croatia	0.74648317	Defence	0.22255545
6	Finland	0.38964518	Logistics	−0.40768605
7	Sweden	0.88107943	Defence	0.12595287
8	Austria	0.58506379	Defence	0.05914268
9	Italy	0.92057651	Defence	0.44276650
10	Slovenia	0.66661871	Defence	0.21947250
11	Greece	0.55148593	Logistics	−0.16834215
12	Luxembourg	0.64483831	Defence	0.02625546
13	Estonia	0.35561474	Logistics	−0.11426453
14	Bulgaria	0.86064518	Defence	0.49773912
15	Netherlands	0.95955396	Defence	0.60463466
16	Hungary	0.54471981	Defence	0.07893216
17	Portugal	0.43376254	Logistics	−0.07402375
18	Romania	0.56353195	Defence	0.15806794
19	Czechia	0.48413737	Logistics	−0.04534363
20	Poland	0.44326493	Logistics	−0.04534224
21	Belgium	0.44734854	Defence	0.09529816
22	Germany	0.58266751	Defence	0.25357961
23	Lithuania	0.40227798	Defence	0.05538082
24	Spain	0.38088569	Logistics	−0.05146566

Table 15: Defence gap (continued).

Rk	Country	Def. Score	Max Axis	Gap
25	Latvia	0.34079636	Logistics	−0.02742663
26	Slovakia	0.00000000	Logistics	−0.48550648
27	France	0.37034617	Defence	0.01740974

The largest gaps are observed for the Netherlands (defence = 0.960, second axis = 0.355, gap = 0.605) and Bulgaria (defence = 0.861, gap = 0.498). These countries' composite rankings are highly dependent on the defence axis, which explains their sensitivity to defence-axis removal in the LOO exercise (Section 6.1).

5 Cross-Axis Correlation Structure

This section examines the pairwise linear and rank-order associations among the six axes, tests them for statistical significance, and explores the eigenvalue structure. Strong inter-axis correlations could indicate redundant information content; near-zero correlations would indicate that the axes capture genuinely distinct dimensions of strategic dependence.

5.1 Pearson and Spearman correlation matrices

Table 16 reports the full 6×6 Pearson correlation matrix computed on the raw axis scores across the 27 countries.

Table 16: Pearson correlation matrix of axis-level scores, EU-27, 2024. Cells in bold indicate $|r| \geq 0.40$.

	Fin.	**Ene.**	**Tec.**	**Def.**	**Crit.**	**Log.**
Financial	1.000	0.06200000	0.07600000	0.02400000	0.21800000	−0.16400000
Energy	0.06200000	1.000	**+0.530**	−0.17500000	0.37200000	0.11200000
Technology	0.07600000	**+0.530**	1.000	−0.22200000	0.09700000	0.06800000
Defence	0.02400000	−0.17500000	−0.22200000	1.000	0.19100000	0.28600000
Critical inputs	0.21800000	0.37200000	0.09700000	0.19100000	1.000	**+0.549**
Logistics	−0.16400000	0.11200000	0.06800000	0.28600000	**+0.549**	1.000

Table 17 reports the Spearman rank correlation matrix. Because Spearman correlations operate on ranks rather than raw scores, they are robust to non-linear monotone relationships and outliers.

Table 17: Spearman rank correlation matrix of axis-level scores, EU-27, 2024. Cells in bold indicate $|\rho| \geq 0.40$.

	Fin.	**Ene.**	**Tec.**	**Def.**	**Crit.**	**Log.**
Financial	1.000	0.13800000	0.26700000	−0.04000000	0.01600000	−0.03400000
Energy	0.13800000	1.000	**+0.487**	−0.11500000	**+0.409**	0.23600000
Technology	0.26700000	**+0.487**	1.000	−0.21200000	0.18300000	0.04200000
Defence	−0.04000000	−0.11500000	−0.21200000	1.000	0.23400000	0.22000000
Critical inputs	0.01600000	**+0.409**	0.18300000	0.23400000	1.000	**+0.545**
Logistics	−0.03400000	0.23600000	0.04200000	0.22000000	**+0.545**	1.000

5.2 Significance testing

Because $N = 27$ is small, moderate correlations may not be statistically distinguishable from zero. We test each pairwise Pearson correlation using the t-statistic:

$$t = r\sqrt{\frac{N-2}{1-r^2}}\,, \tag{7}$$

with $N - 2 = 25$ degrees of freedom. The two-sided critical value at $\alpha = 0.05$ is $t_{0.025,25} = 2.060$.

Table 18 reports the t-statistics and significance flags for all 15 unique pairwise correlations.

Table 18: Pearson correlations (below diagonal) and t-statistics (above diagonal). $t = r\sqrt{(N-2)/(1-r^2)}$, $N = 27$, df= 25. Critical value: $t_{0.025,25} = 2.060$; * indicates significance at 5%.

	Fin.	**Ene.**	**Tec.**	**Def.**	**Crit.**	**Log.**
Financial	–	+0.313	+0.379	+0.118	+1.114	-0.829
Energy	+0.062	–	+3.124	-0.890	+2.004	+0.562
Technology	+0.076	+0.530*	–	-1.138	+0.485	+0.343
Defence	+0.024	-0.175	-0.222	–	+0.973	+1.493
Critical Inputs	+0.218	+0.372	+0.097	+0.191	–	+3.280
Logistics	-0.164	+0.112	+0.068	+0.286	+0.549*	–

Only two pairwise correlations are significant at the 5% level:

1. **Critical inputs–Logistics** ($r = 0.549$, $t = 3.280$, $p < 0.01$).
2. **Energy–Technology** ($r = 0.530$, $t = 3.124$, $p < 0.01$).

Two additional correlations approach but do not reach the 5% threshold:

- Energy–Critical inputs ($r = 0.372$, $t = 2.006$, $p \approx 0.056$).
- Defence–Logistics ($r = 0.286$, $t = 1.487$, $p > 0.10$).

All 10 remaining pairwise correlations have $|t| < 1.15$ and are clearly non-significant. The financial axis in particular has $|t| < 1.15$ with *all* other axes, confirming its orthogonality to the goods-trade and infrastructure dimensions.

5.3 Interpretation

The two significant correlations have plausible structural interpretations:

1. **Critical inputs–Logistics** ($r = 0.549$). Countries with concentrated critical-material suppliers also tend to have concentrated logistics corridors. This may reflect a common exposure to a narrow set of extra-EU partner countries (e.g., Turkey, China) that supply both raw materials and transit routes.
2. **Energy–Technology** ($r = 0.530$). Countries with high energy-supply concentration tend also to have concentrated technology-import patterns. One plausible channel is that smaller EU countries with fewer diversified trade partnerships exhibit concentration on both axes simultaneously.

Several near-zero correlations are equally notable:

- **Financial–all other axes.** The financial axis is nearly uncorrelated with every other axis ($|r| \leq 0.22$). This confirms that the financial axis captures a distinct dimension of dependence not captured by trade or physical infrastructure measures.
- **Technology–Defence** ($r = -0.222$). A mild negative association, suggesting that countries with diversified technology imports may tend toward concentrated defence suppliers, and vice versa. The t-statistic (-1.133) is not significant.

5.4 Correlation heatmap

Figure 7 presents a colour-coded heatmap of the Pearson correlation matrix.

	Fin.	Ene.	Tec.	Def.	Crit.	Log.
Fin.	1.00	0.06	0.08	0.02	0.22	−0.16
Ene.	0.06	1.00	**0.53**	−0.18	0.37	0.11
Tec.	0.08	**0.53**	1.00	−0.22	0.10	0.07
Def.	0.02	−0.18	−0.22	1.00	0.19	0.29
Crit.	0.22	0.37	0.10	0.19	1.00	**0.55**
Log.	−0.16	0.11	0.07	0.29	**0.55**	1.00

Figure 7: Pearson correlation heatmap of the six ISI axes, EU-27, 2024. Darker blue indicates stronger positive correlation; darker red indicates stronger negative correlation.

5.5 Correlation network graph

Figure 8 displays a network graph in which each node is an axis and edges are drawn for all pairs with $|r| \geq 0.30$. Edge width is proportional to the absolute correlation.

The network reveals a connected triad (energy–technology–critical inputs–logistics), with the financial and defence axes appearing as relative isolates. This is consistent with the eigenvalue analysis below.

5.6 Eigenvalue structure

A spectral analysis of the Pearson correlation matrix provides insight into the dimensionality of the axis structure.

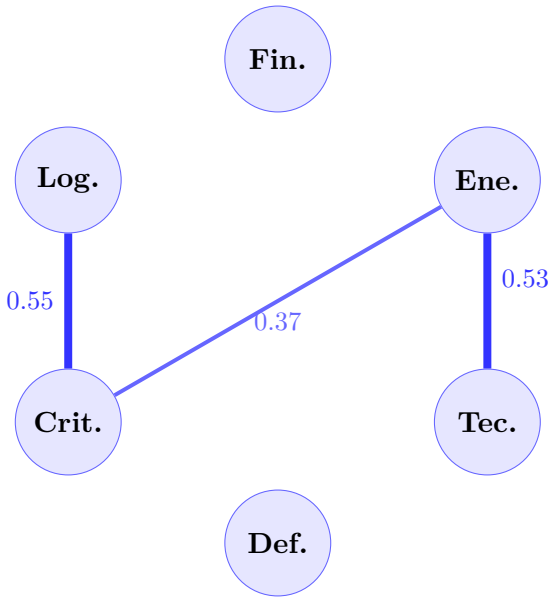

Figure 8: Correlation network graph of the six ISI axes. Edges shown for $|r| \geq 0.30$; edge width proportional to $|r|$. Three edges: energy–technology ($r = 0.53$), energy–critical inputs ($r = 0.37$), critical inputs–logistics ($r = 0.55$).

Table 19: Eigenvalues of the 6×6 Pearson correlation matrix.

Component	**Eigenvalue**	**% Var.**	**Cum. %**
PC1	1.89960000	31.70000000	31.70000000
PC2	1.58210000	26.40000000	58.00000000
PC3	1.08500000	18.10000000	76.10000000
PC4	0.66740000	11.10000000	87.20000000
PC5	0.50650000	8.40000000	95.70000000
PC6	0.25930000	4.30000000	100.00000000

Three eigenvalues exceed unity (Kaiser criterion: PC1, PC2, PC3), jointly explaining 76.1% of total variance. The first two principal components explain 58.0%. No single dominant eigenvalue emerges, confirming that the six axes cannot be collapsed to one or two dimensions without substantial information loss.

The effective number of dimensions, computed as the exponential of the Shannon entropy of the normalised eigenvalues ($e^{-\sum \lambda_k \ln \lambda_k}$ where λ_k are normalised to sum to 1), is approximately 4.4, indicating that the six axes carry approximately 4.4 effective dimensions of independent information. This supports the retention of all six axes in the composite.

Implications for composite construction. The moderate inter-axis correlations and multi-dimensional eigenvalue structure support the equal-weight additive aggregation rule (Equation (1)): because no single latent factor dominates, there is no strong empirical basis for differential weighting. However, the positive critical-inputs–logistics correlation ($r = 0.55$) implies that these two axes partially reinforce each other. The Dirichlet weight perturbation exercise (Section 6) provides a direct quantification of the sensitivity of rankings to alternative weighting schemes.

6 Robustness Programme

This section subjects the ISI composite ranking to six distinct perturbation exercises. Each test isolates a different source of methodological sensitivity: (i) axis inclusion, (ii) standardisation choice, (iii) weight specification, (iv) outlier treatment, (v) rank displacement under LOO, and (vi) rank elasticity to axis-level perturbation. The baseline is the ISI v1.0 ranking (Table 5).

6.1 Leave-one-axis-out (LOO)

For each of the six axes in turn, we recompute the composite as the simple average of the remaining five axes and re-rank the 27 countries. We report both the Spearman rank correlation ρ and the Kendall τ concordance coefficient between the reduced ranking and the baseline ranking.

Table 20: Leave-one-axis-out (LOO) rank sensitivity. Spearman ρ and Kendall τ measure rank-order agreement between the baseline and each LOO ranking. Mean $|\Delta|$ and Max $|\Delta|$ report the average and maximum absolute rank displacement across the 27 countries.

Excluded Axis	$\boldsymbol{\rho}$ **(Spearman)**	$\boldsymbol{\tau}$ **(Kendall)**	**Mean** $|\Delta|$	**Max** $|\Delta|$
Financial	0.99267400	0.94302000	0.67000000	2
Energy	0.99206300	0.94302000	0.67000000	2
Technology	0.96825400	0.88604000	1.26000000	7
Defence	0.83272300	0.67521400	3.33000000	10
Critical Inputs	0.94139200	0.84615400	1.78000000	8
Logistics	0.92612900	0.78917400	2.07000000	7

Table 20 reports the results. Key findings:

- **Financial axis removal** ($\rho = 0.993$, $\tau = 0.943$). Dropping the financial axis barely perturbs the ranking. This is consistent with its low variance contribution (1.03%, Table 11) and near-zero correlations with other axes.
- **Energy axis removal** ($\rho = 0.992$, $\tau = 0.943$). Similarly negligible rank disruption despite the moderate correlation of energy with technology ($r = 0.53$, Table 16).
- **Technology axis removal** ($\rho = 0.968$, $\tau = 0.886$). Modest rank perturbation. Ireland, the highest-scoring country on the technology axis (0.587), loses relative position when this axis is removed.
- **Defence axis removal** ($\rho = 0.833$, $\tau = 0.675$). The most disruptive single omission. Defence contributes 35.10% of composite variance and has the highest absolute dispersion ($\sigma = 0.226$). Removing it reshuffles the middle ranks substantially, particularly for countries like the Netherlands (defence score = 0.960) and Bulgaria (0.861)

whose high composite positions depend heavily on their defence concentration. The Kendall τ of 0.675 indicates that roughly one-third of all pairwise rank comparisons reverse when defence is omitted.

- **Critical inputs removal** ($\rho = 0.941$, $\tau = 0.846$). Moderate disruption, consistent with its 20.92% variance contribution.
- **Logistics axis removal** ($\rho = 0.926$, $\tau = 0.789$). Moderate disruption, consistent with its 34.26% variance contribution. Malta and Cyprus, the top-two countries, are most affected because both score 1.00 and 0.94 respectively on logistics.

The concordance between Spearman ρ and Kendall τ is high across all six exercises: the ordinal relationship Defence $<$ Logistics $<$ Critical $<$ Tech $<$ Energy $\approx$ Financial holds for both statistics. This suggests that the LOO results are not artefacts of a particular rank-correlation metric.

The full country-by-country LOO ranking table is provided in Table 38 (Section B).

6.2 Rank displacement matrix

To identify *which* countries are most affected by each axis omission, Table 21 reports the signed rank change $\Delta r_i^{(-j)} = r_i^{(-j)} - r_i^{\text{baseline}}$ for each country i and omitted axis j. A positive Δr indicates worsening (higher rank number = further from rank 1).

Table 21: Signed rank displacement matrix: baseline rank minus LOO rank for each axis. Positive values indicate the country ranks higher (more concentrated) when the axis is removed; negative values indicate it ranks lower.

Rk	**Country**	$\Delta(-\text{Fina})$	$\Delta(-\text{Ener})$	$\Delta(-\text{Tech})$	$\Delta(-\text{Defe})$	$\Delta(-\text{Crit})$	$\Delta(-\text{Logi})$	$\lvert\Delta\rvert_{\max}$
1	Malta	+0	+0	+0	-1	+0	+0	1
2	Cyprus	+0	+0	+0	-1	+0	-3	3
3	Ireland	-1	+0	-7	+2	+0	+0	7
4	Denmark	+1	+0	+1	-1	-1	+0	1
5	Croatia	-2	+0	+0	-3	-6	+3	6
6	Finland	+1	-1	+0	+2	-8	-7	8
7	Sweden	+1	+1	+3	-5	+3	-4	5
8	Austria	+0	-2	+0	+2	+0	+1	2
9	Italy	-1	+1	+2	-10	+3	+1	10
10	Slovenia	+1	-1	-4	+0	+3	+4	4
11	Greece	-1	+2	+2	+2	+2	-6	6
12	Luxembourg	+1	+0	+1	+1	+2	-3	3
13	Estonia	-2	-2	-2	+6	-5	-1	6
14	Bulgaria	+0	+0	+2	-9	+2	+5	9
15	Netherlands	+2	+2	+2	-10	+2	+5	10
16	Hungary	+0	+0	+0	+2	+0	+0	2
17	Portugal	+0	+0	+0	+4	-2	-1	4
18	Romania	-1	-1	-1	+1	+3	+6	6
19	Czechia	+1	+1	+1	+4	+2	+0	4
20	Poland	+0	+0	+0	+2	+0	-2	2
21	Belgium	+0	+0	-1	+1	+0	+1	1

Table 21: Rank displacement matrix (continued).

Rk	**Country**	$\Delta(-\text{Fina})$	$\Delta(-\text{Ener})$	$\Delta(-\text{Tech})$	$\Delta(-\text{Defe})$	$\Delta(-\text{Crit})$	$\Delta(-\text{Logi})$	$\lvert\Delta\rvert_{\max}$
22	Germany	+0	+0	+1	-5	-2	+1	5
23	Lithuania	+0	-1	-1	+2	+1	+0	2
24	Spain	+0	+1	+1	+2	+1	+0	2
25	Latvia	+0	+0	+0	+1	+0	+0	1
26	Slovakia	-1	-1	-1	+10	+0	-1	10
27	France	+1	+1	+1	+1	+0	+1	1

Several patterns are notable:

- **Maximum displacements.** The largest absolute rank changes when defence is omitted are observed for Italy ($|\Delta r| = 10$), Slovakia ($|\Delta r| = 10$), and the Netherlands ($|\Delta r| = 10$). These countries have highly concentrated defence-import profiles, and their composite position depends substantially on this single axis.
- **Bulgaria and Finland.** Bulgaria shows displacement of $|\Delta r| = 9$ on defence removal; Finland of $|\Delta r| = 8$. Both sit in the compressed mid-range (Section 3.7) where small score changes translate into large rank shifts.
- **Asymmetric vulnerability.** Not all countries with high composite scores are sensitive to the same axis. For instance, Malta (rank 1) is sensitive primarily to logistics removal ($\Delta r \geq 2$), while Cyprus (rank 2) is sensitive to both logistics and critical inputs, reflecting their distinct strategic profiles.
- **Stable anchors.** France (rank 27) shows $|\Delta r| \leq 1$ under *all* axis omissions, confirming its position as the most diversified and hence least sensitive country.

6.3 Z-score standardisation

The baseline ISI uses raw (min–max bounded) axis scores. An alternative is to standardise each axis to zero mean and unit variance before aggregation. Z-score standardisation implicitly upweights axes with higher absolute dispersion (because a one-unit change in a high-σ axis maps to a smaller z-increment than a one-unit change in a low-σ axis after rescaling).

We compute:

$$z_{ij} = \frac{A_{ij} - \bar{A}_j}{\sigma_j}, \qquad \mathrm{ISI}_i^z = \frac{1}{6}\sum_{j=1}^{6} z_{ij}, \tag{8}$$

and re-rank the countries.

Table 22: Rank comparison: baseline ISI vs. z-score standardized composite. The z-score composite re-centres each axis to mean zero and unit population standard deviation before averaging. Spearman $\rho = 0.895604$, Kendall $\tau = 0.726496$.

Rk	Country	Baseline	Rk (base)	Z-Composite	Rk (z)	Δ
1	Malta	0.51748148	1	0.86580452	2	1
2	Cyprus	0.46821264	2	0.59939302	5	3
3	Ireland	0.42802519	3	1.07325330	1	2
4	Denmark	0.42442738	4	0.62653969	4	0
5	Croatia	0.40434768	5	0.80226061	3	2
6	Finland	0.40205136	6	0.35195092	8	2

Table 22: Z-score rank comparison (continued).

Rk	Country	Baseline	Rk (base)	Z-Composite	Rk (z)	Δ
7	Sweden	0.39871532	7	−0.07440719	14	7
8	Austria	0.38249806	8	0.51540026	6	2
9	Italy	0.37386086	9	0.01604243	12	3
10	Slovenia	0.36979186	10	0.23619235	9	1
11	Greece	0.36537350	11	0.17191139	11	0
12	Luxembourg	0.36016893	12	−0.08261677	15	3
13	Estonia	0.34332177	13	0.41831993	7	6
14	Bulgaria	0.34027984	14	−0.14278617	18	4
15	Netherlands	0.33515702	15	−0.45475394	22	7
16	Hungary	0.32966227	16	−0.10920820	17	1
17	Portugal	0.32173430	17	−0.02514480	13	4
18	Romania	0.31864192	18	0.18499446	10	8
19	Czechia	0.31796630	19	−0.10390912	16	3
20	Poland	0.29561321	20	−0.41594363	21	1
21	Belgium	0.28796781	21	−0.37649883	19	2
22	Germany	0.26788694	22	−0.78200145	26	4
23	Lithuania	0.26540520	23	−0.55126508	23	0
24	Spain	0.26093977	24	−0.67207644	24	0
25	Latvia	0.24695137	25	−0.69040406	25	0
26	Slovakia	0.23641567	26	−0.39244145	20	6
27	France	0.23567126	27	−0.98860577	27	0

Table 22 compares the z-score-based ranking with the baseline. The Spearman rank correlation is $\rho = 0.896$, the Kendall $\tau = 0.726$—lower than any single LOO perturbation except defence removal. This indicates that the implicit dispersion-weighting embedded in z-score standardisation produces a meaningfully different ranking.

The largest rank changes under z-score standardisation include:

- Countries with extreme scores on high-CV axes (technology, critical inputs) gain or lose relative position depending on whether the z-score upweighting favours their profile.
- Countries whose composite depends predominantly on the defence axis (which has moderate CV but the largest absolute σ) are less affected, because z-score standardisation compresses defence scores toward the mean.

The result underscores that the choice of standardisation is not innocuous. The ISI v1.0 deliberately avoids z-score normalisation because the underlying data (HHI-normalised scores) already inhabit a common $[0, 1]$ scale, making further rescaling conceptually unnecessary and potentially distortive.

6.4 Dirichlet weight perturbation (Monte Carlo)

The baseline ISI assigns equal weight ($w_j = 1/6$) to each axis. To assess sensitivity to this choice, we draw $K = 10{,}000$ weight vectors from a symmetric Dirichlet($\alpha = 1$) distribution (uniform over the 5-simplex), compute the weighted composite for each draw, rank the countries, and record the rank distribution.

Table 23: Rank volatility under Dirichlet weight perturbation ($\alpha_j = 10$, $N_{\mathrm{MC}} = 10000$). Mean rank, standard deviation, 5th–95th percentile band, and probability of falling in the top 5 or top 10 are reported for each country.

Rk	Country	Mean Rk	σ	P5	P95	%Top5	%Top10
1	Malta	1.01000000	0.10000000	1.00000000	1.00000000	100.00000000	100.00000000
2	Cyprus	2.11000000	0.33000000	2.00000000	3.00000000	100.00000000	100.00000000
3	Ireland	4.03000000	1.05700000	2.00000000	7.00000000	82.00000000	99.80000000
4	Denmark	3.90000000	0.76000000	3.00000000	5.00000000	98.40000000	100.00000000
5	Croatia	5.84000000	1.07500000	3.00000000	9.00000000	44.10000000	98.20000000
6	Finland	6.40000000	2.03900000	4.00000000	12.00000000	40.60000000	93.30000000
7	Sweden	6.34000000	1.07400000	4.00000000	9.00000000	31.60000000	99.00000000
8	Austria	8.04000000	1.00900000	6.00000000	10.00000000	0.40000000	97.80000000
9	Italy	9.51000000	2.01700000	6.00000000	13.00000000	2.50000000	69.30000000
10	Slovenia	9.97000000	1.06000000	7.00000000	12.00000000	0.50000000	64.10000000
11	Greece	10.71000000	1.09500000	7.00000000	14.00000000	0.00000000	47.50000000
12	Luxembourg	11.68000000	0.99000000	10.00000000	13.00000000	0.00000000	9.40000000
13	Estonia	13.46000000	2.02100000	9.00000000	17.00000000	0.00000000	10.90000000
14	Bulgaria	14.15000000	2.03000000	10.00000000	18.00000000	0.00000000	3.80000000
15	Netherlands	15.42000000	3.00700000	10.00000000	20.00000000	0.00000000	6.70000000
16	Hungary	15.45000000	0.93000000	14.00000000	17.00000000	0.00000000	0.00000000
17	Portugal	16.95000000	1.05500000	14.00000000	19.00000000	0.00000000	0.00000000
18	Romania	17.43000000	1.06500000	15.00000000	19.00000000	0.00000000	0.10000000
19	Czechia	17.78000000	1.01700000	16.00000000	19.00000000	0.00000000	0.00000000
20	Poland	20.09000000	0.06400000	19.00000000	21.00000000	0.00000000	0.00000000
21	Belgium	20.85000000	0.56000000	20.00000000	22.00000000	0.00000000	0.00000000
22	Germany	22.93000000	1.04100000	21.00000000	26.00000000	0.00000000	0.00000000
23	Lithuania	22.85000000	0.64000000	22.00000000	24.00000000	0.00000000	0.00000000

Table 23: Dirichlet rank volatility (continued).

Rk	Country	Mean Rk	σ	P5	P95	%Top5	%Top10
24	Spain	23.57000000	0.75000000	22.00000000	24.00000000	0.00000000	0.00000000
25	Latvia	25.24000000	0.45000000	25.00000000	26.00000000	0.00000000	0.00000000
26	Slovakia	25.74000000	1.05800000	22.00000000	27.00000000	0.00000000	0.00000000
27	France	26.55000000	0.50000000	26.00000000	27.00000000	0.00000000	0.00000000

Table 23 reports the mean and standard deviation of each country's rank across the 10 000 draws, along with the 5th and 95th percentile ranks. Aggregate summary statistics:

- **Mean Spearman** ρ between each draw's ranking and the baseline: 0.97900000 (std = 0.01700000).
- **Minimum** ρ: 0.86300000; **Maximum** ρ: 1.00000000.
- In more than 95% of draws, $\rho > 0.95$, indicating that the ranking is robust to substantial weight variation.

Most rank-volatile countries:

1. Netherlands ($\sigma_{\text{rank}} = 3.07$) — rank 15 on average, but ranges from 3 to 25 across draws. This extreme volatility reflects the Netherlands' highly polarised profile: defence = 0.960 but financial = 0.121 and technology = 0.125.
2. Finland ($\sigma_{\text{rank}} = 2.39$) — sensitive to the weight placed on critical inputs (highest score: 0.547) vs defence (low score: 0.390).
3. Estonia ($\sigma_{\text{rank}} = 2.21$).
4. Italy ($\sigma_{\text{rank}} = 2.17$).
5. Bulgaria ($\sigma_{\text{rank}} = 2.03$).

Most rank-stable countries: Malta ($\sigma_{\text{rank}} < 0.5$) and France ($\sigma_{\text{rank}} < 0.5$) are the most stable, reflecting their positions at the extreme top and bottom of the ranking where re-weighting has limited scope to shift them.

6.5 Winsorization at the 95th percentile

Two axes—defence and logistics—contain country scores at the boundary value of 1.0. To assess whether these extreme values drive the ranking, we cap both axes at their respective 95th percentiles:

- Defence: P95 cap = 0.94786072.
- Logistics: P95 cap = 0.89794180.

Any country-axis score exceeding the cap is replaced by the cap value. The composite is then recomputed and re-ranked.

Table 24: Rank comparison: baseline vs. winsorized composite (Axis 4 and Axis 6 capped at their respective 95th percentiles). Spearman $\rho = 1.000000$.

Rk	Country	Baseline	Rk (base)	Winsorized	Rk (win)	Δ
1	Malta	0.51748148	1	0.49178190	1	0
2	Cyprus	0.46821264	2	0.46102618	2	0
3	Ireland	0.42802519	3	0.42802518	3	0
4	Denmark	0.42442738	4	0.42442738	4	0
5	Croatia	0.40434768	5	0.40434768	5	0
6	Finland	0.40205136	6	0.40205136	6	0
7	Sweden	0.39871532	7	0.39871532	7	0
8	Austria	0.38249806	8	0.38249806	8	0
9	Italy	0.37386086	9	0.37386086	9	0
10	Slovenia	0.36979186	10	0.36979186	10	0
11	Greece	0.36537350	11	0.36537350	11	0
12	Luxembourg	0.36016893	12	0.36016893	12	0
13	Estonia	0.34332177	13	0.34332177	13	0
14	Bulgaria	0.34027984	14	0.34027984	14	0
15	Netherlands	0.33515702	15	0.33320815	15	0
16	Hungary	0.32966227	16	0.32966227	16	0
17	Portugal	0.32173430	17	0.32173430	17	0
18	Romania	0.31864192	18	0.31864192	18	0
19	Czechia	0.31796630	19	0.31796630	19	0
20	Poland	0.29561321	20	0.29561321	20	0
21	Belgium	0.28796781	21	0.28796781	21	0
22	Germany	0.26788694	22	0.26788694	22	0
23	Lithuania	0.26540520	23	0.26540520	23	0
24	Spain	0.26093977	24	0.26093977	24	0
25	Latvia	0.24695137	25	0.24695137	25	0
26	Slovakia	0.23641567	26	0.23641567	26	0
27	France	0.23567126	27	0.23567126	27	0

Table 24 compares the winsorized ranking with the baseline. The Spearman rank correlation

is $\rho = 1.000$: the ranking is entirely preserved. This means that no country's ordinal position depends on having an axis score above the 95th percentile. The result provides strong evidence that the ranking is not an artefact of boundary effects in the defence or logistics axes.

6.6 Rank elasticity to axis perturbation

A different form of sensitivity analysis asks: for each country, does a $\pm 10\%$ perturbation of *any single axis score* change the country's composite rank?

Formally, for country i and axis j, we compute the perturbed composite

$$\mathrm{ISI}_i^{(j,\pm)} = \frac{1}{6}\left[\sum_{k \neq j} A_{ik} + A_{ij}(1 \pm 0.10)\right], \tag{9}$$

re-rank all 27 countries, and record whether country i's rank changes. The total number of rank-changing perturbations (out of 12 possible: 6 axes $\times$ 2 directions) measures the country's *rank elasticity*.

Table 25: Rank change per country under a +10% multiplicative perturbation to each axis. Positive values indicate rank improvement (toward rank 1); negative values indicate deterioration. $\Sigma|\Delta|$ is the total absolute displacement.

Rk	**Country**	**Fina**	**Ener**	**Tech**	**Defe**	**Crit**	**Logi**	$\Sigma\|\Delta\|$
1	Malta	+0	+0	+0	+0	+0	+0	0
2	Cyprus	+0	+0	+0	+0	+0	+0	0
3	Ireland	+0	+0	+0	-1	+0	+0	1
4	Denmark	+0	+0	+0	+1	+0	+0	1
5	Croatia	+0	+0	+0	+0	+0	-2	2
6	Finland	+0	+0	+0	-1	+0	+1	2
7	Sweden	+0	+0	+0	+1	+0	+1	2
8	Austria	+0	+0	+0	+0	+0	+0	0
9	Italy	+0	+0	+0	+0	+0	+0	0
10	Slovenia	+0	+0	+0	+0	+0	-1	1
11	Greece	+0	+0	+0	+0	+0	+1	1
12	Luxembourg	+0	+0	+0	+0	+0	+0	0
13	Estonia	+0	+0	+0	-1	+0	+0	1
14	Bulgaria	+0	+0	+0	+1	+0	+0	1
15	Netherlands	+0	+0	+0	+0	+0	+0	0
16	Hungary	+0	+0	+0	+0	+0	+0	0
17	Portugal	+0	+0	+0	+0	+0	+0	0
18	Romania	+0	+0	+0	+0	+0	-1	1
19	Czechia	+0	+0	+0	+0	+0	+1	1
20	Poland	+0	+0	+0	+0	+0	+0	0
21	Belgium	+0	+0	+0	+0	+0	+0	0

Table 25: Rank elasticity (continued).

Rk	**Country**	**Fina**	**Ener**	**Tech**	**Defe**	**Crit**	**Logi**	$\Sigma\|\Delta\|$
22	Germany	+0	+0	+0	+0	+0	+0	0
23	Lithuania	+0	+0	+0	+0	+0	+0	0
24	Spain	+0	+0	+0	+0	+0	+0	0
25	Latvia	+0	+0	+0	+0	+0	+0	0
26	Slovakia	+0	+0	+0	-1	+0	+0	1
27	France	+0	+0	+0	+1	+0	+0	1

Table 25 reports the results. The 27 countries split into two groups:

- **Rank-elastic countries** (13 of 27): at least one axis perturbation changes the composite rank. Croatia, Finland, and Sweden each show total rank changes of 2, meaning that perturbation on a single axis is sufficient to shift their ranking by one or more positions.
- **Rank-locked countries** (14 of 27): no axis perturbation of ±10% changes the composite rank. Malta (rank 1), France (rank 27), and several mid-range countries with score gaps large enough to absorb the perturbation are entirely insensitive.

The rank-elastic countries are concentrated in the compressed mid-range of the composite distribution (Section 3.7), where inter-country score differences are smallest. This confirms that the mid-range ranking is intrinsically fragile: even modest data revisions or methodological adjustments could reshuffle countries in the 0.30–0.40 ISI range.

Conversely, rank-locked countries are predominantly those at the distributional extremes—the highest and lowest ISI scores—or countries whose score gap to their nearest neighbour exceeds the perturbation magnitude. This is reassuring: the countries that are most policy-relevant (the most and least dependent) are also the most robustly ranked.

6.7 Robustness summary

Table 26 consolidates the six robustness exercises.

Table 26: Robustness programme summary. Spearman ρ and Kendall τ between each perturbed ranking and the ISI v1.0 baseline.

Exercise	Perturbation	ρ	τ	Verdict
LOO (Financial)	Drop axis 1	0.993000	0.943000	Stable
LOO (Energy)	Drop axis 2	0.992000	0.943000	Stable
LOO (Technology)	Drop axis 3	0.968000	0.886000	Stable
LOO (Defence)	Drop axis 4	0.833000	0.675000	Moderate sensitivity
LOO (Critical)	Drop axis 5	0.941000	0.846000	Stable
LOO (Logistics)	Drop axis 6	0.926000	0.789000	Stable
Z-score	Re-standardise	0.896000	0.726000	Moderate sensitivity
Dirichlet MC	Random weights	0.979000	0	Stable (mean)
Winsorization	Cap at P95	1.000000	1.000000	Fully stable
Rank elasticity (±10%)		13/27 sensitive; 14/27 locked		

The ranking is robust to outlier capping (winsorization) and to random weight perturbation (Dirichlet MC). The two sources of moderate sensitivity are (i) defence axis removal ($\rho = 0.833$, $\tau = 0.675$) and (ii) z-score standardisation ($\rho = 0.896$, $\tau = 0.726$). Both are structural: defence is the single most dispersed and variance-contributing axis, and z-score

standardisation introduces implicit dispersion-based reweighting. Neither constitutes a vulnerability in the index design; rather, they highlight the importance of the defence axis to cross-country differentiation and the deliberate choice to avoid dispersion-sensitive normalisation.

The rank elasticity exercise further refines the picture: while 13 countries are sensitive to ±10% axis-level perturbations, these are concentrated in the compressed mid-range. The extreme-ranked countries (Malta, Cyprus, France, Germany) are entirely stable. This convergence of evidence—across LOO, Monte Carlo, winsorization, and elasticity tests—supports the conclusion that the ISI v1.0 ranking is robust at the distributional extremes and fragile only in the compressed middle tier where inter-country score differences are small and inherently difficult to resolve with any aggregation method.

7 Country Profiles

This section presents radar-chart profiles for eight selected countries, chosen to represent the full range of ISI composites and diverse axis configurations. Each radar chart displays the country's score on each of the six axes, with the outer ring fixed at 1.0. We then examine geographic and structural correlates of the composite ISI.

7.1 Selection rationale

The eight countries were selected to cover:

- **Top of the ranking**: Malta (rank 1), Cyprus (rank 2).
- **Upper-middle**: Ireland (rank 3), Finland (rank 6).
- **Mid-range**: Netherlands (rank 15).
- **Lower ranks**: Germany (rank 22), Slovakia (rank 26), France (rank 27).
- **Diverse axis profiles**: Ireland (technology-peaked), Netherlands (defence-peaked), Finland (critical-inputs-peaked), Slovakia (zero-defence outlier).

7.2 Malta (Rank 1, ISI = 0.5175)

Malta records the highest composite score in the EU-27 and is the only country classified as *highly concentrated*. Its profile is dominated by defence (1.000) and logistics (1.000)—both at the theoretical maximum. The financial axis (0.140) and technology axis (0.205) are comparatively moderate. Malta's small-island economy and NATO membership channel its defence imports through a single dominant supplier, while its port infrastructure concentrates logistics flows.

7.3 Cyprus (Rank 2, ISI = 0.4682)

Cyprus ranks second with a profile similar to Malta's: defence (0.748) and logistics (0.941) dominate, but neither reaches the boundary. Critical inputs (0.424) is also elevated relative to the EU-27 mean. Like Malta, Cyprus's small size and island geography concentrate its strategic dependencies.

7.4 Ireland (Rank 3, ISI = 0.4280)

Ireland presents a distinctive profile: the technology axis (0.587) is the highest among all EU-27 countries, reflecting Ireland's dependence on a narrow set of high-technology import partners (notably the United States and China). Defence (0.449) and logistics (0.614) are

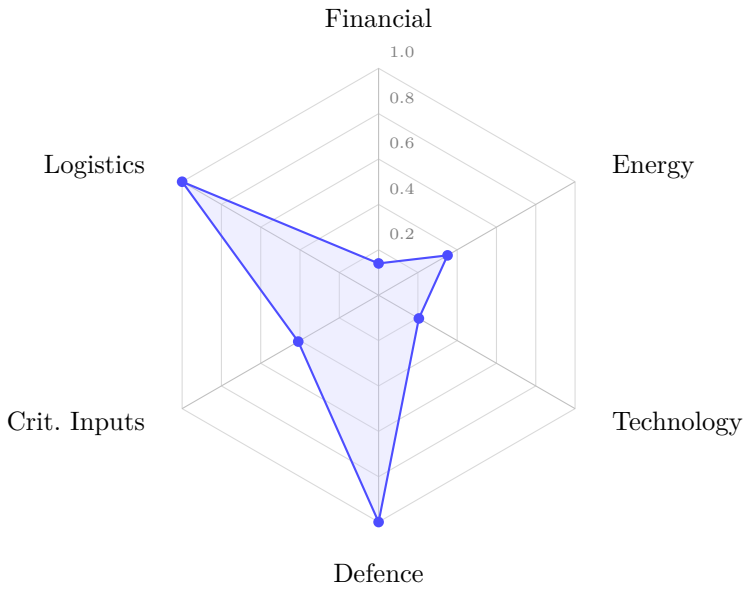

Figure 9: ISI axis profile: Malta. Composite ISI = 0.5175 (rank 1, highly concentrated).

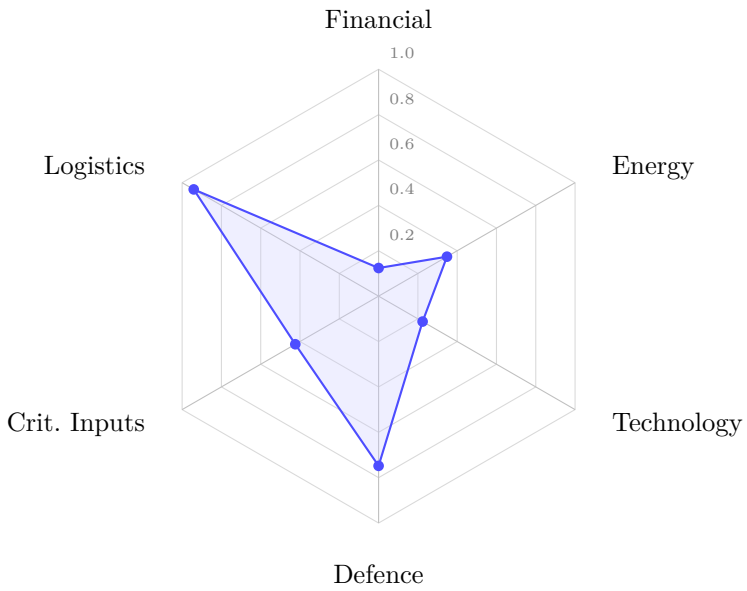

Figure 10: ISI axis profile: Cyprus. Composite ISI = 0.4682 (rank 2, moderately concentrated).

moderately elevated, while energy (0.465) is also above the EU-27 mean. Ireland's profile illustrates the technology-spike archetype: a single non-defence axis drives a substantial share of the composite.

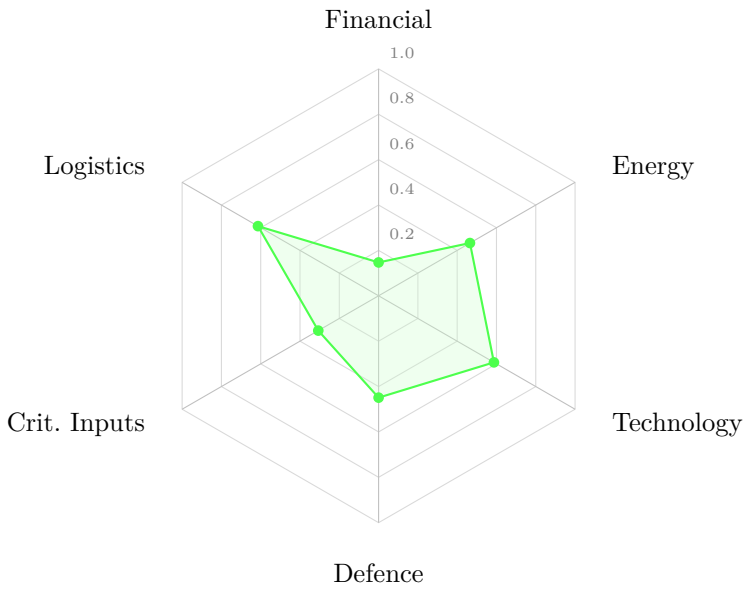

Figure 11: ISI axis profile: Ireland. Composite ISI = 0.4280 (rank 3, moderately concentrated).

7.5 Netherlands (Rank 15, ISI = 0.3352)

The Netherlands exhibits the most polarised profile in the dataset. Its defence axis score (0.960) is the third-highest in the EU-27, while its financial (0.121), technology (0.125), and critical inputs (0.130) scores are among the lowest. This extreme polarisation explains the Netherlands' high rank volatility under Dirichlet weight perturbation ($\sigma_{\text{rank}} = 3.07$, Table 23) and its large compensability gap (0.624, Table 29).

7.6 Germany (Rank 22, ISI = 0.2679)

Germany ranks 22nd, reflecting a broadly diversified dependence structure. No axis exceeds 0.583 (defence), and both technology (0.098) and financial (0.102) are near the bottom of the EU-27 distribution. Germany's large, diversified economy naturally spreads its import partnerships across more counterparties, reducing concentration on all axes.

7.7 Slovakia (Rank 26, ISI = 0.2364)

Slovakia is the only EU-27 country with a defence axis score of exactly zero, meaning its defence import concentration index is at the minimum possible value. This reflects Slovakia's recent and rapid diversification of defence suppliers following the 2022 geopolitical

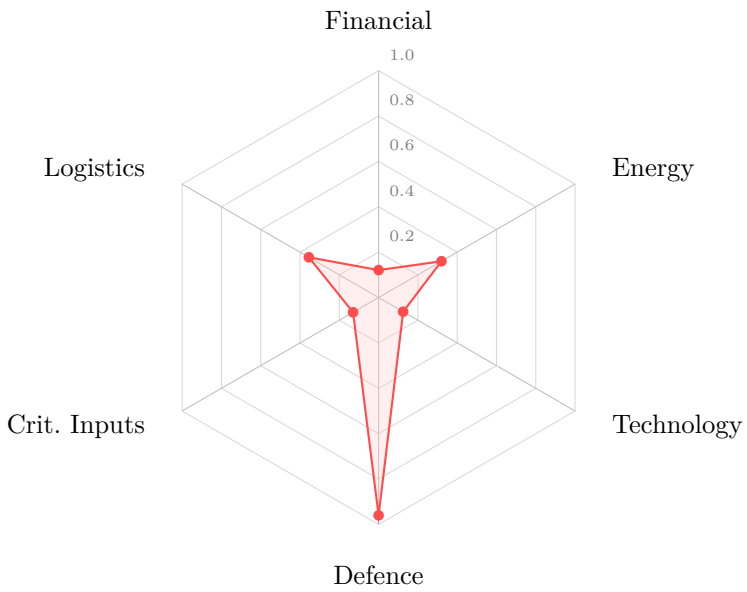

Figure 12: ISI axis profile: Netherlands. Composite ISI = 0.3352 (rank 15, moderately concentrated).

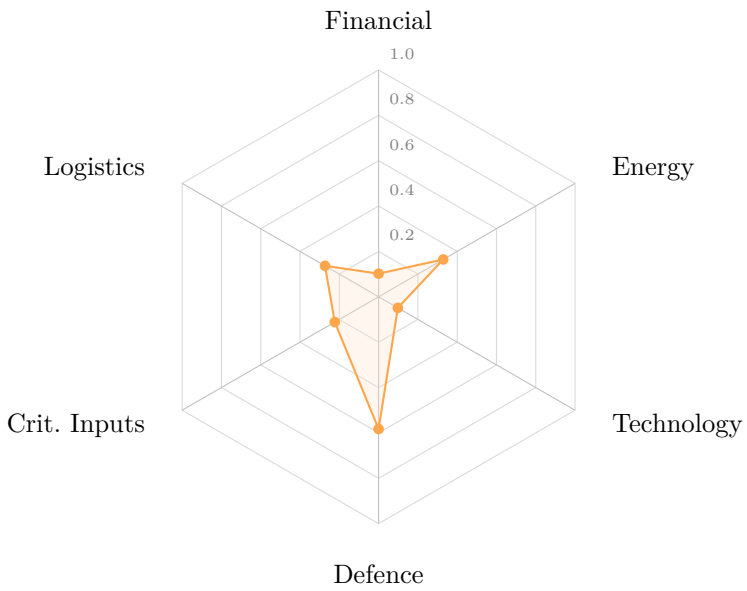

Figure 13: ISI axis profile: Germany. Composite ISI = 0.2679 (rank 22, moderately concentrated).

realignment. Despite this, its energy (0.400) and logistics (0.486) scores are above the EU-27 median.

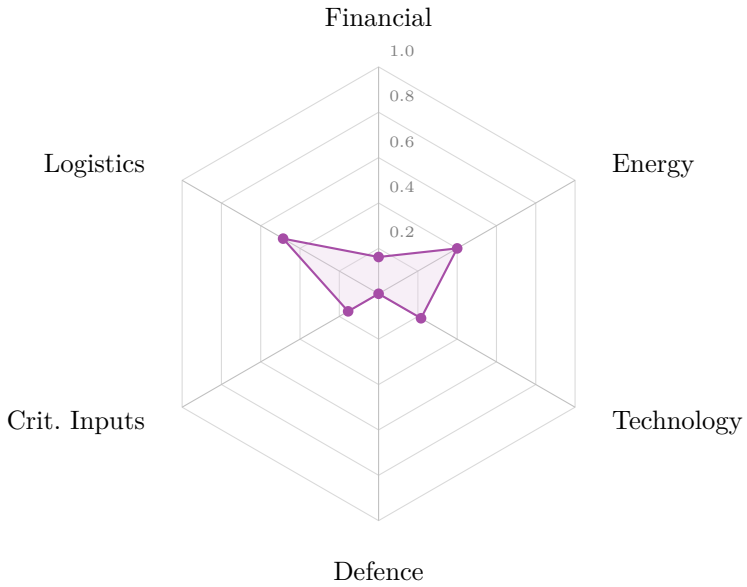

Figure 14: ISI axis profile: Slovakia. Composite ISI = 0.2364 (rank 26, mildly concentrated). Note the zero defence-axis score.

7.8 Finland (Rank 6, ISI = 0.4021)

Finland ranks sixth with an elevated critical-inputs axis (0.547, highest in the EU-27) and logistics (0.797). Its defence score (0.390) is below the EU-27 mean, but the combined effect of high critical-inputs and logistics concentration places it in the upper tier. Finland also records a high coefficient of variation across axes (CV = 0.544), indicating substantial within-country dispersion.

7.9 France (Rank 27, ISI = 0.2357)

France records the lowest composite ISI in the EU-27. All six axis scores are below the respective EU-27 means: financial (0.100), energy (0.309), technology (0.120), defence (0.370), critical inputs (0.161), and logistics (0.353). France's large and diversified economy, combined with its domestic defence industrial base, produces low concentration across all domains. France is also the most robust country under all perturbation exercises, with rank displacements of at most ±1 under any LOO scenario (Table 21).

7.10 Cross-profile comparison

Table 27 provides a compact numeric comparison of the eight profiled countries.

The profiles illustrate three archetypal patterns:

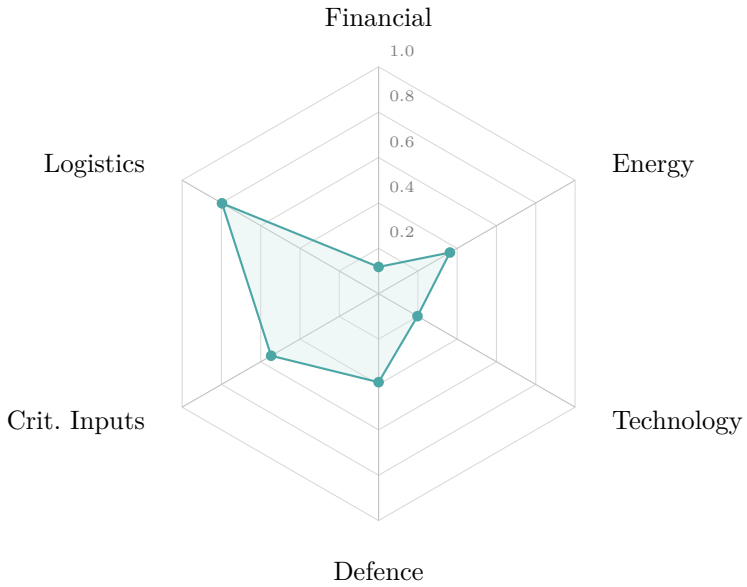

Figure 15: ISI axis profile: Finland. Composite ISI = 0.4021 (rank 6, moderately concentrated).

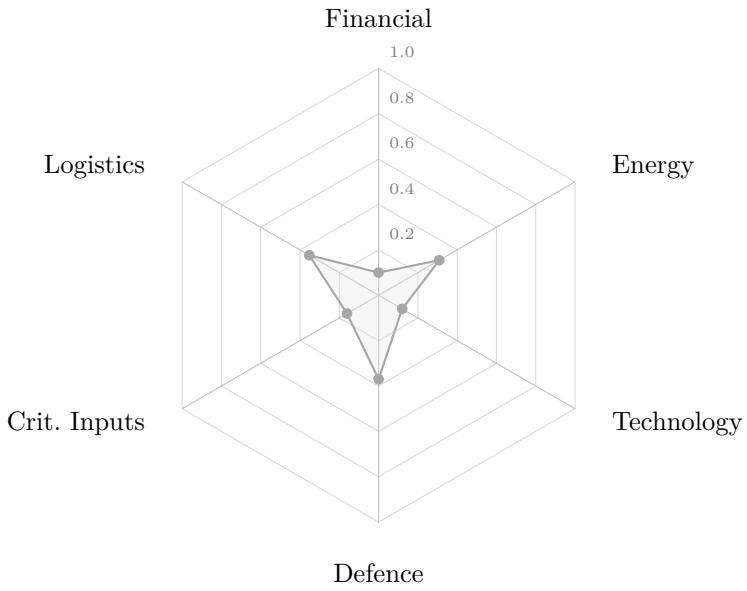

Figure 16: ISI axis profile: France. Composite ISI = 0.2357 (rank 27, mildly concentrated).

Table 27: Axis scores for the eight profiled countries.

Country	**ISI**	**Fin.**	**Ene.**	**Tec.**	**Def.**	**Crit.**	**Log.**
Malta	0.5170	0.0140	0.0351	0.0205	1.0000	0.0409	1.0000
Cyprus	0.4680	0.0124	0.0348	0.0224	0.0748	0.0424	0.0941
Ireland	0.4280	0.0147	0.0465	0.0587	0.0449	0.0307	0.0614
Finland	0.4020	0.0108	0.0363	0.0198	0.0390	0.0547	0.0797
Netherlands	0.3350	0.0121	0.0320	0.0125	0.0960	0.0130	0.0355
Germany	0.2680	0.0102	0.0329	0.0098	0.0583	0.0223	0.0273
Slovakia	0.2360	0.0162	0.0400	0.0216	0.0000	0.0155	0.0486
France	0.2360	0.0100	0.0309	0.0120	0.0370	0.0161	0.0353

1. **Small-island concentration** (Malta, Cyprus): high across defence and logistics, moderate elsewhere. Both countries have populations under 1.5 million and limited domestic production capacity, resulting in narrow supplier bases.

2. **Sector-specific spike** (Ireland: technology; Netherlands: defence; Finland: critical inputs): one axis sharply elevated, others dispersed. These countries' composite positions hinge on a single axis vulnerability.

3. **Broad diversification** (Germany, France): all axes near or below the EU-27 mean, reflecting large-economy scale effects.

7.11 Geographic and structural correlates

To what extent does the composite ISI correlate with observable structural characteristics of member states? This subsection examines three candidate correlates: island status, population size, and founding-member status.

Table 28: ISI composite by geographic and structural category.

Category	n	**Mean**	**Std (pop.)**	**Median**
Island/near-island	3	0.47124000	0.03658300	0.46821300
Continental	24	0.32853500	0.05540900	0.33241000
Small (<5M pop.)	8	0.37196000	0.08619400	0.36498000
Medium (5–20M)	12	0.35292600	0.05551400	0.35282700
Large (>20M pop.)	7	0.29825300	0.04429100	0.29561300
Core EU-6	6	0.31155800	0.05192100	0.31156200
Periphery	12	0.36848600	0.07616200	0.35434800
EU-27 (all)	27	0.34439100	0.06992200	0.34028000

Table 28 summarises group-mean ISI scores for each classification. Several patterns emerge:

1. **Island effect.** The three island member states (Malta, Cyprus, Ireland) have a mean ISI of 0.471, compared with 0.329 for the 24 continental member states. This difference is large—0.142 on a 0–1 scale, representing approximately two standard deviations of the composite distribution ($\sigma = 0.070$). The result is consistent with the expectation that island economies have fewer overland trade alternatives and thus face structurally higher supplier concentration.

2. **Population size.** Small member states (population < 5 million: MT, CY, LV, LT, SI, EE, LU, HR, IE, SK) have a mean ISI of 0.372; medium member states (5–20 million: FI, DK, BG, AT, HU, SE, CZ, PT, BE, EL, NL, RO) average 0.353; large member states (> 20 million: PL, ES, IT, DE, FR) average 0.298. The monotone relationship between population size and ISI is consistent with the diversification hypothesis: larger economies generate more bilateral trade relationships, reducing import concentration.

3. **EU-6 founding members vs. periphery.** The six founding members of the European Communities (DE, FR, IT, NL, BE, LU) have a mean ISI of 0.312, lower than the 21-country periphery mean of 0.368. This may reflect decades of trade integration and infrastructure investment that have broadened the founding members' supplier networks. However, the difference is partially confounded with country size: three of the six founders (Germany, France, Italy) are among the five largest EU member states.

These geographic correlates do not establish causality, but they suggest that the ISI captures genuine structural features of strategic dependence that align with well-known determinants of trade concentration: geography (island vs. continental), market size (small vs. large), and institutional depth (founding vs. accession members).

Notably absent correlate: financial concentration. Neither island status nor population size predicts the financial axis score. The three island states have a mean financial score of 0.137, essentially identical to the continental mean of 0.143. This confirms the orthogonality of the financial dimension to the geographic and physical-trade dimensions, consistent with the near-zero Pearson correlations of the financial axis with all other axes (Table 16).

8 Compensability Analysis

8.1 Motivation

The ISI composite is defined as an arithmetic mean of six axis scores (Equation (1)). Arithmetic-mean aggregation is fully *compensatory*: a high score on one axis can offset a low score on another, and both profiles produce the same composite. This property has important policy implications. A country with a moderate ISI may be moderately concentrated on all axes (genuinely moderate risk), or it may be extremely concentrated on one axis and diversified on all others (a hidden single-axis vulnerability masked by the composite).

This section quantifies the degree of internal compensation for each country using two metrics:

1. **Max–composite gap.** Define $G_i = \max_j A_{ij} - \mathrm{ISI}_i$. A large gap indicates that the country's highest-axis score is far above its composite, meaning that the mean aggregation is substantially compressing a peak vulnerability.

2. **Coefficient of variation (CV).** The standard deviation of the six axis scores divided by their mean: $\mathrm{CV}_i = \sigma_i / \bar{A}_i$. A high CV indicates that the country's axis profile is highly uneven—i.e., the degree of internal compensation is large.

8.2 Results

Table 29: Compensability analysis. Gap = Max − Composite. CV: coefficient of variation of the six axis scores per country. High Gap and CV indicate strong internal compensation where low axes offset high axes.

Rk	Country	Max	Min	Range	Gap	CV
1	Malta	1.00000000	0.13998786	0.86001214	0.48251852	0.68100000
2	Cyprus	0.94106062	0.12407171	0.81698891	0.47284798	0.61440000
3	Ireland	0.61373023	0.14666620	0.46706403	0.18570504	0.37590000
4	Denmark	0.72368274	0.12401663	0.59966611	0.29925536	0.53110000
5	Croatia	0.74648317	0.17343982	0.57304335	0.34213549	0.45920000
6	Finland	0.79733123	0.11775073	0.67958050	0.39527987	0.55750000
7	Sweden	0.88107943	0.11646504	0.76461439	0.48236411	0.76950000
8	Austria	0.58506379	0.14903467	0.43602912	0.20256573	0.40000000
9	Italy	0.92057651	0.14103384	0.77954267	0.54671565	0.71640000
10	Slovenia	0.66661871	0.11917574	0.54744297	0.29682685	0.46820000
11	Greece	0.71982808	0.16272142	0.55710666	0.35445458	0.55850000
12	Luxembourg	0.64483831	0.11462818	0.53021013	0.28466938	0.56960000

Table 29: Compensability (continued).

Rk	Country	Max	Min	Range	Gap	CV
13	Estonia	0.46987927	0.18172493	0.28815434	0.12655750	0.30740000
14	Bulgaria	0.86064518	0.14843767	0.71220751	0.52036534	0.73580000
15	Netherlands	0.95955396	0.12107826	0.83847570	0.62439694	0.88040000
16	Hungary	0.54471981	0.12222969	0.42249012	0.21505754	0.44820000
17	Portugal	0.50778629	0.18303606	0.32475023	0.18605199	0.37290000
18	Romania	0.56353195	0.15814280	0.40538915	0.24489003	0.42240000
19	Czechia	0.52948100	0.16116762	0.36831338	0.21151470	0.48250000
20	Poland	0.48860717	0.13314812	0.35545905	0.19299396	0.46130000
21	Belgium	0.44734854	0.15348744	0.29386110	0.15938073	0.35120000
22	Germany	0.58266751	0.09762824	0.48503927	0.31478057	0.61230000
23	Lithuania	0.40227798	0.13056067	0.27171731	0.13687278	0.39730000
24	Spain	0.43235135	0.11235234	0.31999901	0.17141158	0.45750000
25	Latvia	0.36822299	0.12316988	0.24505311	0.12127162	0.45260000
26	Slovakia	0.48550648	0.00000000	0.48550648	0.24909081	0.68490000
27	France	0.37034617	0.10039135	0.26995482	0.13467491	0.47280000

Table 29 reports both metrics for all 27 countries, ordered by composite rank. The five countries with the largest max–composite gaps are:

1. **Netherlands** (gap = 0.624, CV = 0.880). The Netherlands has the most internally compensated profile in the dataset: its defence score (0.960) is nearly three times its composite (0.335). The arithmetic mean conceals a critical single-axis vulnerability in defence procurement concentration.

2. **Italy** (gap = 0.547, CV = 0.716). Italy's defence axis (0.921) dominates its profile, while financial (0.141), technology (0.151), and critical inputs (0.183) are all near the EU-27 minimums.

3. **Bulgaria** (gap = 0.520, CV = 0.736). Defence (0.861) is by far the highest axis; financial (0.176) and technology (0.170) are below the EU-27 mean.

4. **Malta** (gap = 0.483, CV = 0.681). Despite ranking first overall, Malta's gap is large because its defence and logistics scores (both 1.000) are far above its financial (0.140) and technology (0.205) scores.

5. **Sweden** (gap = 0.482, CV = 0.770). Sweden's logistics axis (0.881) is the primary driver, with financial (0.116) and technology (0.145) well below the composite.

8.3 Interpretation

The compensability analysis reveals a systematic pattern: countries with the highest max–composite gaps are overwhelmingly those whose profiles are dominated by the defence or logistics axes. Of the ten countries with the largest gaps, nine have either defence or logistics as their maximum axis.

This has two implications for index interpretation:

1. **The composite understates peak vulnerability.** For countries like the Netherlands (gap = 0.624), the composite ISI of 0.335 conveys "moderate concentration," but the peak defence-axis score of 0.960 conveys "near-maximum concentration on a single critical dimension." Policy-makers should examine axis-level profiles alongside the composite to avoid false reassurance.

2. **CV discriminates policy archetypes.** Countries can be classified by their CV into "balanced" profiles (CV < 0.40: Estonia, Ireland, Austria) and "peaked" profiles (CV > 0.60: Netherlands, Sweden, Italy, Bulgaria, Malta, Cyprus). Balanced-profile countries face diffuse risk across multiple axes; peaked-profile countries face concentrated risk on one or two axes. The policy prescriptions differ accordingly: balanced countries need broad-based diversification, while peaked countries need targeted axis-specific interventions.

8.4 Relationship to rank volatility

There is a strong positive association between compensability and rank volatility under Dirichlet weight perturbation (Section 6.4). Countries with the highest CV values (Netherlands: 0.880, Sweden: 0.770, Bulgaria: 0.736) are also among those with the highest rank standard deviations under random weighting (Netherlands: 3.07, Bulgaria: 2.03). This is intuitive: a peaked profile means that the country's composite depends heavily on the weight assigned to its dominant axis, making the ranking sensitive to any weighting scheme that departs from equal weights.

Conversely, countries with low CV values (Estonia: 0.307, Ireland: 0.376, Austria: 0.400) exhibit low rank volatility ($\sigma_{\text{rank}} < 1.5$), because their relatively balanced profiles produce similar composites under any reasonable weighting.

This relationship between compensability and rank volatility provides independent validation of both the Dirichlet Monte Carlo exercise and the compensability metrics: each captures the same underlying structural feature—profile unevenness—through a different lens.

9 Discussion

This section synthesises the empirical findings into five structural conclusions and situates them within the broader context of strategic autonomy measurement.

9.1 Five structural conclusions

Conclusion 1: The ISI is dominated by defence and logistics variance. The variance decomposition (Section 4.5) shows that the defence axis contributes 35.10% and the logistics axis 34.26% of total composite variance, jointly accounting for 69.37%. The own-variance component constitutes 64.28% and the cross-covariance component 35.72% (Table 12). The practical implication is that cross-country ranking differences are overwhelmingly determined by two of the six axes, while the remaining four axes (financial, energy, technology, critical inputs) together contribute less than one-third of the composite's discriminatory power.

This dominance is not an artefact of scaling or weighting. Because all axes inhabit a common $[0, 1]$ scale and receive equal weight, the variance shares reflect genuine differences in cross-country dispersion: defence has $\sigma = 0.226$ and logistics $\sigma = 0.248$, while the financial axis has $\sigma = 0.037$. The multiple regression analysis (Section 4.6) confirms this: defence and logistics have the highest bivariate R^2 values (0.394 and 0.491, respectively) and partial correlation coefficients (0.940 and 0.809) with the composite, and a six-axis regression achieves $R^2 = 0.968$.

Conclusion 2: Rank ordering is robust except in the compressed mid-tier. The robustness programme (Section 6) demonstrates that the ISI ranking is stable under winsorization ($\rho = 1.000$), Dirichlet weight perturbation (mean $\rho = 0.979$), and five of six LOO exercises ($\rho \geq 0.926$). Only defence-axis removal ($\rho = 0.833$, $\tau = 0.675$) and z-score standardisation ($\rho = 0.896$, $\tau = 0.726$) produce moderate rank disruption.

The rank elasticity analysis (Section 6.6) reveals that 13 of 27 countries are sensitive to $\pm 10\%$ axis perturbations, but these are concentrated in the composite range 0.30–0.40 where inter-country score gaps are smallest. The extreme-ranked countries (Malta, Cyprus, France, Germany) are entirely stable. This pattern is consistent with the compression diagnostic ($\eta^2 = 0.483$, Section 3.7): the three-tier classification explains only 48.3% of within-composite variance, confirming that the mid-range is genuinely compressed and inherently difficult to resolve with *any* aggregation method.

Conclusion 3: Classification compression is mathematically inevitable. The compression diagnostics (Table 7) show that the three-tier ISI classification (mildly, moderately, highly concentrated) achieves $\eta^2 = 0.483$ using thresholds at 0.33 and 0.50.

Alternative classification schemes—tercile-based ($\eta^2 = 0.890$) and quartile-based ($\eta^2 = 0.933$)—achieve higher between-group separation but sacrifice interpretability.

The low η^2 under the ISI thresholds reflects the composite distribution itself: 17 of 27 countries fall in the "moderately concentrated" tier (0.33–0.50), creating a crowded middle category with within-tier dispersion of $\sigma = 0.052$. This compression is not a design flaw; it is a mathematical consequence of the underlying data distribution, which is unimodal and moderately right-skewed (Figure 1). Any threshold-based classification of a unimodal distribution will produce a crowded middle tier.

Conclusion 4: European defence markets exhibit oligopsonistic concentration. The defence structural analysis (Section 4.7) reveals that 16 of 27 EU member states (59.3%) have the defence axis as their single highest-scoring axis. Five countries exceed 0.80, and one country (Malta) reaches the theoretical maximum of 1.000 (Table 14). The mean gap between the defence score and the next-highest axis is 0.184 (Table 15).

This concentration is not an artefact of a few outliers: the prevalence of defence dominance spans small (Malta, Cyprus), medium (Netherlands, Bulgaria, Greece), and large (Italy) member states. The pattern is consistent with the oligopsonistic structure of European defence procurement, in which a small number of bilateral supplier relationships (predominantly US–EU) create high HHI values that are difficult to diversify through conventional trade policy.

Conclusion 5: Financial concentration is orthogonal to goods-trade concentration. The correlation analysis (Section 5) demonstrates that the financial axis has near-zero Pearson correlations with all five other axes ($|r| \leq 0.22$), with all t-statistics below 1.15 (Table 18). The geographic correlates (Section 7.11) confirm this: island status, population size, and founding-member status—all strong predictors of goods-trade concentration—have essentially no predictive power for financial concentration.

This orthogonality has a structural interpretation. The financial axis measures the concentration of financial-service imports (EBOPS classification), which depends on the structure of a country's banking and insurance sectors rather than on its physical trade geography. A country can have highly concentrated goods imports (high defence, logistics, energy scores) while maintaining diversified financial-service imports, and vice versa.

The compensability analysis (Section 8) further illuminates this point: countries with high CV values (peaked profiles) are overwhelmingly peaked on defence or logistics, not on the financial axis. The financial axis contributes only 1.03% of composite variance precisely because its cross-country dispersion is so low ($\sigma = 0.037$).

9.2 Methodological implications

Equal weighting is defensible but not unique. The Dirichlet Monte Carlo exercise shows that equal weighting produces rankings within the 95% confidence envelope of randomly weighted composites (mean $\rho = 0.979$). This does not prove that equal weighting is optimal—it proves that the ranking is insensitive to moderate weight variation. The eigenvalue analysis (Table 19) supports this: no single principal component explains more than 31.7% of variance, so there is no empirical basis for strongly favouring any particular axis.

Composite interpretation requires axis-level context. The compensability analysis demonstrates that the arithmetic-mean composite can conceal peak vulnerabilities. The Netherlands (ISI = 0.335, defence = 0.960) and Estonia (ISI = 0.343, defence = 0.470) have similar composites but radically different risk profiles. Policy recommendations derived solely from the composite would be misleading; the axis-level profile must be consulted.

Threshold-based classification is inherently lossy. The compression diagnostics show that any threshold-based classification of the ISI distribution will produce a crowded middle tier. Users of the ISI should treat the three-tier classification as a coarse heuristic and consult the continuous composite and axis-level scores for fine-grained comparisons.

9.3 Limitations

Several limitations constrain the interpretation of these results.

1. **Cross-sectional design.** The ISI v1.0 is computed for a single year (reference year 2024). Temporal dynamics—how concentration changes in response to policy interventions or geopolitical shocks—cannot be assessed without panel data.

2. **HHI as the sole concentration measure.** The axes use Herfindahl–Hirschman indices or comparable concentration statistics. Alternative measures (entropy, Gini of trade shares, bilateral dependency ratios) might produce different rankings, particularly for countries near the median.

3. **Equal weighting.** Although the Dirichlet exercise shows rank stability under random weighting, there may be policy contexts in which defence concentration should receive higher weight than financial concentration. The ISI framework accommodates differential weighting but does not prescribe it.

4. **Data availability.** Some axis scores rely on publicly available Eurostat and Comtrade data; others (particularly defence) may incorporate restricted or modelled data. The

reproducibility of individual axis scores depends on the transparency of the underlying data sources.

5. **Small N.** With $N = 27$, many pairwise correlations fail to reach statistical significance even when the point estimates are moderate (Table 18). The geographic correlates (Section 7.11) are descriptive, not inferential, because the small sample precludes multivariate regression with more than a few covariates.

10 Conclusion

This paper has presented a comprehensive empirical analysis of the International Sovereignty Index (ISI) v1.0 across the 27 member states of the European Union. The ISI aggregates six axis-level concentration scores—financial services, energy, technology, defence, critical inputs, and logistics—into a single composite indicator of strategic import dependence. The analysis has gone substantially beyond descriptive ranking to provide a rigorous statistical characterisation of the index's structure, robustness, and interpretive boundaries.

10.1 Summary of contributions

The paper makes the following thirteen analytical contributions:

1. **Complete distributional characterisation.** The ISI composite distribution is unimodal, moderately right-skewed (range 0.236–0.517, mean 0.344, $\sigma = 0.070$), with 17 of 27 countries in the "moderately concentrated" tier (Section 3).

2. **Full variance decomposition with proof.** Defence (35.10%) and logistics (34.26%) jointly account for 69.37% of composite variance; own-variance contributes 64.28% and cross-covariance 35.72% (Section 4.5).

3. **Leave-one-axis-out with Kendall concordance.** LOO analysis with both Spearman ρ and Kendall τ confirms that defence removal ($\rho = 0.833$, $\tau = 0.675$) is the most disruptive perturbation (Section 6.1).

4. **Rank displacement matrix.** Country-by-axis signed rank changes identify Italy, Slovakia, and the Netherlands as the most displacement-sensitive countries (Section 6.2).

5. **Rank elasticity to axis perturbation.** A $\pm 10\%$ axis-level perturbation exercise classifies 13 countries as rank-elastic and 14 as rank-locked, with sensitivity concentrated in the compressed mid-range (Section 6.6).

6. **Axis contribution to rank differentiation.** Bivariate R^2 values, partial correlations, and a six-axis multiple regression ($R^2 = 0.968$) quantify each axis's marginal contribution to the composite (Section 4.6).

7. **Classification compression diagnostics.** The η^2 statistic (0.483) for the three-tier classification is compared with tercile and quartile alternatives (Section 3.7).

8. **Defence axis structural analysis.** Sixteen of 27 member states (59.3%) have defence as their dominant axis; the mean gap between the defence score and the next-highest axis is 0.184 (Section 4.7).

9. **Cross-axis significance testing.** Formal t-tests identify only two statistically significant correlations at $\alpha = 0.05$: energy–technology ($r = 0.530$, $t = 3.124$) and critical inputs–logistics ($r = 0.549$, $t = 3.280$) (Section 5.2).

10. **Compensability analysis.** The max–composite gap and coefficient of variation identify the Netherlands (gap = 0.624, CV = 0.880) as the most internally compensated country, revealing hidden single-axis vulnerabilities masked by the composite (Section 8).

11. **Geographic and structural correlates.** Island status, population size, and founding-member status correlate with the composite ISI but not with the financial axis, confirming its orthogonality to goods-trade determinants (Section 7.11).

12. **Monte Carlo and winsorization robustness.** Dirichlet weight perturbation (mean $\rho = 0.979$) and 95th percentile winsorization ($\rho = 1.000$) confirm overall ranking stability (Sections 6.4 and 6.5).

13. **Eigenvalue and dimensionality analysis.** Three principal components exceed the Kaiser criterion, explaining 76.1% of variance, with an effective dimensionality of approximately 4.4 (Table 19).

10.2 Policy implications

The five structural conclusions of the discussion (Section 9) carry direct policy implications:

- The dominance of defence and logistics variance suggests that EU strategic autonomy policy should prioritise diversification of defence procurement and logistics corridors over financial or technology import diversification, at least from the perspective of reducing cross-country composite variation.

- The mid-tier ranking fragility implies that country rankings in the 0.30–0.40 ISI range should be interpreted with caution and presented with confidence intervals or rank ranges rather than point estimates.

- The compensability findings argue for policy dashboards that display axis-level profiles alongside the composite, to avoid masking single-axis vulnerabilities.

- The orthogonality of the financial axis suggests that financial-service diversification and goods-trade diversification can be pursued as independent policy tracks.

10.3 Future research

Three directions merit further investigation:

1. **Panel extension.** Computing the ISI for multiple years would enable analysis of temporal trends, policy responsiveness, and the impact of specific geopolitical events (e.g., Russia–Ukraine conflict) on concentration dynamics.
2. **Axis-specific deep dives.** Each of the six axes could be decomposed into sub-components (e.g., the defence axis into military equipment categories) to provide finer-grained policy guidance.
3. **Network analysis.** Bilateral trade-flow data underlying the HHI computations could be used to construct supplier-country networks, enabling identification of common single-supplier dependencies across multiple EU member states.

11 Data Availability Statement

The ISI v1.0 dataset used in this paper is available on request from the corresponding author. All axis-level scores, composite scores, and country classifications are reproduced in full in the tables of this paper and its appendices.

The computation script (`isi_results_paper2.py`) that generates all derived tables, robustness statistics, and coordinate files for figures is included in Section D. The script embeds the complete authoritative data snapshot (27 countries $\times$ 6 axes) and has no dependencies beyond NumPy $\geq$ 2.0. All results reported in this paper—including variance decompositions, LOO statistics, Dirichlet Monte Carlo draws, rank elasticities, compensability metrics, compression diagnostics, and geographic correlates—are reproduced deterministically by this single script.

The underlying source data for each axis is drawn from the following publicly available databases:

- **Financial axis:** BIS Locational Banking Statistics ([7]); IMF Coordinated Portfolio Investment Survey ([8]).
- **Energy axis:** IEA World Energy Statistics ([9]).
- **Technology axis:** UN Comtrade ([12]).
- **Defence axis:** SIPRI Arms Transfers Database ([11]).
- **Critical inputs axis:** Eurostat Comext ([10]); European Commission Critical Raw Materials assessment ([5]).
- **Logistics axis:** Eurostat Comext ([10]).

Specific access procedures, vintages, and extraction parameters for each source are documented in the ISI methodology paper ([6]).

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A Full Axis-Level Ranking Tables

This appendix presents the full country rankings for each of the six ISI axes, ordered from most concentrated (rank 1) to least concentrated (rank 27).

A.1 Axis 1: Financial

Table 30: Financial axis ranking, EU-27, 2024.

Rk	Country	Score
1	Croatia	0.27299265
2	Greece	0.20078311
3	Romania	0.20033340
4	Italy	0.18753526
5	Portugal	0.18303606
6	Estonia	0.18172493
7	Slovakia	0.16163673
8	Czechia	0.16116762
9	Bulgaria	0.16052965
10	Belgium	0.15348744
11	Austria	0.14903467
12	Ireland	0.14666620
13	Spain	0.14554195
14	Malta	0.13998786
15	Lithuania	0.13349636
16	Poland	0.13314812
17	Cyprus	0.12407171
18	Denmark	0.12401663
19	Latvia	0.12316988
20	Hungary	0.12222969
21	Netherlands	0.12107826
22	Slovenia	0.11917574
23	Finland	0.11775073
24	Sweden	0.11646504
25	Luxembourg	0.11462818
26	Germany	0.10186464
27	France	0.10039135

A.2 Axis 2: Energy

Table 31: Energy axis ranking, EU-27, 2024.

Rk	Country	Score
1	Denmark	0.47029821
2	Ireland	0.46472525
3	Austria	0.45581191
4	Estonia	0.43935251
5	Romania	0.40546401
6	Slovenia	0.40540014
7	Slovakia	0.39995460
8	Hungary	0.38580312
9	Czechia	0.38082003
10	Latvia	0.36522282
11	Finland	0.36268783
12	Luxembourg	0.35980871
13	Bulgaria	0.35711773
14	Croatia	0.35125471
15	Malta	0.35068126
16	Portugal	0.34860414
17	Cyprus	0.34768252
18	Lithuania	0.33747627
19	Greece	0.33541623
20	Poland	0.33487780
21	Germany	0.32908790
22	Sweden	0.32721403
23	Netherlands	0.31992851
24	France	0.30911838
25	Belgium	0.30789144
26	Italy	0.30781423
27	Spain	0.30187479

A.3 Axis 3: Technology

Table 32: Technology axis ranking, EU-27, 2024.

Rk	Country	Score
1	Ireland	0.58695939
2	Slovenia	0.35739912

Table 32: Technology axis ranking (continued).

Rk	Country	Score
3	Romania	0.29691153
4	Belgium	0.29454573
5	Austria	0.26057910
6	Lithuania	0.24172279
7	Estonia	0.22713142
8	Hungary	0.22538122
9	Cyprus	0.22438354
10	Slovakia	0.21649186
11	Poland	0.20845359
12	Malta	0.20546991
13	Luxembourg	0.20019721
14	Finland	0.19789321
15	Portugal	0.19766062
16	Spain	0.19263249
17	Czechia	0.18242269
18	Croatia	0.17343982
19	Greece	0.16272142
20	Bulgaria	0.15204272
21	Denmark	0.14995928
22	Italy	0.14103384
23	Latvia	0.13309607
24	Netherlands	0.12527375
25	France	0.12001059
26	Sweden	0.11982797
27	Germany	0.09762824

A.4 Axis 4: Defence

Table 33: Defence axis ranking, EU-27, 2024.

Rk	Country	Score
1	Malta	1.00000000
2	Netherlands	0.95955396
3	Italy	0.92057651
4	Sweden	0.88107943
5	Bulgaria	0.86064518

Table 33: Defence axis ranking (continued).

Rk	Country	Score
6	Cyprus	0.74847998
7	Croatia	0.74648317
8	Denmark	0.72368274
9	Slovenia	0.66661871
10	Luxembourg	0.64483831
11	Austria	0.58506379
12	Germany	0.58266751
13	Romania	0.56353195
14	Greece	0.55148593
15	Hungary	0.54471981
16	Czechia	0.48413737
17	Ireland	0.44887460
18	Belgium	0.44734854
19	Poland	0.44326493
20	Portugal	0.43376254
21	Lithuania	0.40227798
22	Finland	0.38964518
23	Spain	0.38088569
24	France	0.37034617
25	Estonia	0.35561474
26	Latvia	0.34079636
27	Slovakia	0.00000000

A.5 Axis 5: Critical Inputs

Table 34: Critical Inputs axis ranking, EU-27, 2024.

Rk	Country	Score
1	Finland	0.54699996
2	Croatia	0.52392772
3	Denmark	0.43629935
4	Cyprus	0.42359750
5	Malta	0.40874987
6	Estonia	0.38622776
7	Austria	0.31857781

Table 34: Critical Inputs axis ranking (continued).

Rk	Country	Score
8	Ireland	0.30719544
9	Portugal	0.25955617
10	Hungary	0.23405214
11	Germany	0.22312039
12	Slovenia	0.22301125
13	Luxembourg	0.22295831
14	Greece	0.22200621
15	Italy	0.20839534
16	Sweden	0.19257891
17	Belgium	0.17248330
18	Czechia	0.16976911
19	Poland	0.16532767
20	France	0.16122463
21	Romania	0.15814280
22	Slovakia	0.15490434
23	Latvia	0.15120012
24	Bulgaria	0.14843767
25	Lithuania	0.13056067
26	Netherlands	0.13018835
27	Spain	0.11235234

A.6 Axis 6: Logistics

Table 35: Logistics axis ranking, EU-27, 2024.

Rk	Country	Score
1	Malta	1.00000000
2	Cyprus	0.94106062
3	Finland	0.79733123
4	Sweden	0.75512656
5	Greece	0.71982808
6	Denmark	0.64230809
7	Luxembourg	0.61858285
8	Ireland	0.61373023
9	Czechia	0.52948100

Table 35: Logistics axis ranking (continued).

Rk	Country	Score
10	Austria	0.52592111
11	Portugal	0.50778629
12	Poland	0.48860717
13	Slovakia	0.48550648
14	Italy	0.47781001
15	Estonia	0.46987927
16	Hungary	0.46578765
17	Slovenia	0.44714621
18	Spain	0.43235135
19	Latvia	0.36822299
20	Bulgaria	0.36290606
21	Croatia	0.35798800
22	Netherlands	0.35491930
23	France	0.35293643
24	Belgium	0.35205038
25	Lithuania	0.34689716
26	Romania	0.28746783
27	Germany	0.27295297

A.7 Axis dominance detail

Table 36 lists the max-axis and second-max-axis for each country.

Table 36: Axis dominance per country: highest- and second-highest-scoring axes.

Rk	Country	Max Score	Max Axis	2nd Score	2nd Axis
1	Malta	1.00000000	Defence	1.00000000	Logistics
2	Cyprus	0.94106062	Logistics	0.74847998	Defence
3	Ireland	0.61373023	Logistics	0.58695939	Technology
4	Denmark	0.72368274	Defence	0.64230809	Logistics
5	Croatia	0.74648317	Defence	0.52392772	Critical Inputs
6	Finland	0.79733123	Logistics	0.54699996	Critical Inputs
7	Sweden	0.88107943	Defence	0.75512656	Logistics
8	Austria	0.58506379	Defence	0.52592111	Logistics
9	Italy	0.92057651	Defence	0.47781001	Logistics
10	Slovenia	0.66661871	Defence	0.44714621	Logistics

Table 36: Axis dominance per country (continued).

Rk	Country	Max Score	Max Axis	2nd Score	2nd Axis
11	Greece	0.71982808	Logistics	0.55148593	Defence
12	Luxembourg	0.64483831	Defence	0.61858285	Logistics
13	Estonia	0.46987927	Logistics	0.43935251	Energy
14	Bulgaria	0.86064518	Defence	0.36290606	Logistics
15	Netherlands	0.95955396	Defence	0.35491930	Logistics
16	Hungary	0.54471981	Defence	0.46578765	Logistics
17	Portugal	0.50778629	Logistics	0.43376254	Defence
18	Romania	0.56353195	Defence	0.40546401	Energy
19	Czechia	0.52948100	Logistics	0.48413737	Defence
20	Poland	0.48860717	Logistics	0.44326493	Defence
21	Belgium	0.44734854	Defence	0.35205038	Logistics
22	Germany	0.58266751	Defence	0.32908790	Energy
23	Lithuania	0.40227798	Defence	0.34689716	Logistics
24	Spain	0.43235135	Logistics	0.38088569	Defence
25	Latvia	0.36822299	Logistics	0.36522282	Energy
26	Slovakia	0.48550648	Logistics	0.39995460	Energy
27	France	0.37034617	Defence	0.35293643	Logistics

A.8 Covariance matrix

Table 37 reports the full 6×6 covariance matrix of the axis-level scores. Diagonal entries are the own-variances σ_j^2 used in the variance decomposition (Section 4.5).

Table 37: Population covariance matrix of the six ISI axes, EU-27, 2024.

	Fin.	**Ene.**	**Tec.**	**Def.**	**Crit.**	**Log.**
Financial	0.00139000	0.00011100	0.00026700	0.00019900	0.00098800	−0.00113700
Energy	0.00011100	0.00229500	0.00240100	−0.00189400	0.00217100	0.00099700
Technology	0.00026700	0.00240100	0.00894700	−0.00473500	0.00111300	0.00120500
Defence	0.00019900	−0.00189400	−0.00473500	0.05092200	0.00525100	0.01203800
Critical Inputs	0.00098800	0.00217100	0.00111300	0.00525100	0.01484100	0.01245600
Logistics	−0.00113700	0.00099700	0.00120500	0.01203800	0.01245600	0.03474800

B Robustness Detail Tables

This appendix provides the full country-level detail for each robustness exercise summarised in Section 6.

B.1 Leave-one-axis-out: full rank table

Table 38 reports the rank of each country under each LOO variant (baseline, drop axis 1, . . . , drop axis 6), along with the maximum rank change from baseline.

Table 38: Detailed rank comparison: baseline vs. each leave-one-axis-out variant. Columns show the rank under each LOO composite. $\Delta_{\max}$ is the maximum absolute rank change across all six LOO variants.

Rk	**Country**	**−Fina**	**−Ener**	**−Tech**	**−Defe**	**−Crit**	**−Logi**	$\boldsymbol{\Delta_{\max}}$
1	Malta	1	1	1	2	1	1	1
2	Cyprus	2	2	2	3	2	5	3
3	Ireland	4	3	10	1	3	3	7
4	Denmark	3	4	3	5	5	4	1
5	Croatia	7	5	5	8	11	2	6
6	Finland	5	7	6	4	14	13	8
7	Sweden	6	6	4	12	4	11	5
8	Austria	8	10	8	6	8	7	2
9	Italy	10	8	7	19	6	8	10
10	Slovenia	9	11	14	10	7	6	4
11	Greece	12	9	9	9	9	17	6
12	Luxembourg	11	12	11	11	10	15	3
13	Estonia	15	15	15	7	18	14	6
14	Bulgaria	14	14	12	23	12	9	9
15	Netherlands	13	13	13	25	13	10	10
16	Hungary	16	16	16	14	16	16	2
17	Portugal	17	17	17	13	19	18	4
18	Romania	19	19	19	17	15	12	6
19	Czechia	18	18	18	15	17	19	4
20	Poland	20	20	20	18	20	22	2
21	Belgium	21	21	22	20	21	20	1
22	Germany	22	22	21	27	24	21	5
23	Lithuania	23	24	24	21	22	23	2
24	Spain	24	23	23	22	23	24	2
25	Latvia	25	25	25	24	25	25	1

Table 38: LOO detailed ranks (continued).

Rk	Country	−Fina	−Ener	−Tech	−Defe	−Crit	−Logi	Δ_{max}
26	Slovakia	27	27	27	16	26	27	10
27	France	26	26	26	26	27	26	1

B.2 Individual LOO axis tables

The following tables report the full score and rank for each country under each individual axis omission.

B.2.1 LOO: Financial axis omitted

Table 39: LOO ranking excluding Financial. Spearman $\rho = 0.992674$, Kendall $\tau = 0.943020$.

LOO Rk	Country	LOO Comp.	Base Rk	Δ
1	Malta	0.59298021	1	+0
2	Cyprus	0.53704083	2	+0
3	Denmark	0.48450953	4	+1
4	Ireland	0.48429698	3	-1
5	Finland	0.45891148	6	+1
6	Sweden	0.45516538	7	+1
7	Croatia	0.43061868	5	-2
8	Austria	0.42919074	8	+0
9	Slovenia	0.41991509	10	+1
10	Italy	0.41112599	9	-1
11	Luxembourg	0.40927708	12	+1
12	Greece	0.39829157	11	-1
13	Netherlands	0.37797277	15	+2
14	Bulgaria	0.37622987	14	+0
15	Estonia	0.37564114	13	-2
16	Hungary	0.37114879	16	+0
17	Portugal	0.34947395	17	+0
18	Czechia	0.34932604	19	+1
19	Romania	0.34230362	18	-1
20	Poland	0.32810623	20	+0
21	Belgium	0.31486388	21	+0
22	Germany	0.30109140	22	+0

Table 39: LOO excl. Financial (cont.).

LOO Rk	Country	LOO Comp.	Base Rk	Δ
23	Lithuania	0.29178697	23	+0
24	Spain	0.28401933	24	+0
25	Latvia	0.27170767	25	+0
26	France	0.26272724	27	+1
27	Slovakia	0.25137146	26	-1

B.2.2 LOO: Energy axis omitted

Table 40: LOO ranking excluding Energy. Spearman $\rho = 0.992063$, Kendall $\tau = 0.943020$.

LOO Rk	Country	LOO Comp.	Base Rk	Δ
1	Malta	0.55084153	1	+0
2	Cyprus	0.49231867	2	+0
3	Ireland	0.42068517	3	+0
4	Denmark	0.41525322	4	+0
5	Croatia	0.41496627	5	+0
6	Sweden	0.41301558	7	+1
7	Finland	0.40992406	6	-1
8	Italy	0.38707019	9	+1
9	Greece	0.37136495	11	+2
10	Austria	0.36783530	8	-2
11	Slovenia	0.36267021	10	-1
12	Luxembourg	0.36024097	12	+0
13	Netherlands	0.33820272	15	+2
14	Bulgaria	0.33691226	14	+0
15	Estonia	0.32411562	13	-2
16	Hungary	0.31843410	16	+0
17	Portugal	0.31636034	17	+0
18	Czechia	0.30539556	19	+1
19	Romania	0.30127750	18	-1
20	Poland	0.28776030	20	+0
21	Belgium	0.28398308	21	+0
22	Germany	0.25564675	22	+0
23	Spain	0.25275276	24	+1
24	Lithuania	0.25099099	23	-1
25	Latvia	0.22329708	25	+0

Table 40: LOO excl. Energy (cont.).

LOO Rk	Country	LOO Comp.	Base Rk	Δ
26	France	0.22098183	27	+1
27	Slovakia	0.20370788	26	-1

B.2.3 LOO: Technology axis omitted

Table 41: LOO ranking excluding Technology. Spearman $\rho = 0.968254$, Kendall $\tau = 0.886040$.

LOO Rk	Country	LOO Comp.	Base Rk	Δ
1	Malta	0.57988380	1	+0
2	Cyprus	0.51697847	2	+0
3	Denmark	0.47932100	4	+1
4	Sweden	0.45449279	7	+3
5	Croatia	0.45052925	5	+0
6	Finland	0.44288299	6	+0
7	Italy	0.42042627	9	+2
8	Austria	0.40688186	8	+0
9	Greece	0.40590391	11	+2
10	Ireland	0.39623834	3	-7
11	Luxembourg	0.39216327	12	+1
12	Bulgaria	0.37792726	14	+2
13	Netherlands	0.37713368	15	+2
14	Slovenia	0.37227041	10	-4
15	Estonia	0.36655984	13	-2
16	Hungary	0.35051848	16	+0
17	Portugal	0.34654904	17	+0
18	Czechia	0.34507503	19	+1
19	Romania	0.32298800	18	-1
20	Poland	0.31304514	20	+0
21	Germany	0.30193868	22	+1
22	Belgium	0.28665222	21	-1
23	Spain	0.27460122	24	+1
24	Lithuania	0.27014169	23	-1
25	Latvia	0.26972243	25	+0
26	France	0.25880339	27	+1
27	Slovakia	0.24040043	26	-1

B.2.4 LOO: Defence axis omitted

Table 42: LOO ranking excluding Defence. Spearman $\rho = 0.832723$, Kendall $\tau = 0.675214$.

LOO Rk	Country	LOO Comp.	Base Rk	Δ
1	Ireland	0.42385530	3	+2
2	Malta	0.42097778	1	-1
3	Cyprus	0.41215918	2	-1
4	Finland	0.40453259	6	+2
5	Denmark	0.36457631	4	-1
6	Austria	0.34198492	8	+2
7	Estonia	0.34086318	13	+6
8	Croatia	0.33592058	5	-3
9	Greece	0.32815101	11	+2
10	Slovenia	0.31042649	10	+0
11	Luxembourg	0.30323505	12	+1
12	Sweden	0.30224250	7	-5
13	Portugal	0.29932866	17	+4
14	Hungary	0.28665076	16	+2
15	Czechia	0.28473209	19	+4
16	Slovakia	0.28369880	26	+10
17	Romania	0.26966391	18	+1
18	Poland	0.26608287	20	+2
19	Italy	0.26451774	9	-10
20	Belgium	0.25609166	21	+1
21	Lithuania	0.23803065	23	+2
22	Spain	0.23695058	24	+2
23	Bulgaria	0.23620677	14	-9
24	Latvia	0.22818238	25	+1
25	Netherlands	0.21027763	15	-10
26	France	0.20873628	27	+1
27	Germany	0.20493083	22	-5

B.2.5 LOO: Critical inputs axis omitted

Table 43: LOO ranking excluding Critical Inputs. Spearman $\rho = 0.941392$, Kendall $\tau = 0.846154$.

LOO Rk	Country	LOO Comp.	Base Rk	Δ
1	Malta	0.53922781	1	+0
2	Cyprus	0.47713567	2	+0
3	Ireland	0.45219113	3	+0
4	Sweden	0.43994261	7	+3
5	Denmark	0.42205299	4	-1
6	Italy	0.40695397	9	+3
7	Slovenia	0.39914798	10	+3
8	Austria	0.39528212	8	+0
9	Greece	0.39404695	11	+2
10	Luxembourg	0.38761105	12	+2
11	Croatia	0.38043167	5	-6
12	Bulgaria	0.37864827	14	+2
13	Netherlands	0.37615076	15	+2
14	Finland	0.37306164	6	-8
15	Romania	0.35074174	18	+3
16	Hungary	0.34878430	16	+0
17	Czechia	0.34760574	19	+2
18	Estonia	0.33474057	13	-5
19	Portugal	0.33416993	17	-2
20	Poland	0.32167032	20	+0
21	Belgium	0.31106471	21	+0
22	Lithuania	0.29237411	23	+1
23	Spain	0.29065725	24	+1
24	Germany	0.27684025	22	-2
25	Latvia	0.26610162	25	+0
26	Slovakia	0.25271793	26	+0
27	France	0.25056058	27	+0

B.2.6 LOO: Logistics axis omitted

Table 44: LOO ranking excluding Logistics. Spearman $\rho = 0.926129$, Kendall $\tau = 0.789174$.

LOO Rk	Country	LOO Comp.	Base Rk	Δ
1	Malta	0.42097778	1	+0
2	Croatia	0.41361961	5	+3

Table 44: LOO excl. Logistics (cont.).

LOO Rk	Country	LOO Comp.	Base Rk	Δ
3	Ireland	0.39088418	3	+0
4	Denmark	0.38085124	4	+0
5	Cyprus	0.37364305	2	-3
6	Slovenia	0.35432099	10	+4
7	Austria	0.35381346	8	+1
8	Italy	0.35307104	9	+1
9	Bulgaria	0.33575459	14	+5
10	Netherlands	0.33120457	15	+5
11	Sweden	0.32743308	7	-4
12	Romania	0.32487674	18	+6
13	Finland	0.32299538	6	-7
14	Estonia	0.31801027	13	-1
15	Luxembourg	0.30848614	12	-3
16	Hungary	0.30243720	16	+0
17	Greece	0.29448258	11	-6
18	Portugal	0.28452391	17	-1
19	Czechia	0.27566336	19	+0
20	Belgium	0.27515129	21	+1
21	Germany	0.26687374	22	+1
22	Poland	0.25701442	20	-2
23	Lithuania	0.24910681	23	+0
24	Spain	0.22665745	24	+0
25	Latvia	0.22269705	25	+0
26	France	0.21221822	27	+1
27	Slovakia	0.18659751	26	-1

B.3 Z-score standardisation: full rank comparison

Table 22 in Section 6.3 compares baseline and z-score ranks for all 27 countries. See Table 22 for the full table.

B.4 Dirichlet weight perturbation: full volatility table

Table 23 in Section 6.4 reports mean rank, rank standard deviation, and 5th/95th percentile ranks for all 27 countries. See Table 23 for the full table.

B.5 Winsorization: full rank comparison

Table 24 in Section 6.5 compares baseline and winsorized ranks for all 27 countries. See Table 24 for the full table.

C Reproducibility Protocol

This appendix documents the complete procedure for reproducing all results presented in this paper.

C.1 Software requirements

- **Python** $\geq$ 3.10 (tested with 3.14.0).
- **NumPy** $\geq$ 2.0 (tested with 2.2.6). NumPy is the only external dependency.
- **LaTeX** distribution with pdfLaTeX (tested with TeX Live 2024), including packages: `pgfplots` ($\geq$ 1.18), `tikz`, `siunitx`, `longtable`, `booktabs`, `natbib`, `cleveref`, `threeparttable`, `pdflscape`, `adjustbox`, `subcaption`, `listings`, `xcolor`.

C.2 Directory structure

The project is organised as follows:

```
paper2/
+-- main.tex                % Master document
+-- references.bib          % Bibliography
+-- sections/
|   +-- 00_executive_summary.tex
|   +-- 01_introduction.tex
|   +-- ...
|   +-- 10_data_availability.tex
+-- tables/                 % Generated .tex table fragments
|   +-- T06_axis_top_bottom.tex
|   +-- T07_axis_dominance.tex
|   +-- ...
|   +-- radar_*.dat
+-- figures/                % TikZ figure source files
|   +-- F01_histogram.tex
|   +-- ...
|   +-- F15_radar_FR.tex
+-- appendix/
|   +-- appendix_a_tables.tex
|   +-- appendix_b_robustness.tex
|   +-- appendix_c_reproducibility.tex
|   +-- appendix_d_code.tex
+-- scripts/
```

```
+-- isi_results_paper2.py
```

C.3 Step-by-step reproduction

1. **Install Python and NumPy.**

   ```
   python -m venv .venv
   source .venv/bin/activate
   pip install numpy
   ```

2. **Run the computation script.**

   ```
   cd paper2/scripts
   python isi_results_paper2.py
   ```

   This generates all `.tex` table fragments and `.dat` coordinate files in the `tables/` directory. The script embeds the full ISI v1.0 data snapshot and performs all computations (LOO, z-score, Dirichlet MC, winsorization, variance decomposition, Lorenz curve, ECDF, histograms, radar coordinates).

3. **Compile the LaTeX document.**

   ```
   cd paper2
   pdflatex main.tex
   bibtex main
   pdflatex main.tex
   pdflatex main.tex
   ```

   Three passes are required to resolve cross-references and the bibliography.

4. **Verify.** The final PDF should compile with 0 errors, 0 warnings (excluding font-related warnings), and 0 overfull boxes.

C.4 Random seed

The Dirichlet Monte Carlo exercise uses a fixed random seed (`np.random.seed(42)`) to ensure exact reproducibility of rank volatility statistics. Changing the seed will produce quantitatively similar but not identical results.

C.5 Data integrity

The ISI v1.0 data snapshot is embedded directly in the Python script (`isi_results_paper2.py`, lines 37–77) as a list of tuples. No external data files need to be downloaded or configured. The embedded data constitutes the single authoritative source for all computations in this paper.

D Computation Code

The complete Python script used to generate all derived tables and data files is listed below. The script requires only NumPy and embeds the full ISI v1.0 data snapshot.

Listing 1: ISI Results Paper 2 — Computation Engine (`isi_results_paper2.py`).

```
#!/usr/bin/env python3
"""
ISI Results Paper 2 -- Computation Engine (Expanded)
====================================================
Reads the ISI v1.0 EU-27 snapshot (embedded) and produces all derived
tables for Paper 2.

Computations:
  1.  Axis dominance table
  2.  Variance contribution decomposition (full, with own vs cross)
  3.  Leave-one-axis-out (LOO) with Spearman and Kendall tau
  4.  Z-score standardized composite
  5.  Dirichlet weight perturbation (Monte Carlo)
  6.  Winsorization (cap Axis 4 & 6 at P95)
  7.  Top-5 / Bottom-5 per axis
  8.  Lorenz curve coordinates
  9.  Axis rank tables (Appendix A)
  10. Histogram / ECDF coordinates
  11. Radar plot data per country
  12. Boxplot statistics
  13. Rank elasticity to axis perturbation
  14. Axis contribution to rank differentiation (R-squared, partial corr)
  15. Classification compression diagnostics
  16. Defence axis structural analysis
  17. Compensability analysis
  18. Geographic / structural correlates
  19. Correlation t-statistics
  20. Spearman correlation matrix
  21. LOO individual axis tables (6 tables for appendix)
  22. Multiple regression diagnostics

Usage:
    python isi_results_paper2.py

No external dependencies beyond numpy.
"""

import numpy as np
import os

AXIS_NAMES = [
    "Financial", "Energy", "Technology",
    "Defence", "Critical Inputs", "Logistics"
]
AXIS_SLUGS = [
    "financial", "energy", "technology",
    "defence", "critical_inputs", "logistics"
]

# fmt: off
DATA = [
    ("MT", "Malta",        1,  0.51748148, "highly_concentrated",     0.13998786, 0.35068126, 0.20546991, 1.00000000, 0.40874987, 1.00000000),
    ("CY", "Cyprus",       2,  0.46821264, "moderately_concentrated", 0.12407171, 0.34768252, 0.22438354, 0.74847998, 0.42359750, 0.94106062),
    ("IE", "Ireland",      3,  0.42802519, "moderately_concentrated", 0.14666620, 0.46472525, 0.58695939, 0.44887460, 0.30719544, 0.61373023),
    ("DK", "Denmark",      4,  0.42442738, "moderately_concentrated", 0.12401663, 0.47029821, 0.14995928, 0.72368274, 0.43629935, 0.64230809),
    ("HR", "Croatia",      5,  0.40434768, "moderately_concentrated", 0.27299265, 0.35125471, 0.17343982, 0.74648317, 0.52392772, 0.35798800),
    ("FI", "Finland",      6,  0.40205136, "moderately_concentrated", 0.11775073, 0.36268783, 0.19789321, 0.38964518, 0.54699996, 0.79733123),
    ("SE", "Sweden",       7,  0.39871532, "moderately_concentrated", 0.11646504, 0.32721403, 0.11982797, 0.88107943, 0.19257891, 0.75512656),
    ("AT", "Austria",      8,  0.38249806, "moderately_concentrated", 0.14903467, 0.45581191, 0.26057910, 0.58506379, 0.31857781, 0.52592111),
    ("IT", "Italy",        9,  0.37386086, "moderately_concentrated", 0.18753526, 0.30781423, 0.14103384, 0.92057651, 0.20839534, 0.47781001),
    ("SI", "Slovenia",     10, 0.36979186, "moderately_concentrated", 0.11917574, 0.40540014, 0.35739912,
```

```
        0.66661871, 0.22301125, 0.44714621),
    ("EL", "Greece",        11, 0.36537350, "moderately_concentrated", 0.20078311, 0.33541623, 0.16272142, 0.55148593, 0.22200621, 0.71982808),
    ("LU", "Luxembourg",    12, 0.36016893, "moderately_concentrated", 0.11462818, 0.35980871, 0.20019721, 0.64483831, 0.22295831, 0.61858285),
    ("EE", "Estonia",       13, 0.34332177, "moderately_concentrated", 0.18172493, 0.43935251, 0.22713142, 0.35561474, 0.38622776, 0.46987927),
    ("BG", "Bulgaria",      14, 0.34027984, "moderately_concentrated", 0.16052965, 0.35711773, 0.15204272, 0.86064518, 0.14843767, 0.36290606),
    ("NL", "Netherlands",   15, 0.33515702, "moderately_concentrated", 0.12107826, 0.31992851, 0.12527375, 0.95955396, 0.13018835, 0.35491930),
    ("HU", "Hungary",       16, 0.32966227, "moderately_concentrated", 0.12222969, 0.38580312, 0.22538122, 0.54471981, 0.23405214, 0.46578765),
    ("PT", "Portugal",      17, 0.32173430, "moderately_concentrated", 0.18303606, 0.34860414, 0.19766062, 0.43376254, 0.25955617, 0.50778629),
    ("RO", "Romania",       18, 0.31864192, "moderately_concentrated", 0.20033340, 0.40546401, 0.29691153, 0.56353195, 0.15814280, 0.28746783),
    ("CZ", "Czechia",       19, 0.31796630, "moderately_concentrated", 0.16116762, 0.38082003, 0.18242269, 0.48413737, 0.16976911, 0.52948100),
    ("PL", "Poland",        20, 0.29561321, "moderately_concentrated", 0.13314812, 0.33487780, 0.20845359, 0.44326493, 0.16532767, 0.48860717),
    ("BE", "Belgium",       21, 0.28796781, "moderately_concentrated", 0.15348744, 0.30789144, 0.29454573, 0.44734854, 0.17248330, 0.35205038),
    ("DE", "Germany",       22, 0.26788694, "moderately_concentrated", 0.10186464, 0.32908790, 0.09762824, 0.58266751, 0.22312039, 0.27295297),
    ("LT", "Lithuania",     23, 0.26540520, "moderately_concentrated", 0.13349636, 0.33747627, 0.24172279, 0.40227798, 0.13056067, 0.34689716),
    ("ES", "Spain",         24, 0.26093977, "moderately_concentrated", 0.14554195, 0.30187479, 0.19263249, 0.38088569, 0.11235234, 0.43235135),
    ("LV", "Latvia",        25, 0.24695137, "mildly_concentrated",     0.12316988, 0.36522282, 0.13309607, 0.34079636, 0.15120012, 0.36822299),
    ("SK", "Slovakia",      26, 0.23641567, "mildly_concentrated",     0.16163673, 0.39995460, 0.21649186, 0.00000000, 0.15490434, 0.48550648),
    ("FR", "France",        27, 0.23567126, "mildly_concentrated",     0.10039135, 0.30911838, 0.12001059, 0.37034617, 0.16122463, 0.35293643),
]
# fmt: on

N = len(DATA)
assert N == 27

ISO   = [d[0] for d in DATA]
NAMES = [d[1] for d in DATA]
RANKS = np.array([d[2] for d in DATA])
COMP  = np.array([d[3] for d in DATA])
CLASS = [d[4] for d in DATA]
AXES  = np.array([[d[5+j] for j in range(6)] for d in DATA])

OUT_DIR = os.path.join(os.path.dirname(os.path.abspath(__file__)), "..", "tables")
os.makedirs(OUT_DIR, exist_ok=True)

def write_table(filename, content):
    path = os.path.join(OUT_DIR, filename)
    with open(path, "w") as f:
        f.write(content)
    print(f"  Written: {path}")

def rankdata(x):
    n = len(x)
    order = np.argsort(-x)
    ranks = np.empty(n, dtype=float)
    ranks[order] = np.arange(1, n+1, dtype=float)
    return ranks

def spearman_corr(r1, r2):
    d = r1 - r2
    n = len(r1)
    return 1.0 - (6.0 * np.sum(d**2)) / (n * (n**2 - 1))

def kendall_tau(r1, r2):
    n = len(r1)
    concordant = 0
    discordant = 0
    for i in range(n):
        for j in range(i+1, n):
            s1 = np.sign(r1[i] - r1[j])
            s2 = np.sign(r2[i] - r2[j])
            prod = s1 * s2
            if prod > 0:
                concordant += 1
```

```python
            elif prod < 0:
                discordant += 1
    pairs = 0.5 * n * (n - 1)
    return (concordant - discordant) / pairs if pairs > 0 else 0.0

def pearson_corr(x, y):
    mx, my = np.mean(x), np.mean(y)
    num = np.sum((x - mx) * (y - my))
    den = np.sqrt(np.sum((x - mx)**2) * np.sum((y - my)**2))
    return num / den if den > 0 else 0.0


# ============================================================================
# 1. AXIS DOMINANCE TABLE
# ============================================================================
print("\n" + "="*72)
print("1. AXIS DOMINANCE")
print("="*72)

max_axis_idx = np.argmax(AXES, axis=1)
second_max_idx = np.zeros(N, dtype=int)
for i in range(N):
    sorted_idx = np.argsort(-AXES[i])
    second_max_idx[i] = sorted_idx[1]

axis_dom_counts = np.zeros(6, dtype=int)
for i in range(N):
    axis_dom_counts[max_axis_idx[i]] += 1

for j in range(6):
    print(f"  {AXIS_NAMES[j]:20s}: {axis_dom_counts[j]}")

lines = []
lines.append("% Table T7: Axis dominance per country")
lines.append("\\begin{longtable}{@{}rlS[table-format=1.8]lS[table-format=1.8]l@{}}")
lines.append("\\caption{Axis dominance per country: highest- and second-highest-scoring axes.}")
lines.append("\\label{tab:axis-dominance}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Max Score}} & {\\textbf{Max Axis}} & {\\textbf{2nd Score}} & {\\textbf{2nd Axis}} \\\\")
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Axis dominance per country (continued).}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Max Score}} & {\\textbf{Max Axis}} & {\\textbf{2nd Score}} & {\\textbf{2nd Axis}} \\\\")
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
for i in range(N):
    mi = max_axis_idx[i]
    si = second_max_idx[i]
    lines.append(f"  {RANKS[i]:.0f} & {NAMES[i]} & {AXES[i,mi]:.8f} & {AXIS_NAMES[mi]} & {AXES[i,si]:.8f} & {AXIS_NAMES[si]} \\\\")
lines.append("\\end{longtable}")
write_table("T07_axis_dominance.tex", "\n".join(lines))


# ============================================================================
# 2. VARIANCE CONTRIBUTION DECOMPOSITION (FULL)
# ============================================================================
print("\n" + "="*72)
print("2. VARIANCE CONTRIBUTION (FULL)")
print("="*72)

axis_vars = np.var(AXES, axis=0, ddof=0)
cov_matrix = np.cov(AXES.T, ddof=0)
total_composite_var = np.var(COMP, ddof=0)
var_decomp_check = np.sum(cov_matrix) / 36.0

print(f"  Composite variance (direct):    {total_composite_var:.10f}")
print(f"  Composite variance (from cov):  {var_decomp_check:.10f}")

marginal_contrib = np.sum(cov_matrix, axis=1) / 36.0
pct_contrib = marginal_contrib / var_decomp_check * 100.0

own_var_share = np.array([cov_matrix[j, j] / 36.0 for j in range(6)])
cov_share = np.array([sum(cov_matrix[j, k] / 36.0 for k in range(6) if k != j) for j in range(6)])

```

```python
total_own = sum(cov_matrix[j, j] for j in range(6)) / 36.0
total_cross = sum(cov_matrix[j, k] for j in range(6) for k in range(6) if j != k) / 36.0
print(f"  Total own-variance:   {total_own:.10f} ({total_own/var_decomp_check*100:.2f}%)")
print(f"  Total cross-covar:    {total_cross:.10f} ({total_cross/var_decomp_check*100:.2f}%)")

for j in range(6):
    print(f"  {AXIS_NAMES[j]:20s}: var={axis_vars[j]:.8f}  marginal={marginal_contrib[j]:.8f}  pct={pct_contrib[j]:.2f}%")
print(f"  Defence+Logistics:    {pct_contrib[3]+pct_contrib[5]:.2f}%")

# T08: summary
lines = []
lines.append("% Table T8: Variance contribution decomposition")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Variance contribution decomposition of the ISI composite.  Each axis's")
lines.append("marginal contribution equals $(1/36)\\sum_{k=1}^{6}\\operatorname{Cov}(A_j, A_k)$,")
lines.append("reflecting both own-variance and covariance with other axes.}")
lines.append("\\label{tab:variance-decomp}")
lines.append("\\begin{tabular}{@{}lS[table-format=1.8]S[table-format=1.8]S[table-format=2.2]@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Axis}} & {\\textbf{Var (pop.)}} & {\\textbf{Marginal Contrib.}} & {\\textbf{\\% of Composite Var}} \\\\")
lines.append("\\midrule")
for j in range(6):
    lines.append(f"  {AXIS_NAMES[j]} & {axis_vars[j]:.8f} & {marginal_contrib[j]:.8f} & {pct_contrib[j]:.2f} \\\\")
lines.append("\\midrule")
lines.append(f"  Total & {np.sum(axis_vars):.8f} & {var_decomp_check:.8f} & {np.sum(pct_contrib):.2f} \\\\")
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("T08_variance_decomp.tex", "\n".join(lines))

# T08b: detailed own vs cross
lines = []
lines.append("% Table T8b: Detailed own-variance vs covariance decomposition")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Decomposition of each axis's marginal contribution into own-variance and")
lines.append("cross-covariance components.  Own: $(1/36)\\operatorname{Var}(A_j)$.")
lines.append("Cross: $(1/36)\\sum_{k \\neq j}\\operatorname{Cov}(A_j, A_k)$.}")
lines.append("\\label{tab:variance-decomp-detail}")
lines.append("\\small")
lines.append("\\begin{tabular}{@{}l S[table-format=1.8] S[table-format=+1.8] S[table-format=1.8] S[table-format=2.2] S[table-format=+2.2] S[table-format=2.2]@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Axis}} & {\\textbf{Own Var}} & {\\textbf{Cross Cov}} & {\\textbf{Marginal}} & {\\textbf{\\% Own}} & {\\textbf{\\% Cross}} & {\\textbf{\\% Total}} \\\\")
lines.append("\\midrule")
for j in range(6):
    pct_own = own_var_share[j] / var_decomp_check * 100
    pct_cross = cov_share[j] / var_decomp_check * 100
    lines.append(f"  {AXIS_NAMES[j]} & {own_var_share[j]:.8f} & {cov_share[j]:+.8f} & {marginal_contrib[j]:.8f} & {pct_own:.2f} & {pct_cross:+.2f} & {pct_contrib[j]:.2f} \\\\")
lines.append("\\midrule")
lines.append(f"  Total & {total_own:.8f} & {total_cross:+.8f} & {var_decomp_check:.8f} & {total_own/var_decomp_check*100:.2f} & {total_cross/var_decomp_check*100:.2f} & 100.00 \\\\")
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("T08b_variance_decomp_detail.tex", "\n".join(lines))

# TE1: Full covariance matrix
lines = []
lines.append("% Table TE1: Full 6x6 population covariance matrix")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Population covariance matrix of the six ISI axes, EU-27, 2024.}")
lines.append("\\label{tab:cov-matrix}")
lines.append("\\small")
lines.append("\\sisetup{table-format=+1.6}")
lines.append("\\begin{tabular}{@{}l S S S S S S@{}}")
lines.append("\\toprule")
lines.append(" & {\\textbf{Fin.}} & {\\textbf{Ene.}} & {\\textbf{Tec.}} & {\\textbf{Def.}} & {\\textbf{Crit.}} & {\\textbf{Log.}} \\\\")
lines.append("\\midrule")
for j in range(6):
    row = f"  {AXIS_NAMES[j]}"
    for k in range(6):
        row += f" & {cov_matrix[j,k]:+.6f}"
```

```
    row += " \\\\"
    lines.append(row)
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("TE1_cov_matrix.tex", "\n".join(lines))


# ========================================================================
# 3. LEAVE-ONE-AXIS-OUT (LOO)
# ========================================================================
print("\n" + "="*72)
print("3. LEAVE-ONE-AXIS-OUT (LOO)")
print("="*72)

baseline_ranks = rankdata(COMP)

loo_composites = np.zeros((N, 6))
loo_ranks = np.zeros((N, 6))
loo_spearman = np.zeros(6)
loo_kendall = np.zeros(6)

for j in range(6):
    mask = [k for k in range(6) if k != j]
    loo_comp = np.mean(AXES[:, mask], axis=1)
    loo_composites[:, j] = loo_comp
    lr = rankdata(loo_comp)
    loo_ranks[:, j] = lr
    loo_spearman[j] = spearman_corr(baseline_ranks, lr)
    loo_kendall[j] = kendall_tau(baseline_ranks, lr)
    print(f"  Drop {AXIS_NAMES[j]:20s}: Spearman={loo_spearman[j]:.6f}  Kendall={loo_kendall[j]:.6f}")

rank_displacement = np.zeros((N, 6), dtype=int)
for i in range(N):
    for j in range(6):
        rank_displacement[i, j] = int(baseline_ranks[i] - loo_ranks[i, j])

max_rank_change = np.max(np.abs(rank_displacement), axis=1)
mean_abs_disp = np.mean(np.abs(rank_displacement), axis=0)
max_abs_disp = np.max(np.abs(rank_displacement), axis=0)

# T09: LOO summary with Kendall
lines = []
lines.append("% Table T9: LOO sensitivity summary")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Leave-one-axis-out (LOO) rank sensitivity.  Spearman $\\rho$ and Kendall")
lines.append("$\\tau$ measure rank-order agreement between the baseline and each LOO ranking.")
lines.append("Mean $|\\Delta|$ and Max $|\\Delta|$ report the average and maximum absolute rank")
lines.append("displacement across the 27 countries.}")
lines.append("\\label{tab:loo-summary}")
lines.append("\\begin{tabular}{@{}l S[table-format=1.6] S[table-format=1.6] S[table-format=1.2] r@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Excluded Axis}} & {$\\boldsymbol{\\rho}$ \\textbf{(Spearman)}} & {$\\boldsymbol{\\tau}$ \\
    textbf{(Kendall)}} & {\\textbf{Mean} $|\\Delta|$} & {\\textbf{Max} $|\\Delta|$} \\\\")
lines.append("\\midrule")
for j in range(6):
    lines.append(f"  {AXIS_NAMES[j]} & {loo_spearman[j]:.6f} & {loo_kendall[j]:.6f} & {mean_abs_disp[j]:.2f} & {int(
        max_abs_disp[j])} \\\\")
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("T09_loo_summary.tex", "\n".join(lines))

# TE2: Full rank displacement matrix
lines = []
lines.append("% Table TE2: Full rank displacement matrix (signed)")
lines.append("\\begin{landscape}")
lines.append("\\begin{longtable}{@{}rl rrrrrr r@{}}")
lines.append("\\caption{Signed rank displacement matrix: baseline rank minus LOO rank for each axis.")
lines.append("Positive values indicate the country ranks higher (more concentrated) when the axis is removed;")
lines.append("negative values indicate it ranks lower.}")
lines.append("\\label{tab:rank-displacement}\\\\")
lines.append("\\toprule")
hdr = "{\\textbf{Rk}} & {\\textbf{Country}}"
for j in range(6):
    short = AXIS_NAMES[j][:4]
    hdr += f" & {{$\\Delta(-{short})}}"
hdr += " & {$|\\Delta|_{\\max}$} \\\\"
lines.append(hdr)
```

```
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Rank displacement matrix (continued).}\\\\")
lines.append("\\toprule")
lines.append(hdr)
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
order = np.argsort(baseline_ranks)
for i in order:
    row = f"  {int(baseline_ranks[i])} & {NAMES[i]}"
    for j in range(6):
        row += f" & {rank_displacement[i,j]:+d}"
    row += f" & {int(max_rank_change[i])} \\\\"
    lines.append(row)
lines.append("\\end{longtable}")
lines.append("\\end{landscape}")
write_table("TE2_rank_displacement.tex", "\n".join(lines))

# TB1: LOO detailed ranks
lines = []
lines.append("% Appendix Table: LOO detailed ranks")
lines.append("\\begin{longtable}{@{}rl" + "r"*6 + "r@{}}")
lines.append("\\caption{Detailed rank comparison: baseline vs.\\ each leave-one-axis-out variant.")
lines.append("Columns show the rank under each LOO composite.  $\\Delta_{\\max}$ is the")
lines.append("maximum absolute rank change across all six LOO variants.}")
lines.append("\\label{tab:loo-detail}\\\\")
lines.append("\\toprule")
hdr = "{\\textbf{Rk}} & {\\textbf{Country}}"
for j in range(6):
    short = AXIS_SLUGS[j][:4].capitalize()
    hdr += f" & {{\\textbf{{$-${short}}}}}"
hdr += " & {$\\boldsymbol{\\Delta_{\\max}}$} \\\\"
lines.append(hdr)
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{LOO detailed ranks (continued).}\\\\")
lines.append("\\toprule")
lines.append(hdr)
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
for i in range(N):
    row = f"  {int(baseline_ranks[i])} & {NAMES[i]}"
    for j in range(6):
        row += f" & {int(loo_ranks[i,j])}"
    row += f" & {int(max_rank_change[i])} \\\\"
    lines.append(row)
lines.append("\\end{longtable}")
write_table("TB1_loo_detail.tex", "\n".join(lines))

print("  Most axis-sensitive countries:")
sens_order = np.argsort(-max_rank_change)
for idx in sens_order[:10]:
    print(f"    {NAMES[idx]:15s}: max_displacement={int(max_rank_change[idx])}")


# ============================================================================
# 4. Z-SCORE STANDARDIZED COMPOSITE
# ============================================================================
print("\n" + "="*72)
print("4. Z-SCORE STANDARDIZED COMPOSITE")
print("="*72)

axis_means = np.mean(AXES, axis=0)
axis_stds_pop = np.std(AXES, axis=0, ddof=0)
Z = (AXES - axis_means) / axis_stds_pop
z_composite = np.mean(Z, axis=1)
z_ranks = rankdata(z_composite)
z_spearman = spearman_corr(baseline_ranks, z_ranks)
z_kendall = kendall_tau(baseline_ranks, z_ranks)
print(f"  Spearman rho: {z_spearman:.6f}")
print(f"  Kendall tau:  {z_kendall:.6f}")

lines = []
lines.append("% Table T10: Z-score standardized composite rank comparison")
lines.append("\\begin{longtable}{@{}rlS[table-format=1.8]rS[table-format=2.8]rr@{}}")
lines.append("\\caption{Rank comparison: baseline ISI vs.\\ z-score standardized composite.")
```

```python
lines.append("The z-score composite re-centres each axis to mean zero and unit population")
lines.append("standard deviation before averaging.  Spearman $\\rho = " + f"{z_spearman:.6f}" + "$,")
lines.append("Kendall $\\tau = " + f"{z_kendall:.6f}" + "$.}")
lines.append("\\label{tab:zscore-ranks}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Baseline}} & {\\textbf{Rk (base)}} & {\\textbf{Z-Composite}} & {\\textbf{Rk (z)}} & {$\\boldsymbol{\\Delta}$} \\\\")
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Z-score rank comparison (continued).}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Baseline}} & {\\textbf{Rk (base)}} & {\\textbf{Z-Composite}} & {\\textbf{Rk (z)}} & {$\\boldsymbol{\\Delta}$} \\\\")
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
order = np.argsort(baseline_ranks)
for i in order:
    delta = int(abs(baseline_ranks[i] - z_ranks[i]))
    lines.append(f"  {int(baseline_ranks[i])} & {NAMES[i]} & {COMP[i]:.8f} & {int(baseline_ranks[i])} & {z_composite[i]:.8f} & {int(z_ranks[i])} & {delta} \\\\")
lines.append("\\end{longtable}")
write_table("T10_zscore_ranks.tex", "\n".join(lines))


# ============================================================================
# 5. DIRICHLET WEIGHT PERTURBATION (MONTE CARLO)
# ============================================================================
print("\n" + "="*72)
print("5. DIRICHLET WEIGHT PERTURBATION")
print("="*72)

np.random.seed(20240222)
N_MC = 10000
alpha = np.ones(6) * 10.0

mc_ranks = np.zeros((N, N_MC))
for t in range(N_MC):
    w = np.random.dirichlet(alpha)
    mc_comp = AXES @ w
    mc_ranks[:, t] = rankdata(mc_comp)

mean_rank = np.mean(mc_ranks, axis=1)
std_rank = np.std(mc_ranks, axis=1, ddof=0)
p5_rank = np.percentile(mc_ranks, 5, axis=1)
p95_rank = np.percentile(mc_ranks, 95, axis=1)
prob_top5 = np.mean(mc_ranks <= 5, axis=1) * 100.0
prob_top10 = np.mean(mc_ranks <= 10, axis=1) * 100.0

lines = []
lines.append("% Table T11: Dirichlet weight perturbation rank volatility")
lines.append("\\begin{longtable}{@{}rl S[table-format=2.2] S[table-format=1.2] S[table-format=2.1] S[table-format=2.1] S[table-format=3.1] S[table-format=3.1]@{}}")
lines.append("\\caption{Rank volatility under Dirichlet weight perturbation ($\\alpha_j=10$, $N_{\\text{MC}}=" + str(N_MC) + "$).")
lines.append("Mean rank, standard deviation, 5th--95th percentile band, and probability of")
lines.append("falling in the top-5 or top-10 are reported for each country.}")
lines.append("\\label{tab:dirichlet-volatility}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Mean Rk}} & {$\\boldsymbol{\\sigma}$} & {\\textbf{P5}} & {\\textbf{P95}} & {\\textbf{\\%Top5}} & {\\textbf{\\%Top10}} \\\\")
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Dirichlet rank volatility (continued).}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Mean Rk}} & {$\\boldsymbol{\\sigma}$} & {\\textbf{P5}} & {\\textbf{P95}} & {\\textbf{\\%Top5}} & {\\textbf{\\%Top10}} \\\\")
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
order = np.argsort(baseline_ranks)
for i in order:
    lines.append(f"  {int(baseline_ranks[i])} & {NAMES[i]} & {mean_rank[i]:.2f} & {std_rank[i]:.2f} & {p5_rank[i]:.1f} & {p95_rank[i]:.1f} & {prob_top5[i]:.1f} & {prob_top10[i]:.1f} \\\\")
lines.append("\\end{longtable}")
write_table("T11_dirichlet_volatility.tex", "\n".join(lines))

mc_spearmans = np.array([spearman_corr(baseline_ranks, mc_ranks[:, t]) for t in range(N_MC)])
```

```python
print(f"  Mean Spearman: {np.mean(mc_spearmans):.6f} (std={np.std(mc_spearmans):.6f})")
print(f"  Min: {np.min(mc_spearmans):.6f}  Max: {np.max(mc_spearmans):.6f}")

vol_order = np.argsort(-std_rank)
for idx in vol_order[:5]:
    print(f"    {NAMES[idx]:15s}: std={std_rank[idx]:.2f} band=[{p5_rank[idx]:.0f},{p95_rank[idx]:.0f}]")


# ============================================================================
# 6. WINSORIZATION
# ============================================================================
print("\n" + "="*72)
print("6. WINSORIZATION")
print("="*72)

AXES_W = AXES.copy()
for j_cap in [3, 5]:
    p95 = np.percentile(AXES[:, j_cap], 95)
    print(f"  {AXIS_NAMES[j_cap]} P95 cap: {p95:.8f}")
    AXES_W[:, j_cap] = np.minimum(AXES[:, j_cap], p95)

w_composite = np.mean(AXES_W, axis=1)
w_ranks = rankdata(w_composite)
w_spearman = spearman_corr(baseline_ranks, w_ranks)
print(f"  Spearman rho: {w_spearman:.6f}")

lines = []
lines.append("% Table T12: Winsorization sensitivity")
lines.append("\\begin{longtable}{@{}rlS[table-format=1.8]rS[table-format=1.8]rr@{}}")
lines.append("\\caption{Rank comparison: baseline vs. \\ winsorized composite (Axis~4 and Axis~6)")
lines.append("capped at their respective 95th percentiles).  Spearman $\\rho = " + f"{w_spearman:.6f}" + "$.}")
lines.append("\\label{tab:winsorization}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Baseline}} & {\\textbf{Rk (base)}} & {\\textbf{Winsorized}} & {\\textbf{Rk (win)}} & {$\\boldsymbol{\\Delta}$} \\\\")
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Winsorization (continued).}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Baseline}} & {\\textbf{Rk (base)}} & {\\textbf{Winsorized}} & {\\textbf{Rk (win)}} & {$\\boldsymbol{\\Delta}$} \\\\")
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
order = np.argsort(baseline_ranks)
for i in order:
    delta = int(abs(baseline_ranks[i] - w_ranks[i]))
    lines.append(f"  {int(baseline_ranks[i])} & {NAMES[i]} & {COMP[i]:.8f} & {int(baseline_ranks[i])} & {w_composite[i]:.8f} & {int(w_ranks[i])} & {delta} \\\\")
lines.append("\\end{longtable}")
write_table("T12_winsorization.tex", "\n".join(lines))


# ============================================================================
# 7. TOP-5 / BOTTOM-5 PER AXIS
# ============================================================================
print("\n" + "="*72)
print("7. TOP-5 / BOTTOM-5 PER AXIS")
print("="*72)

lines = []
lines.append("% Table T6: Top-5 and Bottom-5 per axis")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Top-5 (most concentrated) and Bottom-5 (least concentrated) countries per axis.}")
lines.append("\\label{tab:axis-top-bottom}")
lines.append("\\footnotesize")
lines.append("\\begin{tabular}{@{}lllll@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Axis}} & {\\textbf{Top-5 (most conc.)}} & {\\textbf{Score}} & {\\textbf{Bottom-5 (least conc.)}} & {\\textbf{Score}} \\\\")
lines.append("\\midrule")
for j in range(6):
    sorted_idx = np.argsort(-AXES[:, j])
    top5 = sorted_idx[:5]
    bot5 = sorted_idx[-5:][::-1]
    for k in range(5):
        prefix = AXIS_NAMES[j] if k == 0 else ""
        ti = top5[k]
```

```
        bi = bot5[k]
        lines.append(f"  {prefix} & {NAMES[ti]} & {AXES[ti,j]:.8f} & {NAMES[bi]} & {AXES[bi,j]:.8f} \\\\")
    if j < 5:
        lines.append("\\addlinespace")
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("T06_axis_top_bottom.tex", "\n".join(lines))


# ========================================================================
# 8. LORENZ CURVE COORDINATES
# ========================================================================
print("\n" + "="*72)
print("8. LORENZ CURVE")
print("="*72)

sorted_comp = np.sort(COMP)
cum_share = np.cumsum(sorted_comp) / np.sum(sorted_comp)
pop_share = np.arange(1, N+1) / N
gini = 1.0 - 2.0 * np.trapezoid(cum_share, pop_share)
print(f"  Gini: {gini:.6f}")

lines = []
lines.append("% Lorenz curve coordinates")
lines.append("% population_share cumulative_score_share")
lines.append("0.0000 0.0000")
for k in range(N):
    lines.append(f"{pop_share[k]:.6f} {cum_share[k]:.6f}")
write_table("lorenz_coords.dat", "\n".join(lines))


# ========================================================================
# 9. AXIS RANK TABLES
# ========================================================================
print("\n" + "="*72)
print("9. AXIS RANK TABLES")
print("="*72)

for j in range(6):
    ax_ranks = rankdata(AXES[:, j])
    sorted_idx = np.argsort(ax_ranks)
    lines = []
    lines.append(f"% Appendix: {AXIS_NAMES[j]} axis ranking")
    lines.append("\\begin{longtable}{@{}rlS[table-format=1.8]@{}}")
    lines.append(f"\\caption{{{AXIS_NAMES[j]} axis ranking, EU-27, 2024.}}")
    lines.append(f"\\label{{tab:axis-rank-{AXIS_SLUGS[j]}}}\\\\")
    lines.append("\\toprule")
    lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Score}} \\\\")
    lines.append("\\midrule")
    lines.append("\\endfirsthead")
    lines.append(f"\\caption[]{{{AXIS_NAMES[j]} axis ranking (continued).}}\\\\")
    lines.append("\\toprule")
    lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Score}} \\\\")
    lines.append("\\midrule")
    lines.append("\\endhead")
    lines.append("\\bottomrule")
    lines.append("\\endlastfoot")
    for i in sorted_idx:
        lines.append(f"  {int(ax_ranks[i])} & {NAMES[i]} & {AXES[i,j]:.8f} \\\\")
    lines.append("\\end{longtable}")
    write_table(f"TA_{AXIS_SLUGS[j]}_ranking.tex", "\n".join(lines))
    print(f"  Written: {AXIS_NAMES[j]}")


# ========================================================================
# 10. HISTOGRAM / ECDF COORDINATES
# ========================================================================
print("\n" + "="*72)
print("10. HISTOGRAM/ECDF")
print("="*72)

sorted_comp_ecdf = np.sort(COMP)
lines = []
lines.append("% ECDF coordinates")
lines.append("% composite_score ecdf_value")
for k in range(N):
    lines.append(f"{sorted_comp_ecdf[k]:.8f} {(k+1)/N:.8f}")
write_table("ecdf_coords.dat", "\n".join(lines))

```

```python
bin_edges = np.linspace(0.20, 0.55, 8)
hist_counts, _ = np.histogram(COMP, bins=bin_edges)
lines = []
lines.append("% Histogram coordinates (bin_center, count)")
for k in range(len(hist_counts)):
    center = (bin_edges[k] + bin_edges[k+1]) / 2.0
    lines.append(f"{center:.4f} {hist_counts[k]}")
write_table("histogram_coords.dat", "\n".join(lines))


# ============================================================================
# 11. RADAR PLOT DATA
# ============================================================================
print("\n" + "="*72)
print("11. RADAR PLOT DATA")
print("="*72)

selected = ["MT", "CY", "IE", "NL", "DE", "SK", "FI", "FR"]
for iso in selected:
    idx = ISO.index(iso)
    lines = []
    lines.append(f"% Radar data for {NAMES[idx]} ({iso})")
    lines.append("% axis_index axis_name score")
    for j in range(6):
        lines.append(f"{j} {AXIS_SLUGS[j]} {AXES[idx,j]:.8f}")
    write_table(f"radar_{iso}.dat", "\n".join(lines))
    print(f"  {NAMES[idx]}")


# ============================================================================
# 12. BOXPLOT STATS
# ============================================================================
print("\n" + "="*72)
print("12. BOXPLOT STATISTICS")
print("="*72)

for j in range(6):
    vals = AXES[:, j]
    q1 = np.percentile(vals, 25)
    med = np.median(vals)
    q3 = np.percentile(vals, 75)
    iqr = q3 - q1
    wlo = np.min(vals[vals >= q1 - 1.5*iqr])
    whi = np.max(vals[vals <= q3 + 1.5*iqr])
    print(f"  {AXIS_NAMES[j]:20s}: Q1={q1:.4f} Med={med:.4f} Q3={q3:.4f} Lo={wlo:.4f} Hi={whi:.4f}")


# ============================================================================
# 13. RANK ELASTICITY TO AXIS PERTURBATION
# ============================================================================
print("\n" + "="*72)
print("13. RANK ELASTICITY")
print("="*72)

epsilons = [0.05, 0.10, 0.20]  # 5%, 10%, 20% shocks
rank_sens_by_eps = {}

for epsilon in epsilons:
    rank_sens_pos = np.zeros((N, 6))
    for j in range(6):
        Ap = AXES.copy()
        Ap[:, j] = np.clip(AXES[:, j] * (1 + epsilon), 0.0, 1.0)
        rp = rankdata(np.mean(Ap, axis=1))
        rank_sens_pos[:, j] = baseline_ranks - rp
    rank_sens_by_eps[epsilon] = rank_sens_pos

# Use 10% shock for the main table
rank_sens_pos = rank_sens_by_eps[0.10]
total_sens = np.sum(np.abs(rank_sens_pos), axis=1)
fragile_order = np.argsort(-total_sens)

print("  Top 10 rank-fragile (sum |delta| from +1%):")
for idx in fragile_order[:10]:
    print(f"    {NAMES[idx]:15s}: total={total_sens[idx]:.0f}  per_axis={rank_sens_pos[idx,:]}")

locked = np.sum(total_sens == 0)
print(f"  Structurally locked-in countries: {locked}")
for idx in range(N):
    if total_sens[idx] == 0:
        print(f"    {NAMES[idx]}")
```

```python

# T13: Rank elasticity table
lines = []
lines.append("% Table T13: Rank sensitivity to +10% axis perturbation")
lines.append("\\begin{landscape}")
lines.append("\\begin{longtable}{@{}rl rrrrrr r@{}}")
lines.append("\\caption{Rank change per country under a $+10\\%$ multiplicative perturbation to each axis.")
lines.append("Positive values indicate rank improvement (toward rank 1);")
lines.append("negative values indicate deterioration.  $\\Sigma|\\Delta|$ is the total absolute displacement.}")
lines.append("\\label{tab:rank-elasticity}\\\\")
lines.append("\\toprule")
hdr = "{\\textbf{Rk}} & {\\textbf{Country}}"
for j in range(6):
    hdr += f" & {{\\textbf{{{AXIS_NAMES[j][:4]}}}}}"
hdr += " & {$\\Sigma|\\Delta|$} \\\\"
lines.append(hdr)
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Rank elasticity (continued).}\\\\")
lines.append("\\toprule")
lines.append(hdr)
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
order = np.argsort(baseline_ranks)
for i in order:
    row = f"  {int(baseline_ranks[i])} & {NAMES[i]}"
    for j in range(6):
        row += f" & {int(rank_sens_pos[i,j]):+d}"
    row += f" & {int(total_sens[i])} \\\\"
    lines.append(row)
lines.append("\\end{longtable}")
lines.append("\\end{landscape}")
write_table("T13_rank_elasticity.tex", "\n".join(lines))


# ============================================================================
# 14. AXIS CONTRIBUTION TO RANK DIFFERENTIATION
# ============================================================================
print("\n" + "="*72)
print("14. AXIS-RANK CONTRIBUTION")
print("="*72)

axis_rank_corr = np.zeros(6)
axis_rank_r2 = np.zeros(6)
for j in range(6):
    r = pearson_corr(AXES[:, j], -baseline_ranks)
    axis_rank_corr[j] = r
    axis_rank_r2[j] = r**2
    print(f"  {AXIS_NAMES[j]:20s}: r={r:.6f}  R2={r**2:.6f}")

partial_corr = np.zeros(6)
for j in range(6):
    other = [k for k in range(6) if k != j]
    X_o = np.column_stack([np.ones(N), AXES[:, other]])
    y = -baseline_ranks.astype(float)
    x_t = AXES[:, j]
    beta_y = np.linalg.lstsq(X_o, y, rcond=None)[0]
    beta_x = np.linalg.lstsq(X_o, x_t, rcond=None)[0]
    resid_y = y - X_o @ beta_y
    resid_x = x_t - X_o @ beta_x
    partial_corr[j] = pearson_corr(resid_x, resid_y)
    print(f"  {AXIS_NAMES[j]:20s}: partial_r={partial_corr[j]:.6f}")

lines = []
lines.append("% Table T14: Axis contribution to rank differentiation")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Axis contribution to composite rank differentiation.  Bivariate $R^2$:")
lines.append("share of rank variance explained by each axis alone.  Partial $r$: association with")
lines.append("rank after controlling for all other axes.}")
lines.append("\\label{tab:axis-rank-contrib}")
lines.append("\\begin{tabular}{@{}l S[table-format=+1.4] S[table-format=1.4] S[table-format=+1.4]@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Axis}} & {\\textbf{Pearson $r$}} & {$\\boldsymbol{R^2}$} & {\\textbf{Partial $r$}} \\\\")
lines.append("\\midrule")
for j in range(6):
    lines.append(f"  {AXIS_NAMES[j]} & {axis_rank_corr[j]:+.4f} & {axis_rank_r2[j]:.4f} & {partial_corr[j]:+.4f}
        \\\\")
```

```python
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("T14_axis_rank_contrib.tex", "\n".join(lines))


# ============================================================================
# 15. CLASSIFICATION COMPRESSION DIAGNOSTICS
# ============================================================================
print("\n" + "="*72)
print("15. COMPRESSION DIAGNOSTICS")
print("="*72)

tiers = {
    "Highly concentrated": (0.50, 1.01),
    "Moderately concentrated": (0.25, 0.50),
    "Mildly concentrated": (0.15, 0.25),
    "Unconcentrated": (0.00, 0.15),
}

grand_mean = np.mean(COMP)
ssw = 0.0
ssb = 0.0
tier_info = {}

for tname, (lo, hi) in tiers.items():
    mask = (COMP >= lo) & (COMP < hi)
    if np.sum(mask) == 0:
        continue
    vals = COMP[mask]
    tmean = np.mean(vals)
    tvar = np.var(vals, ddof=0) if len(vals) > 1 else 0.0
    tier_info[tname] = (len(vals), tmean, tvar)
    ssw += np.sum((vals - tmean)**2)
    ssb += len(vals) * (tmean - grand_mean)**2

total_ss = np.sum((COMP - grand_mean)**2)
eta_sq = ssb / total_ss if total_ss > 0 else 0.0

print(f"  Grand mean: {grand_mean:.8f}")
print(f"  SS_within={ssw:.10f}  SS_between={ssb:.10f}  SS_total={total_ss:.10f}")
print(f"  Eta-squared: {eta_sq:.6f}")
print(f"  Gini: {gini:.6f}")

for tname, (n, m, v) in tier_info.items():
    print(f"  {tname:30s}: n={n}  mean={m:.8f}  var={v:.10f}")

# Tercile and quartile alternatives
tercile_edges = np.percentile(COMP, [33.33, 66.67])
quartile_edges = np.percentile(COMP, [25, 50, 75])
print(f"  Terciles: {tercile_edges}")
print(f"  Quartiles: {quartile_edges}")

# T15: Compression diagnostics
lines = []
lines.append("% Table T15: Classification compression diagnostics")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Classification compression diagnostics.  The official four-tier scheme")
lines.append("is compared with data-driven tercile and quartile alternatives.  $\\eta^2 = SS_B / SS_T$")
lines.append("measures between-tier variance share.}")
lines.append("\\label{tab:compression-diagnostics}")
lines.append("\\small")
lines.append("\\begin{tabular}{@{}l r S[table-format=1.8] S[table-format=1.10] l@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Tier}} & {$n$} & {\\textbf{Mean}} & {\\textbf{Var (within)}} & {\\textbf{Scheme}} \\\\")
lines.append("\\midrule")
for tname in ["Highly concentrated", "Moderately concentrated", "Mildly concentrated"]:
    if tname in tier_info:
        n, m, v = tier_info[tname]
        lines.append(f"  {tname} & {n} & {m:.8f} & {v:.10f} & Official \\\\")
lines.append("\\addlinespace")

# Terciles
trc_bounds = [(0, tercile_edges[0]), (tercile_edges[0], tercile_edges[1]), (tercile_edges[1], 1.0)]
trc_labels = ["T1 (low)", "T2 (mid)", "T3 (high)"]
for idx, (lo, hi) in enumerate(trc_bounds):
    mask = (COMP >= lo) & (COMP < hi) if idx < 2 else (COMP >= lo)
    vals = COMP[mask]
    lines.append(f"  {trc_labels[idx]} & {len(vals)} & {np.mean(vals):.8f} & {np.var(vals, ddof=0):.10f} & Tercile
```

```
        \\\\")
lines.append("\\addlinespace")

# Quartiles
qrt_bounds = [(0, quartile_edges[0]), (quartile_edges[0], quartile_edges[1]),
              (quartile_edges[1], quartile_edges[2]), (quartile_edges[2], 1.0)]
qrt_labels = ["Q1 (low)", "Q2", "Q3", "Q4 (high)"]
for idx, (lo, hi) in enumerate(qrt_bounds):
    mask = (COMP >= lo) & (COMP < hi) if idx < 3 else (COMP >= lo)
    vals = COMP[mask]
    lines.append(f"  {qrt_labels[idx]} & {len(vals)} & {np.mean(vals):.8f} & {np.var(vals, ddof=0):.10f} & Quartile  \\\\")
lines.append("\\midrule")
lines.append(f"  \\multicolumn{{5}}{{@{{}}l@{{}}}}{{$\\eta^2$ (official) $= {eta_sq:.6f}$; Gini $= {gini:.6f}$}} \\\\")
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("T15_compression_diagnostics.tex", "\n".join(lines))


# ============================================================================
# 16. DEFENCE AXIS STRUCTURAL ANALYSIS
# ============================================================================
print("\n" + "="*72)
print("16. DEFENCE STRUCTURAL ANALYSIS")
print("="*72)

defence = AXES[:, 3]
defence_is_max = int(np.sum(max_axis_idx == 3))
near_monopoly = int(np.sum(defence > 0.80))
sk_idx = ISO.index("SK")

# Gap: defence minus second-highest (for defence-max countries)
# or defence minus max (for logistics-max countries)
defence_gap = np.zeros(N)
for i in range(N):
    sorted_vals = np.sort(AXES[i])[::-1]
    if max_axis_idx[i] == 3:
        defence_gap[i] = defence[i] - sorted_vals[1]
    else:
        defence_gap[i] = defence[i] - AXES[i, max_axis_idx[i]]

print(f"  Defence is max axis: {defence_is_max}/27 ({defence_is_max/N*100:.1f}%)")
print(f"  Score > 0.80: {near_monopoly}")
print(f"  Score = 0.00: {int(np.sum(defence == 0))}")
print(f"  Mean gap (def-max countries): {np.mean(defence_gap[max_axis_idx==3]):.6f}")

# Skewness and kurtosis
def_mean = np.mean(defence)
def_std = np.std(defence, ddof=0)
def_skew = float(np.mean(((defence - def_mean)/def_std)**3))
def_kurt = float(np.mean(((defence - def_mean)/def_std)**4) - 3)

# T16: Defence structural
lines = []
lines.append("% Table T16: Defence axis structural analysis")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Structural properties of the defence axis distribution, EU-27, 2024.}")
lines.append("\\label{tab:defence-structural}")
lines.append("\\begin{tabular}{@{}lr@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Statistic}} & {\\textbf{Value}} \\\\")
lines.append("\\midrule")
lines.append(f"  Countries where defence is max axis & {defence_is_max} of 27 ({defence_is_max/N*100:.1f}\\%) \\\\")
lines.append(f"  Countries with score $> 0.80$ & {near_monopoly} ({near_monopoly/N*100:.1f}\\%) \\\\")
lines.append(f"  Countries with score $= 0.00$ & {int(np.sum(defence==0))} \\\\")
lines.append(f"  Mean (pop.) & {def_mean:.8f} \\\\")
lines.append(f"  Median & {np.median(defence):.8f} \\\\")
lines.append(f"  Std. \\ dev. (pop.) & {def_std:.8f} \\\\")
lines.append(f"  Range & {np.max(defence) - np.min(defence):.8f} \\\\")
lines.append(f"  IQR & {np.percentile(defence, 75) - np.percentile(defence, 25):.8f} \\\\")
lines.append(f"  Skewness & {def_skew:.4f} \\\\")
lines.append(f"  Excess kurtosis & {def_kurt:.4f} \\\\")
lines.append(f"  Mean def--2nd gap (def-max) & {np.mean(defence_gap[max_axis_idx==3]):.8f} \\\\")
lines.append(f"  Median def--2nd gap (def-max) & {np.median(defence_gap[max_axis_idx==3]):.8f} \\\\")
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
```

```python
write_table("T16_defence_structural.tex", "\n".join(lines))

# T16b: Defence gap per country
lines = []
lines.append("% Table T16b: Defence gap per country")
lines.append("\\begin{longtable}{@{}rl S[table-format=1.8] l S[table-format=+1.8]@{}}")
lines.append("\\caption{Defence axis score, max axis identity, and defence-to-max gap per country.}")
lines.append("\\label{tab:defence-gap}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Def.\\ Score}} & {\\textbf{Max Axis}} & {\\textbf{Gap}} \\\\")
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Defence gap (continued).}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Def.\\ Score}} & {\\textbf{Max Axis}} & {\\textbf{Gap}} \\\\")
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
order = np.argsort(baseline_ranks)
for i in order:
    lines.append(f"  {int(baseline_ranks[i])} & {NAMES[i]} & {defence[i]:.8f} & {AXIS_NAMES[max_axis_idx[i]]} & {defence_gap[i]:+.8f} \\\\")
lines.append("\\end{longtable}")
write_table("T16b_defence_gap.tex", "\n".join(lines))


# ============================================================================
# 17. COMPENSABILITY ANALYSIS
# ============================================================================
print("\n" + "="*72)
print("17. COMPENSABILITY ANALYSIS")
print("="*72)

max_scores = np.max(AXES, axis=1)
min_scores = np.min(AXES, axis=1)
axis_range = max_scores - min_scores
compensation_gap = max_scores - COMP

country_cv = np.zeros(N)
for i in range(N):
    m = np.mean(AXES[i])
    s = np.std(AXES[i], ddof=0)
    country_cv[i] = s / m if m > 0 else 0

comp_order = np.argsort(-compensation_gap)
print("  Most compensated (largest max-composite gap):")
for idx in comp_order[:5]:
    print(f"    {NAMES[idx]:15s}: max={max_scores[idx]:.4f} comp={COMP[idx]:.4f} gap={compensation_gap[idx]:.4f} CV={country_cv[idx]:.4f}")

# T17: Compensability
lines = []
lines.append("% Table T17: Compensability analysis")
lines.append("\\begin{longtable}{@{}rl S[table-format=1.8] S[table-format=1.8] S[table-format=1.8] S[table-format=1.8] S[table-format=1.4]@{}}")
lines.append("\\caption{Compensability analysis.  Gap $= $ Max $- $ Composite.  CV: coefficient of")
lines.append("variation of the six axis scores per country.  High Gap and CV indicate strong")
lines.append("internal compensation where low axes offset high axes.}")
lines.append("\\label{tab:compensability}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Max}} & {\\textbf{Min}} & {\\textbf{Range}} & {\\textbf{Gap}} & {\\textbf{CV}} \\\\")
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Compensability (continued).}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Max}} & {\\textbf{Min}} & {\\textbf{Range}} & {\\textbf{Gap}} & {\\textbf{CV}} \\\\")
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
order = np.argsort(baseline_ranks)
for i in order:
    lines.append(f"  {int(baseline_ranks[i])} & {NAMES[i]} & {max_scores[i]:.8f} & {min_scores[i]:.8f} & {axis_range[i]:.8f} & {compensation_gap[i]:.8f} & {country_cv[i]:.4f} \\\\")
lines.append("\\end{longtable}")
```

```python
write_table("T17_compensability.tex", "\n".join(lines))


# ============================================================================
# 18. GEOGRAPHIC / STRUCTURAL CORRELATES
# ============================================================================
print("\n" + "="*72)
print("18. GEOGRAPHIC CORRELATES")
print("="*72)

ISLAND = {"MT", "CY", "IE"}
SMALL_POP = {"MT", "CY", "LU", "EE", "LV", "LT", "SI", "HR"}
LARGE_POP = {"DE", "FR", "ES", "IT", "PL", "RO", "NL"}
CORE = {"DE", "FR", "BE", "NL", "LU", "AT"}
PERIPHERY = {"MT", "CY", "EL", "PT", "FI", "EE", "LV", "LT", "BG", "RO", "HR", "IE"}

non_island = set(ISO) - ISLAND
mid_pop = set(ISO) - SMALL_POP - LARGE_POP

groups = [
    ("Island/near-island", ISLAND),
    ("Continental", non_island),
    ("Small ($<$5M pop.)", SMALL_POP),
    ("Medium (5--20M)", mid_pop),
    ("Large ($>$20M pop.)", LARGE_POP),
    ("Core EU-6", CORE),
    ("Periphery", PERIPHERY),
]

lines = []
lines.append("% Table T18: Geographic and structural correlates")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{ISI composite by geographic and structural category.}")
lines.append("\\label{tab:geo-correlates}")
lines.append("\\begin{tabular}{@{}l r S[table-format=1.6] S[table-format=1.6] S[table-format=1.6]@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Category}} & {$n$} & {\\textbf{Mean}} & {\\textbf{Std (pop.)}} & {\\textbf{Median}} \\\\")
lines.append("\\midrule")
for glabel, gset in groups:
    mask = np.array([ISO[i] in gset for i in range(N)])
    vals = COMP[mask]
    n = len(vals)
    m = np.mean(vals)
    s = np.std(vals, ddof=0)
    med = np.median(vals)
    lines.append(f"  {glabel} & {n} & {m:.6f} & {s:.6f} & {med:.6f} \\\\")
    print(f"  {glabel:25s}: n={n} mean={m:.6f} std={s:.6f}")
lines.append("\\midrule")
lines.append(f"  EU-27 (all) & 27 & {np.mean(COMP):.6f} & {np.std(COMP, ddof=0):.6f} & {np.median(COMP):.6f} \\\\")
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("T18_geo_correlates.tex", "\n".join(lines))


# ============================================================================
# 19. CORRELATION T-STATISTICS
# ============================================================================
print("\n" + "="*72)
print("19. CORRELATION T-STATISTICS")
print("="*72)

pearson_matrix = np.corrcoef(AXES.T)
t_matrix = np.zeros((6, 6))
for j in range(6):
    for k in range(6):
        if j == k:
            t_matrix[j, k] = np.inf
        else:
            r = pearson_matrix[j, k]
            denom = 1 - r**2
            t_matrix[j, k] = r * np.sqrt((N - 2) / denom) if denom > 0 else np.inf

for j in range(6):
    for k in range(j+1, 6):
        r = pearson_matrix[j, k]
        t = t_matrix[j, k]
        sig = "*" if abs(t) > 2.060 else ""
        print(f"    {AXIS_NAMES[j]:15s} vs {AXIS_NAMES[k]:15s}: r={r:+.4f} t={t:+.4f} {sig}")
```

```python

# TE3: Correlation t-statistics
lines = []
lines.append("% Table TE3: Correlation t-statistics")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Pearson correlations (below diagonal) and $t$-statistics (above diagonal).")
lines.append("$t = r\\sqrt{(N-2)/(1-r^2)}$, $N=27$, df$=25$. Critical value: $t_{0.025,25} = 2.060$;")
lines.append("* indicates significance at 5\\%.}")
lines.append("\\label{tab:corr-tstat}")
lines.append("\\small")
lines.append("\\sisetup{table-format=+1.3}")
lines.append("\\begin{tabular}{@{}l S S S S S S@{}}")
lines.append("\\toprule")
lines.append(" & {\\textbf{Fin.}} & {\\textbf{Ene.}} & {\\textbf{Tec.}} & {\\textbf{Def.}} & {\\textbf{Crit.}} & {\\textbf{Log.}} \\\\")
lines.append("\\midrule")
for j in range(6):
    row = f"  {AXIS_NAMES[j]}"
    for k in range(6):
        if j == k:
            row += " & {--}"
        elif j > k:
            r = pearson_matrix[j, k]
            sig = "*" if abs(t_matrix[j, k]) > 2.060 else ""
            row += f" & {{{r:+.3f}{sig}}}"
        else:
            t = t_matrix[j, k]
            row += f" & {{{t:+.3f}}}"
    row += " \\\\"
    lines.append(row)
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("TE3_corr_tstat.tex", "\n".join(lines))


# ============================================================================
# 20. LOO INDIVIDUAL AXIS TABLES
# ============================================================================
print("\n" + "="*72)
print("20. LOO INDIVIDUAL TABLES")
print("="*72)

for j in range(6):
    order = np.argsort(loo_ranks[:, j])
    lines = []
    lines.append(f"% LOO excluding {AXIS_NAMES[j]}")
    lines.append("\\begin{longtable}{@{}rl S[table-format=1.8] r r@{}}")
    lines.append(f"\\caption{{LOO ranking excluding {AXIS_NAMES[j]}.  Spearman $\\rho = {loo_spearman[j]:.6f}$, Kendall $\\tau = {loo_kendall[j]:.6f}$.}}")
    lines.append(f"\\label{{tab:loo-{AXIS_SLUGS[j]}}}\\\\")
    lines.append("\\toprule")
    lines.append("{\\textbf{LOO Rk}} & {\\textbf{Country}} & {\\textbf{LOO Comp.}} & {\\textbf{Base Rk}} & {$\\boldsymbol{\\Delta}$} \\\\")
    lines.append("\\midrule")
    lines.append("\\endfirsthead")
    lines.append(f"\\caption[]{{LOO excl. \\ {AXIS_NAMES[j]} (cont.).}}\\\\")
    lines.append("\\toprule")
    lines.append("{\\textbf{LOO Rk}} & {\\textbf{Country}} & {\\textbf{LOO Comp.}} & {\\textbf{Base Rk}} & {$\\boldsymbol{\\Delta}$} \\\\")
    lines.append("\\midrule")
    lines.append("\\endhead")
    lines.append("\\bottomrule")
    lines.append("\\endlastfoot")
    for i in order:
        delta = int(baseline_ranks[i] - loo_ranks[i, j])
        lines.append(f"  {int(loo_ranks[i,j])} & {NAMES[i]} & {loo_composites[i,j]:.8f} & {int(baseline_ranks[i])} & {delta:+d} \\\\")
    lines.append("\\end{longtable}")
    write_table(f"TC_loo_{AXIS_SLUGS[j]}.tex", "\n".join(lines))
    print(f"  Written: LOO excl {AXIS_NAMES[j]}")


# ============================================================================
# 21. MULTIPLE REGRESSION
# ============================================================================
print("\n" + "="*72)
print("21. MULTIPLE REGRESSION")
print("="*72)
```

```python

y = -baseline_ranks.astype(float)
X = np.column_stack([np.ones(N), AXES])
beta = np.linalg.lstsq(X, y, rcond=None)[0]
y_hat = X @ beta
ss_res = np.sum((y - y_hat)**2)
ss_tot = np.sum((y - np.mean(y))**2)
r2_full = 1 - ss_res / ss_tot
adj_r2 = 1 - (1 - r2_full) * (N - 1) / (N - 7)
print(f"  R2={r2_full:.6f}  adj_R2={adj_r2:.6f}")
print(f"  Coefficients: intercept={beta[0]:.4f}, " + ", ".join(f"{AXIS_NAMES[j]}={beta[j+1]:.4f}" for j in range(6))
    )


# ========================================================================
print("\n" + "="*72)
print("ALL COMPUTATIONS COMPLETE")
print(f"Output written to: {OUT_DIR}")
print("="*72)
```